

EXERCISES 10.6

Symmetries and Polar Graphs

Identify the symmetries of the curves in Exercises 1–12. Then sketch the curves.

1. $r = 1 + \cos \theta$
2. $r = 2 - 2 \cos \theta$
3. $r = 1 - \sin \theta$
4. $r = 1 + \sin \theta$
5. $r = 2 + \sin \theta$
6. $r = 1 + 2 \sin \theta$
7. $r = \sin(\theta/2)$
8. $r = \cos(\theta/2)$
9. $r^2 = \cos \theta$
10. $r^2 = \sin \theta$
11. $r^2 = -\sin \theta$
12. $r^2 = -\cos \theta$

Graph the lemniscates in Exercises 13–16. What symmetries do these curves have?

13. $r^2 = 4 \cos 2\theta$
14. $r^2 = 4 \sin 2\theta$
15. $r^2 = -\sin 2\theta$
16. $r^2 = -\cos 2\theta$

Slopes of Polar Curves

Find the slopes of the curves in Exercises 17–20 at the given points. Sketch the curves along with their tangents at these points.

17. **Cardioid** $r = -1 + \cos \theta$; $\theta = \pm\pi/2$
18. **Cardioid** $r = -1 + \sin \theta$; $\theta = 0, \pi$
19. **Four-leaved rose** $r = \sin 2\theta$; $\theta = \pm\pi/4, \pm3\pi/4$
20. **Four-leaved rose** $r = \cos 2\theta$; $\theta = 0, \pm\pi/2, \pi$

Limaçons

Graph the limaçons in Exercises 21–24. Limaçon (“lee-ma-sahn”) is Old French for “snail.” You will understand the name when you graph the limaçons in Exercise 21. Equations for limaçons have the form $r = a \pm b \cos \theta$ or $r = a \pm b \sin \theta$. There are four basic shapes.

21. Limaçons with an inner loop

$$\text{a. } r = \frac{1}{2} + \cos \theta \quad \text{b. } r = \frac{1}{2} + \sin \theta$$

22. Cardioids

$$\text{a. } r = 1 - \cos \theta \quad \text{b. } r = -1 + \sin \theta$$

23. Dimpled limaçons

$$\text{a. } r = \frac{3}{2} + \cos \theta \quad \text{b. } r = \frac{3}{2} - \sin \theta$$

24. Oval limaçons

$$\text{a. } r = 2 + \cos \theta \quad \text{b. } r = -2 + \sin \theta$$

Graphing Polar Inequalities

25. Sketch the region defined by the inequalities $-1 \leq r \leq 2$ and $-\pi/2 \leq \theta \leq \pi/2$.
26. Sketch the region defined by the inequalities $0 \leq r \leq 2 \sec \theta$ and $-\pi/4 \leq \theta \leq \pi/4$.

In Exercises 27 and 28, sketch the region defined by the inequality.

27. $0 \leq r \leq 2 - 2 \cos \theta$
28. $0 \leq r^2 \leq \cos \theta$

Intersections

29. Show that the point $(2, 3\pi/4)$ lies on the curve $r = 2 \sin \theta$.
30. Show that $(1/2, 3\pi/2)$ lies on the curve $r = -\sin(\theta/3)$.

Find the points of intersection of the pairs of curves in Exercises 31–38.

31. $r = 1 + \cos \theta$, $r = 1 - \cos \theta$
32. $r = 1 + \sin \theta$, $r = 1 - \sin \theta$
33. $r = 2 \sin \theta$, $r = 2 \sin 2\theta$
34. $r = \cos \theta$, $r = 1 - \cos \theta$
35. $r = \sqrt{2}$, $r^2 = 4 \sin \theta$
36. $r^2 = \sqrt{2} \sin \theta$, $r^2 = \sqrt{2} \cos \theta$
37. $r = 1$, $r^2 = 2 \sin 2\theta$
38. $r^2 = \sqrt{2} \cos 2\theta$, $r^2 = \sqrt{2} \sin 2\theta$

T Find the points of intersection of the pairs of curves in Exercises 39–42.

39. $r^2 = \sin 2\theta$, $r^2 = \cos 2\theta$
40. $r = 1 + \cos \frac{\theta}{2}$, $r = 1 - \sin \frac{\theta}{2}$
41. $r = 1$, $r = 2 \sin 2\theta$
42. $r = 1$, $r^2 = 2 \sin 2\theta$

T Grapher Explorations

43. Which of the following has the same graph as $r = 1 - \cos \theta$?

- a. $r = -1 - \cos \theta$ b. $r = 1 + \cos \theta$

Confirm your answer with algebra.

44. Which of the following has the same graph as $r = \cos 2\theta$?

- a. $r = -\sin(2\theta + \pi/2)$ b. $r = -\cos(\theta/2)$

Confirm your answer with algebra.

45. **A rose within a rose** Graph the equation $r = 1 - 2 \sin 3\theta$.

46. **The nephroid of Freeth** Graph the nephroid of Freeth:

$$r = 1 + 2 \sin \frac{\theta}{2}.$$

47. **Roses** Graph the roses $r = \cos m\theta$ for $m = 1/3, 2, 3$, and 7 .

48. **Spirals** Polar coordinates are just the thing for defining spirals. Graph the following spirals.

- a. $r = \theta$ b. $r = -\theta$

c. *A logarithmic spiral:* $r = e^{\theta/10}$

d. *A hyperbolic spiral:* $r = 8/\theta$

e. *An equilateral hyperbola:* $r = \pm 10/\sqrt{\theta}$

(Use different colors for the two branches.)

Theory and Examples

49. (Continuation of Example 5.) The simultaneous solution of the equations

$$r^2 = 4 \cos \theta \quad (1)$$

$$r = 1 - \cos \theta \quad (2)$$

in the text did not reveal the points $(0, 0)$ and $(2, \pi)$ in which their graphs intersected.

- a. We could have found the point $(2, \pi)$, however, by replacing the (r, θ) in Equation (1) by the equivalent $(-r, \theta + \pi)$ to obtain

$$\begin{aligned} r^2 &= 4 \cos \theta \\ (-r)^2 &= 4 \cos(\theta + \pi) \\ r^2 &= -4 \cos \theta. \end{aligned} \quad (3)$$

Solve Equations (2) and (3) simultaneously to show that $(2, \pi)$ is a common solution. (This will still not reveal that the graphs intersect at $(0, 0)$.)

- b. The origin is still a special case. (It often is.) Here is one way to handle it: Set $r = 0$ in Equations (1) and (2) and solve each equation for a corresponding value of θ . Since $(0, \theta)$ is the origin for *any* θ , this will show that both curves pass through the origin even if they do so for different θ -values.

50. If a curve has any two of the symmetries listed at the beginning of the section, can anything be said about its having or not having the third symmetry? Give reasons for your answer.

*51. Find the maximum width of the petal of the four-leaved rose $r = \cos 2\theta$, which lies along the x -axis.

*52. Find the maximum height above the x -axis of the cardioid $r = 2(1 + \cos \theta)$.

10.7

Areas and Lengths in Polar Coordinates

This section shows how to calculate areas of plane regions, lengths of curves, and areas of surfaces of revolution in polar coordinates.

Area in the Plane

The region OTS in Figure 10.48 is bounded by the rays $\theta = \alpha$ and $\theta = \beta$ and the curve $r = f(\theta)$. We approximate the region with n nonoverlapping fan-shaped circular sectors based on a partition P of angle TOS . The typical sector has radius $r_k = f(\theta_k)$ and central angle of radian measure $\Delta\theta_k$. Its area is $\Delta\theta_k/2\pi$ times the area of a circle of radius r_k , or

$$A_k = \frac{1}{2} r_k^2 \Delta\theta_k = \frac{1}{2} (f(\theta_k))^2 \Delta\theta_k.$$

The area of region OTS is approximately

$$\sum_{k=1}^n A_k = \sum_{k=1}^n \frac{1}{2} (f(\theta_k))^2 \Delta\theta_k.$$

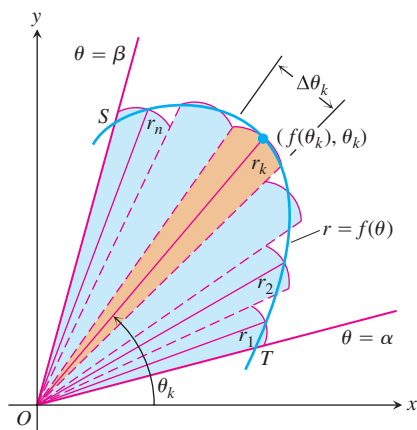


FIGURE 10.48 To derive a formula for the area of region OTS , we approximate the region with fan-shaped circular sectors.

If f is continuous, we expect the approximations to improve as the norm of the partition $\|P\| \rightarrow 0$, and we are led to the following formula for the region's area:

$$\begin{aligned} A &= \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n \frac{1}{2} (f(\theta_k))^2 \Delta\theta_k \\ &= \int_{\alpha}^{\beta} \frac{1}{2} (f(\theta))^2 d\theta. \end{aligned}$$

Area of the Fan-Shaped Region Between the Origin and the Curve
 $r = f(\theta)$, $\alpha \leq \theta \leq \beta$

$$A = \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta.$$

This is the integral of the **area differential** (Figure 10.49)

$$dA = \frac{1}{2} r^2 d\theta = \frac{1}{2} (f(\theta))^2 d\theta.$$

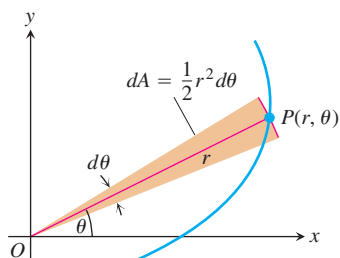


FIGURE 10.49 The area differential dA for the curve $n = f(\theta)$.

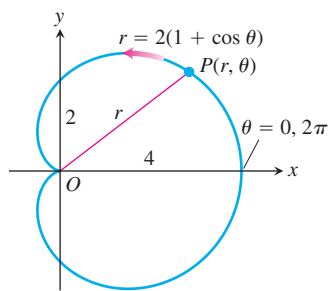


FIGURE 10.50 The cardioid in Example 1.

EXAMPLE 1 Finding Area

Find the area of the region in the plane enclosed by the cardioid $r = 2(1 + \cos \theta)$.

Solution We graph the cardioid (Figure 10.50) and determine that the radius OP sweeps out the region exactly once as θ runs from 0 to 2π . The area is therefore

$$\begin{aligned} \int_{\theta=0}^{\theta=2\pi} \frac{1}{2} r^2 d\theta &= \int_0^{2\pi} \frac{1}{2} \cdot 4(1 + \cos \theta)^2 d\theta \\ &= \int_0^{2\pi} 2(1 + 2\cos \theta + \cos^2 \theta) d\theta \\ &= \int_0^{2\pi} \left(2 + 4\cos \theta + 2 \frac{1 + \cos 2\theta}{2} \right) d\theta \\ &= \int_0^{2\pi} (3 + 4\cos \theta + \cos 2\theta) d\theta \\ &= \left[3\theta + 4\sin \theta + \frac{\sin 2\theta}{2} \right]_0^{2\pi} = 6\pi - 0 = 6\pi. \end{aligned}$$

EXAMPLE 2 Finding Area

Find the area inside the smaller loop of the limaçon

$$r = 2 \cos \theta + 1.$$

Solution After sketching the curve (Figure 10.51), we see that the smaller loop is traced out by the point (r, θ) as θ increases from $\theta = 2\pi/3$ to $\theta = 4\pi/3$. Since the curve is symmetric about the x -axis (the equation is unaltered when we replace θ by $-\theta$), we may calculate the area of the shaded half of the inner loop by integrating from $\theta = 2\pi/3$ to $\theta = \pi$. The area we seek will be twice the resulting integral:

$$A = 2 \int_{2\pi/3}^{\pi} \frac{1}{2} r^2 d\theta = \int_{2\pi/3}^{\pi} r^2 d\theta.$$

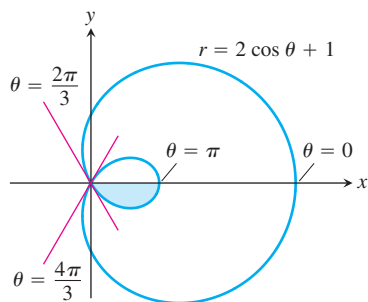


FIGURE 10.51 The limaçon in Example 2. Limaçon (pronounced LEE-ma-sahn) is an old French word for *snail*.

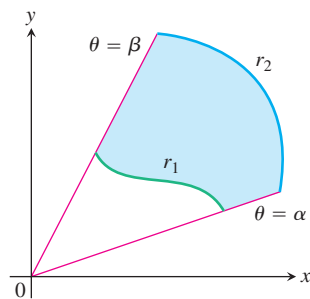


FIGURE 10.52 The area of the shaded region is calculated by subtracting the area of the region between r_1 and the origin from the area of the region between r_2 and the origin.

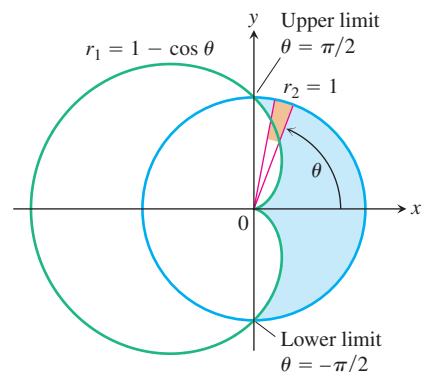


FIGURE 10.53 The region and limits of integration in Example 3.

Since

$$\begin{aligned} r^2 &= (2 \cos \theta + 1)^2 = 4 \cos^2 \theta + 4 \cos \theta + 1 \\ &= 4 \cdot \frac{1 + \cos 2\theta}{2} + 4 \cos \theta + 1 \\ &= 2 + 2 \cos 2\theta + 4 \cos \theta + 1 \\ &= 3 + 2 \cos 2\theta + 4 \cos \theta, \end{aligned}$$

we have

$$\begin{aligned} A &= \int_{2\pi/3}^{\pi} (3 + 2 \cos 2\theta + 4 \cos \theta) d\theta \\ &= \left[3\theta + \sin 2\theta + 4 \sin \theta \right]_{2\pi/3}^{\pi} \\ &= (3\pi) - \left(2\pi - \frac{\sqrt{3}}{2} + 4 \cdot \frac{\sqrt{3}}{2} \right) \\ &= \pi - \frac{3\sqrt{3}}{2}. \end{aligned}$$

To find the area of a region like the one in Figure 10.52, which lies between two polar curves $r_1 = r_1(\theta)$ and $r_2 = r_2(\theta)$ from $\theta = \alpha$ to $\theta = \beta$, we subtract the integral of $(1/2)r_1^2 d\theta$ from the integral of $(1/2)r_2^2 d\theta$. This leads to the following formula.

Area of the Region $0 \leq r_1(\theta) \leq r \leq r_2(\theta)$, $\alpha \leq \theta \leq \beta$

$$A = \int_{\alpha}^{\beta} \frac{1}{2} r_2^2 d\theta - \int_{\alpha}^{\beta} \frac{1}{2} r_1^2 d\theta = \int_{\alpha}^{\beta} \frac{1}{2} (r_2^2 - r_1^2) d\theta \quad (1)$$

EXAMPLE 3 Finding Area Between Polar Curves

Find the area of the region that lies inside the circle $r = 1$ and outside the cardioid $r = 1 - \cos \theta$.

Solution We sketch the region to determine its boundaries and find the limits of integration (Figure 10.53). The outer curve is $r_2 = 1$, the inner curve is $r_1 = 1 - \cos \theta$, and θ runs from $-\pi/2$ to $\pi/2$. The area, from Equation (1), is

$$\begin{aligned} A &= \int_{-\pi/2}^{\pi/2} \frac{1}{2} (r_2^2 - r_1^2) d\theta \\ &= 2 \int_0^{\pi/2} \frac{1}{2} (r_2^2 - r_1^2) d\theta \quad \text{Symmetry} \\ &= \int_0^{\pi/2} (1 - (1 - 2 \cos \theta + \cos^2 \theta)) d\theta \\ &= \int_0^{\pi/2} (2 \cos \theta - \cos^2 \theta) d\theta = \int_0^{\pi/2} \left(2 \cos \theta - \frac{1 + \cos 2\theta}{2} \right) d\theta \\ &= \left[2 \sin \theta - \frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_0^{\pi/2} = 2 - \frac{\pi}{4}. \end{aligned}$$

Length of a Polar Curve

We can obtain a polar coordinate formula for the length of a curve $r = f(\theta)$, $\alpha \leq \theta \leq \beta$, by parametrizing the curve as

$$x = r \cos \theta = f(\theta) \cos \theta, \quad y = r \sin \theta = f(\theta) \sin \theta, \quad \alpha \leq \theta \leq \beta. \quad (2)$$

The parametric length formula, Equation (1) from Section 6.3, then gives the length as

$$L = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta.$$

This equation becomes

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

when Equations (2) are substituted for x and y (Exercise 33).

Length of a Polar Curve

If $r = f(\theta)$ has a continuous first derivative for $\alpha \leq \theta \leq \beta$ and if the point $P(r, \theta)$ traces the curve $r = f(\theta)$ exactly once as θ runs from α to β , then the length of the curve is

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta. \quad (3)$$

EXAMPLE 4 Finding the Length of a Cardioid

Find the length of the cardioid $r = 1 - \cos \theta$.

Solution We sketch the cardioid to determine the limits of integration (Figure 10.54). The point $P(r, \theta)$ traces the curve once, counterclockwise as θ runs from 0 to 2π , so these are the values we take for α and β .

With

$$r = 1 - \cos \theta, \quad \frac{dr}{d\theta} = \sin \theta,$$

we have

$$\begin{aligned} r^2 + \left(\frac{dr}{d\theta}\right)^2 &= (1 - \cos \theta)^2 + (\sin \theta)^2 \\ &= 1 - 2 \cos \theta + \underbrace{\cos^2 \theta + \sin^2 \theta}_1 = 2 - 2 \cos \theta \end{aligned}$$

and

$$\begin{aligned} L &= \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = \int_0^{2\pi} \sqrt{2 - 2 \cos \theta} d\theta \\ &= \int_0^{2\pi} \sqrt{4 \sin^2 \frac{\theta}{2}} d\theta \quad 1 - \cos \theta = 2 \sin^2 \frac{\theta}{2} \end{aligned}$$

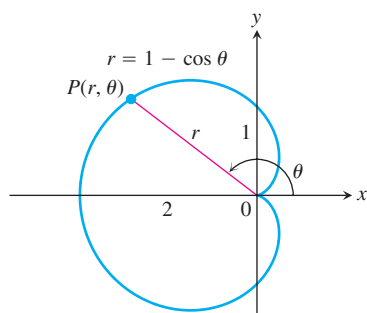


FIGURE 10.54 Calculating the length of a cardioid (Example 4).

$$\begin{aligned}
 &= \int_0^{2\pi} 2 \left| \sin \frac{\theta}{2} \right| d\theta \\
 &= \int_0^{2\pi} 2 \sin \frac{\theta}{2} d\theta \quad \sin \frac{\theta}{2} \geq 0 \text{ for } 0 \leq \theta \leq 2\pi \\
 &= \left[-4 \cos \frac{\theta}{2} \right]_0^{2\pi} = 4 + 4 = 8.
 \end{aligned}$$

Area of a Surface of Revolution

To derive polar coordinate formulas for the area of a surface of revolution, we parametrize the curve $r = f(\theta)$, $\alpha \leq \theta \leq \beta$, with Equations (2) and apply the surface area equations for parametrized curves in Section 6.5.

Area of a Surface of Revolution of a Polar Curve

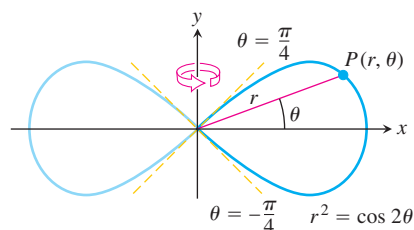
If $r = f(\theta)$ has a continuous first derivative for $\alpha \leq \theta \leq \beta$ and if the point $P(r, \theta)$ traces the curve $r = f(\theta)$ exactly once as θ runs from α to β , then the areas of the surfaces generated by revolving the curve about the x - and y -axes are given by the following formulas:

1. Revolution about the x -axis ($y \geq 0$):

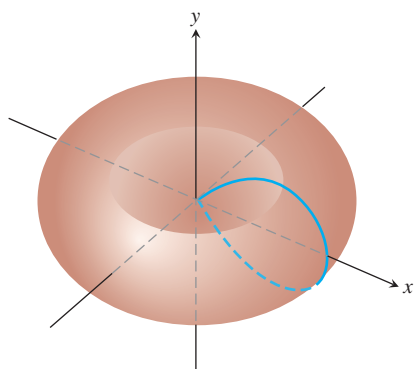
$$S = \int_{\alpha}^{\beta} 2\pi r \sin \theta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \quad (4)$$

2. Revolution about the y -axis ($x \geq 0$):

$$S = \int_{\alpha}^{\beta} 2\pi r \cos \theta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \quad (5)$$



(a)



(b)

FIGURE 10.55 The right-hand half of a lemniscate (a) is revolved about the y -axis to generate a surface (b), whose area is calculated in Example 5.

EXAMPLE 5 Finding Surface Area

Find the area of the surface generated by revolving the right-hand loop of the lemniscate $r^2 = \cos 2\theta$ about the y -axis.

Solution We sketch the loop to determine the limits of integration (Figure 10.55a). The point $P(r, \theta)$ traces the curve once, counterclockwise as θ runs from $-\pi/4$ to $\pi/4$, so these are the values we take for α and β .

We evaluate the area integrand in Equation (5) in stages. First,

$$2\pi r \cos \theta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} = 2\pi \cos \theta \sqrt{r^4 + \left(r \frac{dr}{d\theta}\right)^2}. \quad (6)$$

Next, $r^2 = \cos 2\theta$, so

$$2r \frac{dr}{d\theta} = -2 \sin 2\theta$$

$$r \frac{dr}{d\theta} = -\sin 2\theta$$

$$\left(r \frac{dr}{d\theta}\right)^2 = \sin^2 2\theta.$$

Finally, $r^4 = (r^2)^2 = \cos^2 2\theta$, so the square root on the right-hand side of Equation (6) simplifies to

$$\sqrt{r^4 + \left(r \frac{dr}{d\theta}\right)^2} = \sqrt{\cos^2 2\theta + \sin^2 2\theta} = 1.$$

All together, we have

$$\begin{aligned} S &= \int_{\alpha}^{\beta} 2\pi r \cos \theta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta && \text{Equation (5)} \\ &= \int_{-\pi/4}^{\pi/4} 2\pi \cos \theta \cdot (1) d\theta \\ &= 2\pi \left[\sin \theta \right]_{-\pi/4}^{\pi/4} \\ &= 2\pi \left[\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right] = 2\pi\sqrt{2}. \end{aligned}$$



EXERCISES 10.7

Areas Inside Polar Curves

Find the areas of the regions in Exercises 1–6.

1. Inside the oval limaçon $r = 4 + 2 \cos \theta$
2. Inside the cardioid $r = a(1 + \cos \theta)$, $a > 0$
3. Inside one leaf of the four-leaved rose $r = \cos 2\theta$
4. Inside the lemniscate $r^2 = 2a^2 \cos 2\theta$, $a > 0$
5. Inside one loop of the lemniscate $r^2 = 4 \sin 2\theta$
6. Inside the six-leaved rose $r^2 = 2 \sin 3\theta$

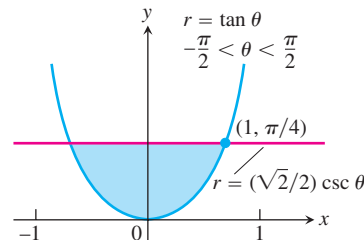
Areas Shared by Polar Regions

Find the areas of the regions in Exercises 7–16.

7. Shared by the circles $r = 2 \cos \theta$ and $r = 2 \sin \theta$
8. Shared by the circles $r = 1$ and $r = 2 \sin \theta$
9. Shared by the circle $r = 2$ and the cardioid $r = 2(1 - \cos \theta)$
10. Shared by the cardioids $r = 2(1 + \cos \theta)$ and $r = 2(1 - \cos \theta)$
11. Inside the lemniscate $r^2 = 6 \cos 2\theta$ and outside the circle $r = \sqrt{3}$
12. Inside the circle $r = 3a \cos \theta$ and outside the cardioid $r = a(1 + \cos \theta)$, $a > 0$
13. Inside the circle $r = -2 \cos \theta$ and outside the circle $r = 1$
14. a. Inside the outer loop of the limaçon $r = 2 \cos \theta + 1$
(See Figure 10.51.)

- b. Inside the outer loop and outside the inner loop of the limaçon $r = 2 \cos \theta + 1$

15. Inside the circle $r = 6$ above the line $r = 3 \csc \theta$
16. Inside the lemniscate $r^2 = 6 \cos 2\theta$ to the right of the line $r = (3/2) \sec \theta$
17. a. Find the area of the shaded region in the accompanying figure.



- b. It looks as if the graph of $r = \tan \theta$, $-\pi/2 < \theta < \pi/2$, could be asymptotic to the lines $x = 1$ and $x = -1$. Is it? Give reasons for your answer.
18. The area of the region that lies inside the cardioid curve $r = \cos \theta + 1$ and outside the circle $r = \cos \theta$ is not

$$\frac{1}{2} \int_0^{2\pi} [(\cos \theta + 1)^2 - \cos^2 \theta] d\theta = \pi.$$

Why not? What *is* the area? Give reasons for your answers.

Lengths of Polar Curves

Find the lengths of the curves in Exercises 19–27.

19. The spiral $r = \theta^2$, $0 \leq \theta \leq \sqrt{5}$
20. The spiral $r = e^{\theta}/\sqrt{2}$, $0 \leq \theta \leq \pi$
21. The cardioid $r = 1 + \cos \theta$
22. The curve $r = a \sin^2(\theta/2)$, $0 \leq \theta \leq \pi$, $a > 0$
23. The parabolic segment $r = 6/(1 + \cos \theta)$, $0 \leq \theta \leq \pi/2$
24. The parabolic segment $r = 2/(1 - \cos \theta)$, $\pi/2 \leq \theta \leq \pi$
25. The curve $r = \cos^3(\theta/3)$, $0 \leq \theta \leq \pi/4$
26. The curve $r = \sqrt{1 + \sin 2\theta}$, $0 \leq \theta \leq \pi\sqrt{2}$
27. The curve $r = \sqrt{1 + \cos 2\theta}$, $0 \leq \theta \leq \pi\sqrt{2}$
28. **Circumferences of circles** As usual, when faced with a new formula, it is a good idea to try it on familiar objects to be sure it gives results consistent with past experience. Use the length formula in Equation (3) to calculate the circumferences of the following circles ($a > 0$):
 - a. $r = a$
 - b. $r = a \cos \theta$
 - c. $r = a \sin \theta$

Surface Area

Find the areas of the surfaces generated by revolving the curves in Exercises 29–32 about the indicated axes.

29. $r = \sqrt{\cos 2\theta}$, $0 \leq \theta \leq \pi/4$, y -axis
30. $r = \sqrt{2}e^{\theta/2}$, $0 \leq \theta \leq \pi/2$, x -axis
31. $r^2 = \cos 2\theta$, x -axis
32. $r = 2a \cos \theta$, $a > 0$, y -axis

Theory and Examples

33. **The length of the curve $r = f(\theta)$, $\alpha \leq \theta \leq \beta$** Assuming that the necessary derivatives are continuous, show how the substitutions

$$x = f(\theta) \cos \theta, \quad y = f(\theta) \sin \theta$$

(Equations 2 in the text) transform

$$L = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$

into

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta.$$

34. **Average value** If f is continuous, the average value of the polar coordinate r over the curve $r = f(\theta)$, $\alpha \leq \theta \leq \beta$, with respect to θ is given by the formula

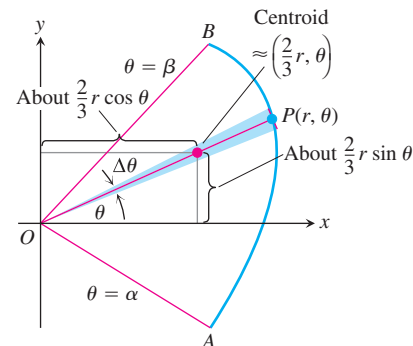
$$r_{\text{av}} = \frac{1}{\beta - \alpha} \int_{\alpha}^{\beta} f(\theta) d\theta.$$

Use this formula to find the average value of r with respect to θ over the following curves ($a > 0$).

- a. The cardioid $r = a(1 - \cos \theta)$
- b. The circle $r = a$
- c. The circle $r = a \cos \theta$, $-\pi/2 \leq \theta \leq \pi/2$
35. **$r = f(\theta)$ vs. $r = 2f(\theta)$** Can anything be said about the relative lengths of the curves $r = f(\theta)$, $\alpha \leq \theta \leq \beta$, and $r = 2f(\theta)$, $\alpha \leq \theta \leq \beta$? Give reasons for your answer.
36. **$r = f(\theta)$ vs. $r = 2f(\theta)$** The curves $r = f(\theta)$, $\alpha \leq \theta \leq \beta$, and $r = 2f(\theta)$, $\alpha \leq \theta \leq \beta$, are revolved about the x -axis to generate surfaces. Can anything be said about the relative areas of these surfaces? Give reasons for your answer.

Centroids of Fan-Shaped Regions

Since the centroid of a triangle is located on each median, two-thirds of the way from the vertex to the opposite base, the lever arm for the moment about the x -axis of the thin triangular region in the accompanying figure is about $(2/3)r \sin \theta$. Similarly, the lever arm for the moment of the triangular region about the y -axis is about $(2/3)r \cos \theta$. These approximations improve as $\Delta\theta \rightarrow 0$ and lead to the following formulas for the coordinates of the centroid of region AOB :



$$\bar{x} = \frac{\int \frac{2}{3} r \cos \theta \cdot \frac{1}{2} r^2 d\theta}{\int \frac{1}{2} r^2 d\theta} = \frac{\frac{2}{3} \int r^3 \cos \theta d\theta}{\int r^2 d\theta},$$

$$\bar{y} = \frac{\int \frac{2}{3} r \sin \theta \cdot \frac{1}{2} r^2 d\theta}{\int \frac{1}{2} r^2 d\theta} = \frac{\frac{2}{3} \int r^3 \sin \theta d\theta}{\int r^2 d\theta},$$

with limits $\theta = \alpha$ to $\theta = \beta$ on all integrals.

37. Find the centroid of the region enclosed by the cardioid $r = a(1 + \cos \theta)$.
38. Find the centroid of the semicircular region $0 \leq r \leq a$, $0 \leq \theta \leq \pi$.

10.8 Conic Sections in Polar Coordinates

Polar coordinates are important in astronomy and astronautical engineering because the ellipses, parabolas, and hyperbolas along which satellites, moons, planets, and comets approximately move can all be described with a single relatively simple coordinate equation. We develop that equation here.

Lines

Suppose the perpendicular from the origin to line L meets L at the point $P_0(r_0, \theta_0)$, with $r_0 \geq 0$ (Figure 10.56). Then, if $P(r, \theta)$ is any other point on L , the points P , P_0 , and O are the vertices of a right triangle, from which we can read the relation

$$r_0 = r \cos(\theta - \theta_0).$$

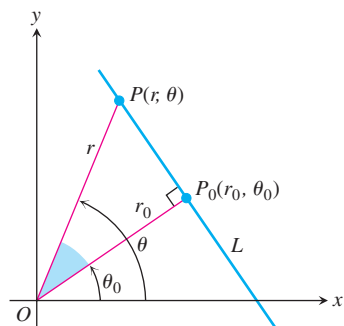


FIGURE 10.56 We can obtain a polar equation for line L by reading the relation $r_0 = r \cos(\theta - \theta_0)$ from the right triangle OP_0P .

The Standard Polar Equation for Lines

If the point $P_0(r_0, \theta_0)$ is the foot of the perpendicular from the origin to the line L , and $r_0 \geq 0$, then an equation for L is

$$r \cos(\theta - \theta_0) = r_0. \quad (1)$$

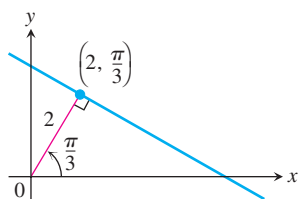


FIGURE 10.57 The standard polar equation of this line converts to the Cartesian equation $x + \sqrt{3}y = 4$ (Example 1).

EXAMPLE 1 Converting a Line's Polar Equation to Cartesian Form

Use the identity $\cos(A - B) = \cos A \cos B + \sin A \sin B$ to find a Cartesian equation for the line in Figure 10.57.

Solution

$$r \cos\left(\theta - \frac{\pi}{3}\right) = 2$$

$$r\left(\cos \theta \cos \frac{\pi}{3} + \sin \theta \sin \frac{\pi}{3}\right) = 2$$

$$\frac{1}{2}r \cos \theta + \frac{\sqrt{3}}{2}r \sin \theta = 2$$

$$\frac{1}{2}x + \frac{\sqrt{3}}{2}y = 2$$

$$x + \sqrt{3}y = 4$$

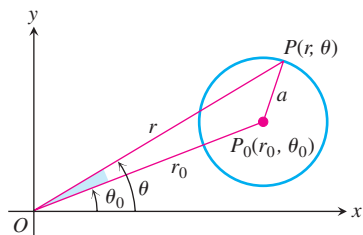


FIGURE 10.58 We can get a polar equation for this circle by applying the Law of Cosines to triangle OP_0P .

Circles

To find a polar equation for the circle of radius a centered at $P_0(r_0, \theta_0)$, we let $P(r, \theta)$ be a point on the circle and apply the Law of Cosines to triangle OP_0P (Figure 10.58). This gives

$$a^2 = r_0^2 + r^2 - 2r_0r \cos(\theta - \theta_0).$$

If the circle passes through the origin, then $r_0 = a$ and this equation simplifies to

$$a^2 = a^2 + r^2 - 2ar \cos(\theta - \theta_0)$$

$$r^2 = 2ar \cos(\theta - \theta_0)$$

$$r = 2a \cos(\theta - \theta_0).$$

If the circle's center lies on the positive x -axis, $\theta_0 = 0$ and we get the further simplification

$$r = 2a \cos \theta$$

(see Figure 10.59a).

If the center lies on the positive y -axis, $\theta = \pi/2$, $\cos(\theta - \pi/2) = \sin \theta$, and the equation $r = 2a \cos(\theta - \theta_0)$ becomes

$$r = 2a \sin \theta$$

(see Figure 10.59b).

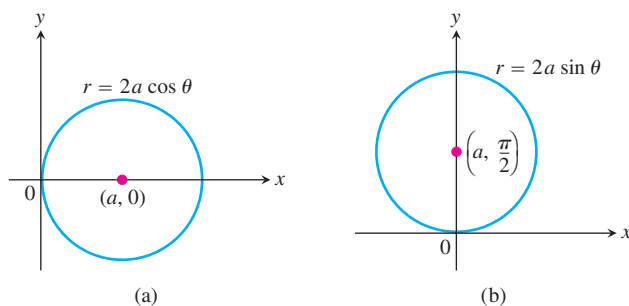


FIGURE 10.59 Polar equation of a circle of radius a through the origin with center on (a) the positive x -axis, and (b) the positive y -axis.

Equations for circles through the origin centered on the negative x - and y -axes can be obtained by replacing r with $-r$ in the above equations (Figure 10.60).

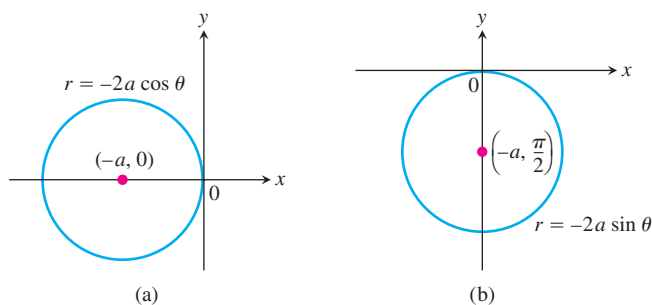


FIGURE 10.60 Polar equation of a circle of radius a through the origin with center on (a) the negative x -axis, and (b) the negative y -axis.

EXAMPLE 2
Circles Through the Origin

Radius	Center (polar coordinates)	Polar equation
3	(3, 0)	$r = 6 \cos \theta$
2	$(2, \pi/2)$	$r = 4 \sin \theta$
$1/2$	$(-1/2, 0)$	$r = -\cos \theta$
1	$(-1, \pi/2)$	$r = -2 \sin \theta$

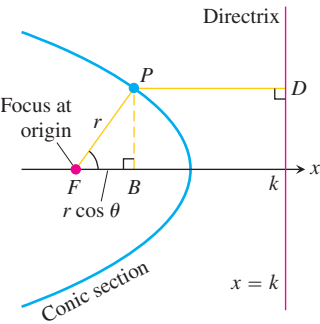


FIGURE 10.61 If a conic section is put in the position with its focus placed at the origin and a directrix perpendicular to the initial ray and right of the origin, we can find its polar equation from the conic’s focus–directrix equation.

Ellipses, Parabolas, and Hyperbolas

To find polar equations for ellipses, parabolas, and hyperbolas, we place one focus at the origin and the corresponding directrix to the right of the origin along the vertical line $x = k$ (Figure 10.61). This makes

$$PF = r$$

and

$$PD = k - FB = k - r \cos \theta.$$

The conic’s focus–directrix equation $PF = e \cdot PD$ then becomes

$$r = e(k - r \cos \theta),$$

which can be solved for r to obtain

Polar Equation for a Conic with Eccentricity e

$$r = \frac{ke}{1 + e \cos \theta}, \tag{2}$$

where $x = k > 0$ is the vertical directrix.

This equation represents an ellipse if $0 < e < 1$, a parabola if $e = 1$, and a hyperbola if $e > 1$. That is, ellipses, parabolas, and hyperbolas all have the same basic equation expressed in terms of eccentricity and location of the directrix.

EXAMPLE 3
Polar Equations of Some Conics

$e = \frac{1}{2} :$ ellipse $r = \frac{k}{2 + \cos \theta}$

$e = 1 :$ parabola $r = \frac{k}{1 + \cos \theta}$

$e = 2 :$ hyperbola $r = \frac{2k}{1 + 2 \cos \theta}$

You may see variations of Equation (2) from time to time, depending on the location of the directrix. If the directrix is the line $x = -k$ to the left of the origin (the origin is still a focus), we replace Equation (2) by

$$r = \frac{ke}{1 - e \cos \theta}.$$

The denominator now has a $(-)$ instead of a $(+)$. If the directrix is either of the lines $y = k$ or $y = -k$, the equations have sines in them instead of cosines, as shown in Figure 10.62.

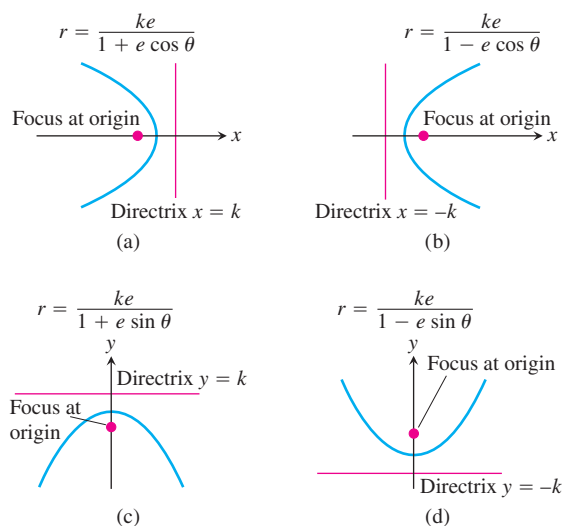


FIGURE 10.62 Equations for conic sections with eccentricity $e > 0$, but different locations of the directrix. The graphs here show a parabola, so $e = 1$.

EXAMPLE 4 Polar Equation of a Hyperbola

Find an equation for the hyperbola with eccentricity $3/2$ and directrix $x = 2$.

Solution We use Equation (2) with $k = 2$ and $e = 3/2$:

$$r = \frac{2(3/2)}{1 + (3/2)\cos \theta} \quad \text{or} \quad r = \frac{6}{2 + 3 \cos \theta}.$$

EXAMPLE 5 Finding a Directrix

Find the directrix of the parabola

$$r = \frac{25}{10 + 10 \cos \theta}.$$

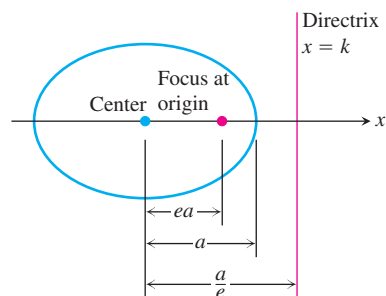


FIGURE 10.63 In an ellipse with semimajor axis a , the focus–directrix distance is $k = (a/e) - ea$, so $ke = a(1 - e^2)$.

Solution We divide the numerator and denominator by 10 to put the equation in standard form:

$$r = \frac{5/2}{1 + \cos \theta}.$$

This is the equation

$$r = \frac{ke}{1 + e \cos \theta}$$

with $k = 5/2$ and $e = 1$. The equation of the directrix is $x = 5/2$. ■

From the ellipse diagram in Figure 10.63, we see that k is related to the eccentricity e and the semimajor axis a by the equation

$$k = \frac{a}{e} - ea.$$

From this, we find that $ke = a(1 - e^2)$. Replacing ke in Equation (2) by $a(1 - e^2)$ gives the standard polar equation for an ellipse.

Polar Equation for the Ellipse with Eccentricity e and Semimajor Axis a

$$r = \frac{a(1 - e^2)}{1 + e \cos \theta} \quad (3)$$

Notice that when $e = 0$, Equation (3) becomes $r = a$, which represents a circle. Equation (3) is the starting point for calculating planetary orbits.

EXAMPLE 6 The Planet Pluto's Orbit

Find a polar equation for an ellipse with semimajor axis 39.44 AU (astronomical units) and eccentricity 0.25. This is the approximate size of Pluto's orbit around the sun.

Solution We use Equation (3) with $a = 39.44$ and $e = 0.25$ to find

$$r = \frac{39.44(1 - (0.25)^2)}{1 + 0.25 \cos \theta} = \frac{147.9}{4 + \cos \theta}.$$

At its point of closest approach (perihelion) where $\theta = 0$, Pluto is

$$r = \frac{147.9}{4 + 1} = 29.58 \text{ AU}$$

from the sun. At its most distant point (aphelion) where $\theta = \pi$, Pluto is

$$r = \frac{147.9}{4 - 1} = 49.3 \text{ AU}$$

from the sun (Figure 10.64). ■

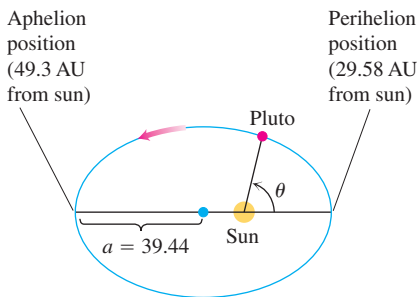


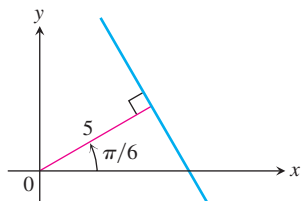
FIGURE 10.64 The orbit of Pluto (Example 6).

EXERCISES 10.8

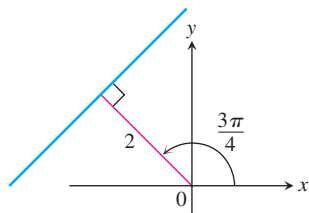
Lines

Find polar and Cartesian equations for the lines in Exercises 1–4.

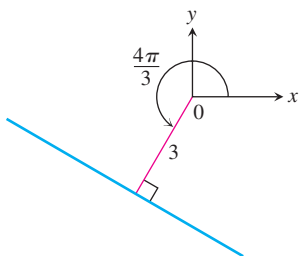
1.



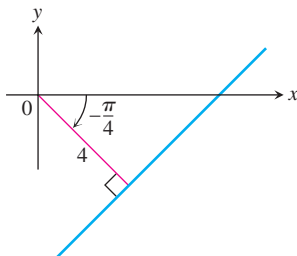
2.



3.



4.



Sketch the lines in Exercises 5–8 and find Cartesian equations for them.

$$5. r \cos \left(\theta - \frac{\pi}{4} \right) = \sqrt{2}$$

$$6. r \cos \left(\theta + \frac{3\pi}{4} \right) = 1$$

$$7. r \cos \left(\theta - \frac{2\pi}{3} \right) = 3$$

$$8. r \cos \left(\theta + \frac{\pi}{3} \right) = 2$$

Find a polar equation in the form $r \cos(\theta - \theta_0) = r_0$ for each of the lines in Exercises 9–12.

$$9. \sqrt{2}x + \sqrt{2}y = 6$$

$$10. \sqrt{3}x - y = 1$$

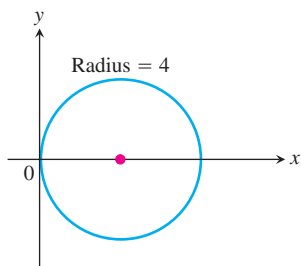
$$11. y = -5$$

$$12. x = -4$$

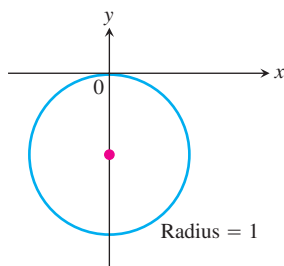
Circles

Find polar equations for the circles in Exercises 13–16.

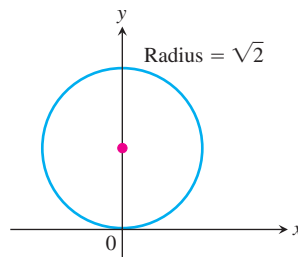
13.



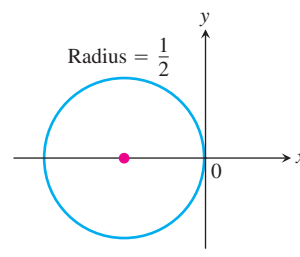
14.



15.



16.



Sketch the circles in Exercises 17–20. Give polar coordinates for their centers and identify their radii.

$$17. r = 4 \cos \theta$$

$$18. r = 6 \sin \theta$$

$$19. r = -2 \cos \theta$$

$$20. r = -8 \sin \theta$$

Find polar equations for the circles in Exercises 21–28. Sketch each circle in the coordinate plane and label it with both its Cartesian and polar equations.

$$21. (x - 6)^2 + y^2 = 36$$

$$22. (x + 2)^2 + y^2 = 4$$

$$23. x^2 + (y - 5)^2 = 25$$

$$24. x^2 + (y + 7)^2 = 49$$

$$25. x^2 + 2x + y^2 = 0$$

$$26. x^2 - 16x + y^2 = 0$$

$$27. x^2 + y^2 + y = 0$$

$$28. x^2 + y^2 - \frac{4}{3}y = 0$$

Conic Sections from Eccentricities and Directrices

Exercises 29–36 give the eccentricities of conic sections with one focus at the origin, along with the directrix corresponding to that focus. Find a polar equation for each conic section.

$$29. e = 1, \quad x = 2$$

$$30. e = 1, \quad y = 2$$

$$31. e = 5, \quad y = -6$$

$$32. e = 2, \quad x = 4$$

$$33. e = 1/2, \quad x = 1$$

$$34. e = 1/4, \quad x = -2$$

$$35. e = 1/5, \quad y = -10$$

$$36. e = 1/3, \quad y = 6$$

Parabolas and Ellipses

Sketch the parabolas and ellipses in Exercises 37–44. Include the directrix that corresponds to the focus at the origin. Label the vertices with appropriate polar coordinates. Label the centers of the ellipses as well.

$$37. r = \frac{1}{1 + \cos \theta}$$

$$38. r = \frac{6}{2 + \cos \theta}$$

$$39. r = \frac{25}{10 - 5 \cos \theta}$$

$$40. r = \frac{4}{2 - 2 \cos \theta}$$

$$41. r = \frac{400}{16 + 8 \sin \theta}$$

$$42. r = \frac{12}{3 + 3 \sin \theta}$$

$$43. r = \frac{8}{2 - 2 \sin \theta}$$

$$44. r = \frac{4}{2 - \sin \theta}$$

Graphing Inequalities

Sketch the regions defined by the inequalities in Exercises 45 and 46.

45. $0 \leq r \leq 2 \cos \theta$ 46. $-3 \cos \theta \leq r \leq 0$

T Grapher Explorations

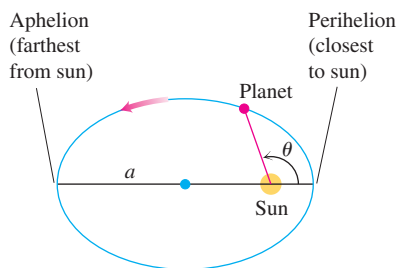
Graph the lines and conic sections in Exercises 47–56.

47. $r = 3 \sec(\theta - \pi/3)$ 48. $r = 4 \sec(\theta + \pi/6)$
 49. $r = 4 \sin \theta$ 50. $r = -2 \cos \theta$
 51. $r = 8/(4 + \cos \theta)$ 52. $r = 8/(4 + \sin \theta)$
 53. $r = 1/(1 - \sin \theta)$ 54. $r = 1/(1 + \cos \theta)$
 55. $r = 1/(1 + 2 \sin \theta)$ 56. $r = 1/(1 + 2 \cos \theta)$

Theory and Examples

57. Perihelion and aphelion A planet travels about its sun in an ellipse whose semimajor axis has length a . (See accompanying figure.)

- Show that $r = a(1 - e)$ when the planet is closest to the sun and that $r = a(1 + e)$ when the planet is farthest from the sun.
- Use the data in the table in Exercise 58 to find how close each planet in our solar system comes to the sun and how far away each planet gets from the sun.



58. Planetary orbits In Example 6, we found a polar equation for the orbit of Pluto. Use the data in the table below to find polar equations for the orbits of the other planets.

Planet	Semimajor axis (astronomical units)	Eccentricity
Mercury	0.3871	0.2056
Venus	0.7233	0.0068
Earth	1.000	0.0167
Mars	1.524	0.0934
Jupiter	5.203	0.0484
Saturn	9.539	0.0543
Uranus	19.18	0.0460
Neptune	30.06	0.0082
Pluto	39.44	0.2481

- Find Cartesian equations for the curves $r = 4 \sin \theta$ and $r = \sqrt{3} \sec \theta$.
 - Sketch the curves together and label their points of intersection in both Cartesian and polar coordinates.
- Repeat Exercise 59 for $r = 8 \cos \theta$ and $r = 2 \sec \theta$.
- Find a polar equation for the parabola with focus $(0, 0)$ and directrix $r \cos \theta = 4$.
- Find a polar equation for the parabola with focus $(0, 0)$ and directrix $r \cos(\theta - \pi/2) = 2$.
- a. The space engineer's formula for eccentricity** The space engineer's formula for the eccentricity of an elliptical orbit is

$$e = \frac{r_{\max} - r_{\min}}{r_{\max} + r_{\min}},$$

where r is the distance from the space vehicle to the attracting focus of the ellipse along which it travels. Why does the formula work?

- b. Drawing ellipses with string** You have a string with a knot in each end that can be pinned to a drawing board. The string is 10 in. long from the center of one knot to the center of the other. How far apart should the pins be to use the method illustrated in Figure 10.5 (Section 10.1) to draw an ellipse of eccentricity 0.2? The resulting ellipse would resemble the orbit of Mercury.
- 64. Halley's comet** (See Section 10.2, Example 1.)
- Write an equation for the orbit of Halley's comet in a coordinate system in which the sun lies at the origin and the other focus lies on the negative x -axis, scaled in astronomical units.
 - How close does the comet come to the sun in astronomical units? In kilometers?
 - What is the farthest the comet gets from the sun in astronomical units? In kilometers?

In Exercises 65–68, find a polar equation for the given curve. In each case, sketch a typical curve.

65. $x^2 + y^2 - 2ay = 0$ 66. $y^2 = 4ax + 4a^2$
 67. $x \cos \alpha + y \sin \alpha = p$ (α, p constant)
 68. $(x^2 + y^2)^2 + 2ax(x^2 + y^2) - a^2y^2 = 0$

COMPUTER EXPLORATIONS

- 69.** Use a CAS to plot the polar equation

$$r = \frac{ke}{1 + e \cos \theta}$$

for various values of k and e , $-\pi \leq \theta \leq \pi$. Answer the following questions.

- Take $k = -2$. Describe what happens to the plots as you take e to be $3/4$, 1 , and $5/4$. Repeat for $k = 2$.

- b. Take $k = -1$. Describe what happens to the plots as you take e to be $7/6, 5/4, 4/3, 3/2, 2, 3, 5, 10$, and 20 . Repeat for $e = 1/2, 1/3, 1/4, 1/10$, and $1/20$.
- c. Now keep $e > 0$ fixed and describe what happens as you take k to be $-1, -2, -3, -4$, and -5 . Be sure to look at graphs for parabolas, ellipses, and hyperbolas.

70. Use a CAS to plot the polar ellipse

$$r = \frac{a(1 - e^2)}{1 + e \cos \theta}$$

for various values of $a > 0$ and $0 < e < 1$, $-\pi \leq \theta \leq \pi$.

- a. Take $e = 9/10$. Describe what happens to the plots as you let a equal $1, 3/2, 2, 3, 5$, and 10 . Repeat with $e = 1/4$.
- b. Take $a = 2$. Describe what happens as you take e to be $9/10, 8/10, 7/10, \dots, 1/10, 1/20$, and $1/50$.

Chapter 10 Additional and Advanced Exercises

Finding Conic Sections

1. Find an equation for the parabola with focus (4, 0) and directrix $x = 3$. Sketch the parabola together with its vertex, focus, and directrix.

2. Find the vertex, focus, and directrix of the parabola

$$x^2 - 6x - 12y + 9 = 0.$$

3. Find an equation for the curve traced by the point $P(x, y)$ if the distance from P to the vertex of the parabola $x^2 = 4y$ is twice the distance from P to the focus. Identify the curve.
4. A line segment of length $a + b$ runs from the x -axis to the y -axis. The point P on the segment lies a units from one end and b units from the other end. Show that P traces an ellipse as the ends of the segment slide along the axes.
5. The vertices of an ellipse of eccentricity 0.5 lie at the points $(0, \pm 2)$. Where do the foci lie?
6. Find an equation for the ellipse of eccentricity $2/3$ that has the line $x = 2$ as a directrix and the point (4, 0) as the corresponding focus.
7. One focus of a hyperbola lies at the point (0, -7) and the corresponding directrix is the line $y = -1$. Find an equation for the hyperbola if its eccentricity is (a) 2, (b) 5.
8. Find an equation for the hyperbola with foci (0, -2) and (0, 2) that passes through the point (12, 7).
9. a. Show that the line

$$b^2xx_1 + a^2yy_1 - a^2b^2 = 0$$

is tangent to the ellipse $b^2x^2 + a^2y^2 - a^2b^2 = 0$ at the point (x_1, y_1) on the ellipse.

- b. Show that the line

$$b^2xx_1 - a^2yy_1 - a^2b^2 = 0$$

is tangent to the hyperbola $b^2x^2 - a^2y^2 - a^2b^2 = 0$ at the point (x_1, y_1) on the hyperbola.

10. Show that the tangent to the conic section

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

at a point (x_1, y_1) on it has an equation that may be written in the form

$$Axx_1 + B\left(\frac{x_1y + xy_1}{2}\right) + Cy_1 + D\left(\frac{x + x_1}{2}\right) + E\left(\frac{y + y_1}{2}\right) + F = 0.$$

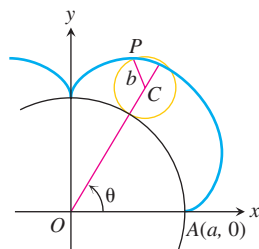
Equations and Inequalities

What points in the xy -plane satisfy the equations and inequalities in Exercises 11–18? Draw a figure for each exercise.

11. $(x^2 - y^2 - 1)(x^2 + y^2 - 25)(x^2 + 4y^2 - 4) = 0$
12. $(x + y)(x^2 + y^2 - 1) = 0$
13. $(x^2/9) + (y^2/16) \leq 1$
14. $(x^2/9) - (y^2/16) \leq 1$
15. $(9x^2 + 4y^2 - 36)(4x^2 + 9y^2 - 16) \leq 0$
16. $(9x^2 + 4y^2 - 36)(4x^2 + 9y^2 - 16) > 0$
17. $x^4 - (y^2 - 9)^2 = 0$
18. $x^2 + xy + y^2 < 3$

Parametric Equations and Cycloids

- 19. Epicycloids** When a circle rolls externally along the circumference of a second, fixed circle, any point P on the circumference of the rolling circle describes an *epicycloid*, as shown here. Let the fixed circle have its center at the origin O and have radius a .



Let the radius of the rolling circle be b and let the initial position of the tracing point P be $A(a, 0)$. Find parametric equations for the epicycloid, using as the parameter the angle θ from the positive x -axis to the line through the circles' centers.

- 20. a.** Find the centroid of the region enclosed by the x -axis and the cycloid arch

$$x = a(t - \sin t), \quad y = a(1 - \cos t); \quad 0 \leq t \leq 2\pi.$$
- b.** Find the first moments about the coordinate axes of the curve

$$x = (2/3)t^{3/2}, \quad y = 2\sqrt{t}; \quad 0 \leq t \leq \sqrt{3}.$$

Polar Coordinates

- 21. a.** Find an equation in polar coordinates for the curve

$$x = e^{2t} \cos t, \quad y = e^{2t} \sin t; \quad -\infty < t < \infty.$$
- b.** Find the length of the curve from $t = 0$ to $t = 2\pi$.
- 22.** Find the length of the curve $r = 2 \sin^3(\theta/3)$, $0 \leq \theta \leq 3\pi$, in the polar coordinate plane.
- 23.** Find the area of the surface generated by revolving the first-quadrant portion of the cardioid $r = 1 + \cos \theta$ about the x -axis. (Hint: Use the identities $1 + \cos \theta = 2 \cos^2(\theta/2)$ and $\sin \theta = 2 \sin(\theta/2) \cos(\theta/2)$ to simplify the integral.)
- 24.** Sketch the regions enclosed by the curves $r = 2a \cos^2(\theta/2)$ and $r = 2a \sin^2(\theta/2)$, $a > 0$, in the polar coordinate plane and find the area of the portion of the plane they have in common.

Exercises 25–28 give the eccentricities of conic sections with one focus at the origin of the polar coordinate plane, along with the directrix for that focus. Find a polar equation for each conic section.

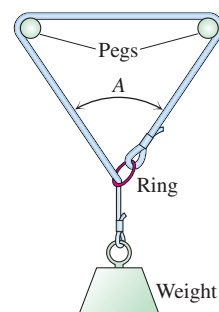
- 25.** $e = 2$, $r \cos \theta = 2$ **26.** $e = 1$, $r \cos \theta = -4$
27. $e = 1/2$, $r \sin \theta = 2$ **28.** $e = 1/3$, $r \sin \theta = -6$

Theory and Examples

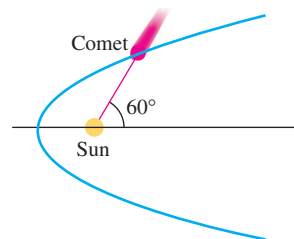
- 29.** A rope with a ring in one end is looped over two pegs in a horizontal line. The free end, after being passed through the ring, has a weight suspended from it to make the rope hang taut. If the rope

slips freely over the pegs and through the ring, the weight will descend as far as possible. Assume that the length of the rope is at least four times as great as the distance between the pegs and that the configuration of the rope is symmetric with respect to the line of the vertical part of the rope.

- a.** Find the angle A formed at the bottom of the loop in the accompanying figure.
- b.** Show that for each fixed position of the ring on the rope, the possible locations of the ring in space lie on an ellipse with foci at the pegs.
- c.** Justify the original symmetry assumption by combining the result in part (b) with the assumption that the rope and weight will take a rest position of minimal potential energy.

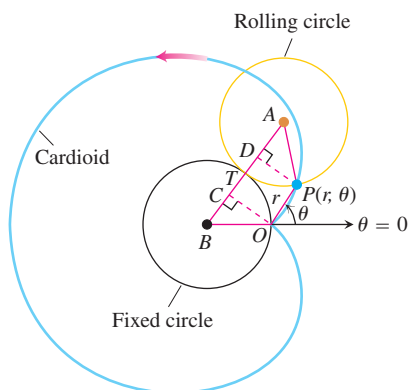


- 30.** Two radar stations lie 20 km apart along an east–west line. A low-flying plane traveling from west to east is known to have a speed of v_0 km/sec. At $t = 0$ a signal is sent from the station at $(-10, 0)$, bounces off the plane, and is received at $(10, 0)$ $30/c$ seconds later (c is the velocity of the signal). When $t = 10/v_0$, another signal is sent out from the station at $(-10, 0)$, reflects off the plane, and is once again received $30/c$ seconds later by the other station. Find the position of the plane when it reflects the second signal under the assumption that v_0 is much less than c .
- 31.** A comet moves in a parabolic orbit with the sun at the focus. When the comet is 4×10^7 miles from the sun, the line from the comet to the sun makes a 60° angle with the orbit's axis, as shown here. How close will the comet come to the sun?

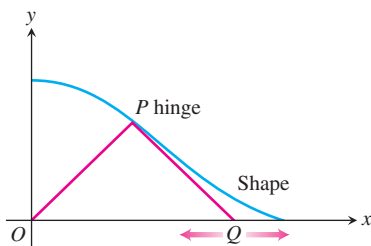


- 32.** Find the points on the parabola $x = 2t, y = t^2$, $-\infty < t < \infty$, closest to the point $(0, 3)$.
- 33.** Find the eccentricity of the ellipse $x^2 + xy + y^2 = 1$ to the nearest hundredth.

34. Find the eccentricity of the hyperbola $xy = 1$.
35. Is the curve $\sqrt{x} + \sqrt{y} = 1$ part of a conic section? If so, what kind of conic section? If not, why not?
36. Show that the curve $2xy - \sqrt{2}y + 2 = 0$ is a hyperbola. Find the hyperbola's center, vertices, foci, axes, and asymptotes.
37. Find a polar coordinate equation for
- the parabola with focus at the origin and vertex at $(a, \pi/4)$;
 - the ellipse with foci at the origin and $(2, 0)$ and one vertex at $(4, 0)$;
 - the hyperbola with one focus at the origin, center at $(2, \pi/2)$, and a vertex at $(1, \pi/2)$.
38. Any line through the origin will intersect the ellipse $r = 3/(2 + \cos \theta)$ in two points P_1 and P_2 . Let d_1 be the distance between P_1 and the origin and let d_2 be the distance between P_2 and the origin. Compute $(1/d_1) + (1/d_2)$.
39. **Generating a cardioid with circles** Cardioids are special epicycloids (Exercise 18). Show that if you roll a circle of radius a about another circle of radius a in the polar coordinate plane, as in the accompanying figure, the original point of contact P will trace a cardioid. (*Hint*: Start by showing that angles OBC and PAD both have measure θ .)



40. **A bifold closet door** A bifold closet door consists of two 1-ft-wide panels, hinged at point P . The outside bottom corner of one panel rests on a pivot at O (see the accompanying figure). The outside bottom corner of the other panel, denoted by Q , slides along a straight track, shown in the figure as a portion of the x -axis. Assume that as Q moves back and forth, the bottom of the door rubs against a thick carpet. What shape will the door sweep out on the surface of the carpet?

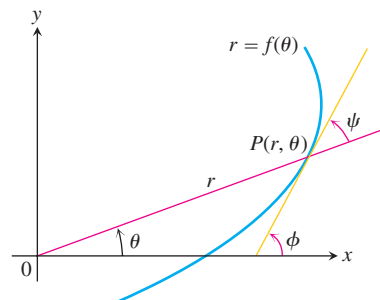


The Angle Between the Radius Vector and the Tangent Line to a Polar Coordinate Curve

In Cartesian coordinates, when we want to discuss the direction of a curve at a point, we use the angle ϕ measured counterclockwise from the positive x -axis to the tangent line. In polar coordinates, it is more convenient to calculate the angle ψ from the *radius vector* to the tangent line (see the accompanying figure). The angle ϕ can then be calculated from the relation

$$\phi = \theta + \psi, \quad (1)$$

which comes from applying the Exterior Angle Theorem to the triangle in the accompanying figure.



Suppose the equation of the curve is given in the form $r = f(\theta)$, where $f(\theta)$ is a differentiable function of θ . Then

$$x = r \cos \theta \quad \text{and} \quad y = r \sin \theta \quad (2)$$

are differentiable functions of θ with

$$\begin{aligned} \frac{dx}{d\theta} &= -r \sin \theta + \cos \theta \frac{dr}{d\theta}, \\ \frac{dy}{d\theta} &= r \cos \theta + \sin \theta \frac{dr}{d\theta}. \end{aligned} \quad (3)$$

Since $\psi = \phi - \theta$ from (1),

$$\tan \psi = \tan (\phi - \theta) = \frac{\tan \phi - \tan \theta}{1 + \tan \phi \tan \theta}.$$

Furthermore,

$$\tan \phi = \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

because $\tan \phi$ is the slope of the curve at P . Also,

$$\tan \theta = \frac{y}{x}.$$

Hence

$$\tan \psi = \frac{\frac{dy/d\theta}{dx/d\theta} - \frac{y}{x}}{1 + \frac{y}{x} \frac{dy/d\theta}{dx/d\theta}} = \frac{x \frac{dy}{d\theta} - y \frac{dx}{d\theta}}{x \frac{dx}{d\theta} + y \frac{dy}{d\theta}}. \quad (4)$$

The numerator in the last expression in Equation (4) is found from Equations (2) and (3) to be

$$x \frac{dy}{d\theta} - y \frac{dx}{d\theta} = r^2.$$

Similarly, the denominator is

$$x \frac{dx}{d\theta} + y \frac{dy}{d\theta} = r \frac{dr}{d\theta}.$$

When we substitute these into Equation (4), we obtain

$$\tan \psi = \frac{r}{dr/d\theta}. \quad (5)$$

This is the equation we use for finding ψ as a function of θ .

41. Show, by reference to a figure, that the angle β between the tangents to two curves at a point of intersection may be found from the formula

$$\tan \beta = \frac{\tan \psi_2 - \tan \psi_1}{1 + \tan \psi_2 \tan \psi_1}. \quad (6)$$

When will the two curves intersect at right angles?

42. Find the value of $\tan \psi$ for the curve $r = \sin^4(\theta/4)$.
43. Find the angle between the radius vector to the curve $r = 2a \sin 3\theta$ and its tangent when $\theta = \pi/6$.
- T** 44. **a.** Graph the hyperbolic spiral $r\theta = 1$. What appears to happen to ψ as the spiral winds in around the origin?
- b.** Confirm your finding in part (a) analytically.
45. The circles $r = \sqrt{3} \cos \theta$ and $r = \sin \theta$ intersect at the point $(\sqrt{3}/2, \pi/3)$. Show that their tangents are perpendicular there.
46. Sketch the cardioid $r = a(1 + \cos \theta)$ and circle $r = 3a \cos \theta$ in one diagram and find the angle between their tangents at the point of intersection that lies in the first quadrant.
47. Find the points of intersection of the parabolas

$$r = \frac{1}{1 - \cos \theta} \quad \text{and} \quad r = \frac{3}{1 + \cos \theta}$$

and the angles between their tangents at these points.

48. Find points on the cardioid $r = a(1 + \cos \theta)$ where the tangent line is **(a)** horizontal, **(b)** vertical.
49. Show that parabolas $r = a/(1 + \cos \theta)$ and $r = b/(1 - \cos \theta)$ are orthogonal at each point of intersection ($ab \neq 0$).
50. Find the angle at which the cardioid $r = a(1 - \cos \theta)$ crosses the ray $\theta = \pi/2$.
51. Find the angle between the line $r = 3 \sec \theta$ and the cardioid $r = 4(1 + \cos \theta)$ at one of their intersections.
52. Find the slope of the tangent line to the curve $r = a \tan(\theta/2)$ at $\theta = \pi/2$.
53. Find the angle at which the parabolas $r = 1/(1 - \cos \theta)$ and $r = 1/(1 - \sin \theta)$ intersect in the first quadrant.
54. The equation $r^2 = 2 \csc 2\theta$ represents a curve in polar coordinates.
- a.** Sketch the curve.
- b.** Find an equivalent Cartesian equation for the curve.
- c.** Find the angle at which the curve intersects the ray $\theta = \pi/4$.
55. Suppose that the angle ψ from the radius vector to the tangent line of the curve $r = f(\theta)$ has the constant value α .
- a.** Show that the area bounded by the curve and two rays $\theta = \theta_1$, $\theta = \theta_2$, is proportional to $r_2^2 - r_1^2$, where (r_1, θ_1) and (r_2, θ_2) are polar coordinates of the ends of the arc of the curve between these rays. Find the factor of proportionality.
- b.** Show that the length of the arc of the curve in part (a) is proportional to $r_2 - r_1$, and find the proportionality constant.
56. Let P be a point on the hyperbola $r^2 \sin 2\theta = 2a^2$. Show that the triangle formed by OP , the tangent at P , and the initial line is isosceles.

Chapter 10

Practice Exercises

Graphing Conic Sections

Sketch the parabolas in Exercises 1–4. Include the focus and directrix in each sketch.

1. $x^2 = -4y$

2. $x^2 = 2y$

3. $y^2 = 3x$

4. $y^2 = -(8/3)x$

Find the eccentricities of the ellipses and hyperbolas in Exercises 5–8. Sketch each conic section. Include the foci, vertices, and asymptotes (as appropriate) in your sketch.

5. $16x^2 + 7y^2 = 112$

6. $x^2 + 2y^2 = 4$

7. $3x^2 - y^2 = 3$

8. $5y^2 - 4x^2 = 20$

Shifting Conic Sections

Exercises 9–14 give equations for conic sections and tell how many units up or down and to the right or left each curve is to be shifted. Find an equation for the new conic section and find the new foci, vertices, centers, and asymptotes, as appropriate. If the curve is a parabola, find the new directrix as well.

9. $x^2 = -12y$, right 2, up 3
10. $y^2 = 10x$, left $1/2$, down 1
11. $\frac{x^2}{9} + \frac{y^2}{25} = 1$, left 3, down 5
12. $\frac{x^2}{169} + \frac{y^2}{144} = 1$, right 5, up 12
13. $\frac{y^2}{8} - \frac{x^2}{2} = 1$, right 2, up $2\sqrt{2}$
14. $\frac{x^2}{36} - \frac{y^2}{64} = 1$, left 10, down 3

Identifying Conic Sections

Identify the conic sections in Exercises 15–22 and find their foci, vertices, centers, and asymptotes (as appropriate). If the curve is a parabola, find its directrix as well.

15. $x^2 - 4x - 4y^2 = 0$
16. $4x^2 - y^2 + 4y = 8$
17. $y^2 - 2y + 16x = -49$
18. $x^2 - 2x + 8y = -17$
19. $9x^2 + 16y^2 + 54x - 64y = -1$
20. $25x^2 + 9y^2 - 100x + 54y = 44$
21. $x^2 + y^2 - 2x - 2y = 0$
22. $x^2 + y^2 + 4x + 2y = 1$

Using the Discriminant

What conic sections or degenerate cases do the equations in Exercises 23–28 represent? Give a reason for your answer in each case.

23. $x^2 + xy + y^2 + x + y + 1 = 0$
24. $x^2 + 4xy + 4y^2 + x + y + 1 = 0$
25. $x^2 + 3xy + 2y^2 + x + y + 1 = 0$
26. $x^2 + 2xy - 2y^2 + x + y + 1 = 0$
27. $x^2 - 2xy + y^2 = 0$
28. $x^2 - 3xy + 4y^2 = 0$

Rotating Conic Sections

Identify the conic sections in Exercises 29–32. Then rotate the coordinate axes to find a new equation for the conic section that has no cross product term. (The new equations will vary with the size and direction of the rotations used.)

29. $2x^2 + xy + 2y^2 - 15 = 0$
30. $3x^2 + 2xy + 3y^2 = 19$
31. $x^2 + 2\sqrt{3}xy - y^2 + 4 = 0$
32. $x^2 - 3xy + y^2 = 5$

Identifying Parametric Equations in the Plane

Exercises 33–36 give parametric equations and parameter intervals for the motion of a particle in the xy -plane. Identify the particle's path by

finding a Cartesian equation for it. Graph the Cartesian equation and indicate the direction of motion and the portion traced by the particle.

33. $x = (1/2)\tan t$, $y = (1/2)\sec t$; $-\pi/2 < t < \pi/2$
34. $x = -2\cos t$, $y = 2\sin t$; $0 \leq t \leq \pi$
35. $x = -\cos t$, $y = \cos^2 t$; $0 \leq t \leq \pi$
36. $x = 4\cos t$, $y = 9\sin t$; $0 \leq t \leq 2\pi$

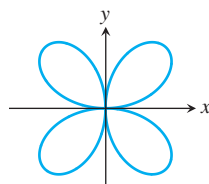
Graphs in the Polar Plane

Sketch the regions defined by the polar coordinate inequalities in Exercises 37 and 38.

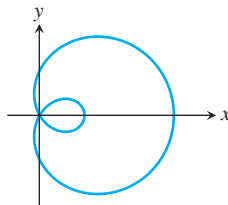
37. $0 \leq r \leq 6\cos\theta$
38. $-4\sin\theta \leq r \leq 0$

Match each graph in Exercises 39–46 with the appropriate equation (a)–(l). There are more equations than graphs, so some equations will not be matched.

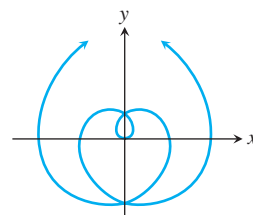
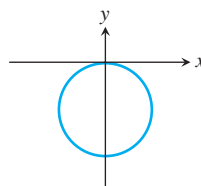
- | | |
|-------------------------------------|---------------------------|
| a. $r = \cos 2\theta$ | b. $r \cos \theta = 1$ |
| c. $r = \frac{6}{1 - 2\cos \theta}$ | d. $r = \sin 2\theta$ |
| e. $r = \theta$ | f. $r^2 = \cos 2\theta$ |
| g. $r = 1 + \cos \theta$ | h. $r = 1 - \sin \theta$ |
| i. $r = \frac{2}{1 - \cos \theta}$ | j. $r^2 = \sin 2\theta$ |
| k. $r = -\sin \theta$ | l. $r = 2\cos \theta + 1$ |
39. Four-leaved rose
 40. Spiral



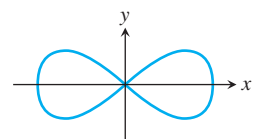
41. Limaçon



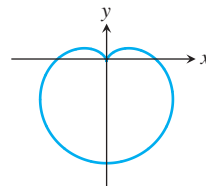
43. Circle



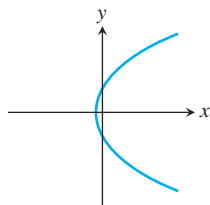
42. Lemniscate



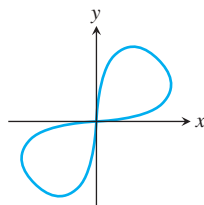
44. Cardioid



45. Parabola



46. Lemniscate



Intersections of Graphs in the Polar Plane

Find the points of intersection of the curves given by the polar coordinate equations in Exercises 47–54.

47. $r = \sin \theta$, $r = 1 + \sin \theta$ 48. $r = \cos \theta$, $r = 1 - \cos \theta$
 49. $r = 1 + \cos \theta$, $r = 1 - \cos \theta$
 50. $r = 1 + \sin \theta$, $r = 1 - \sin \theta$
 51. $r = 1 + \sin \theta$, $r = -1 + \sin \theta$
 52. $r = 1 + \cos \theta$, $r = -1 + \cos \theta$
 53. $r = \sec \theta$, $r = 2 \sin \theta$ 54. $r = -2 \csc \theta$, $r = -4 \cos \theta$

Polar to Cartesian Equations

Sketch the lines in Exercises 55–60. Also, find a Cartesian equation for each line.

55. $r \cos \left(\theta + \frac{\pi}{3} \right) = 2\sqrt{3}$ 56. $r \cos \left(\theta - \frac{3\pi}{4} \right) = \frac{\sqrt{2}}{2}$
 57. $r = 2 \sec \theta$ 58. $r = -\sqrt{2} \sec \theta$
 59. $r = -(3/2) \csc \theta$ 60. $r = (3\sqrt{3}) \csc \theta$

Find Cartesian equations for the circles in Exercises 61–64. Sketch each circle in the coordinate plane and label it with both its Cartesian and polar equations.

61. $r = -4 \sin \theta$ 62. $r = 3\sqrt{3} \sin \theta$
 63. $r = 2\sqrt{2} \cos \theta$ 64. $r = -6 \cos \theta$

Cartesian to Polar Equations

Find polar equations for the circles in Exercises 65–68. Sketch each circle in the coordinate plane and label it with both its Cartesian and polar equations.

65. $x^2 + y^2 + 5y = 0$ 66. $x^2 + y^2 - 2y = 0$
 67. $x^2 + y^2 - 3x = 0$ 68. $x^2 + y^2 + 4x = 0$

Conic Sections in Polar Coordinates

Sketch the conic sections whose polar coordinate equations are given in Exercises 69–72. Give polar coordinates for the vertices and, in the case of ellipses, for the centers as well.

69. $r = \frac{2}{1 + \cos \theta}$ 70. $r = \frac{8}{2 + \cos \theta}$
 71. $r = \frac{6}{1 - 2 \cos \theta}$ 72. $r = \frac{12}{3 + \sin \theta}$

Exercises 73–76 give the eccentricities of conic sections with one focus at the origin of the polar coordinate plane, along with the directrix for that focus. Find a polar equation for each conic section.

73. $e = 2$, $r \cos \theta = 2$ 74. $e = 1$, $r \cos \theta = -4$
 75. $e = 1/2$, $r \sin \theta = 2$ 76. $e = 1/3$, $r \sin \theta = -6$

Area, Length, and Surface Area in the Polar Plane

Find the areas of the regions in the polar coordinate plane described in Exercises 77–80.

77. Enclosed by the limaçon $r = 2 - \cos \theta$
 78. Enclosed by one leaf of the three-leaved rose $r = \sin 3\theta$
 79. Inside the “figure eight” $r = 1 + \cos 2\theta$ and outside the circle $r = 1$
 80. Inside the cardioid $r = 2(1 + \sin \theta)$ and outside the circle $r = 2 \sin \theta$

Find the lengths of the curves given by the polar coordinate equations in Exercises 81–84.

81. $r = -1 + \cos \theta$
 82. $r = 2 \sin \theta + 2 \cos \theta$, $0 \leq \theta \leq \pi/2$
 83. $r = 8 \sin^3(\theta/3)$, $0 \leq \theta \leq \pi/4$
 84. $r = \sqrt{1 + \cos 2\theta}$, $-\pi/2 \leq \theta \leq \pi/2$

Find the areas of the surfaces generated by revolving the polar coordinate curves in Exercises 85 and 86 about the indicated axes.

85. $r = \sqrt{\cos 2\theta}$, $0 \leq \theta \leq \pi/4$, x -axis
 86. $r^2 = \sin 2\theta$, y -axis

Theory and Examples

87. Find the volume of the solid generated by revolving the region enclosed by the ellipse $9x^2 + 4y^2 = 36$ about (a) the x -axis, (b) the y -axis.
 88. The “triangular” region in the first quadrant bounded by the x -axis, the line $x = 4$, and the hyperbola $9x^2 - 4y^2 = 36$ is revolved about the x -axis to generate a solid. Find the volume of the solid.
 89. A ripple tank is made by bending a strip of tin around the perimeter of an ellipse for the wall of the tank and soldering a flat bottom onto this. An inch or two of water is put in the tank and you drop a marble into it, right at one focus of the ellipse. Ripples radiate outward through the water, reflect from the strip around the edge of the tank, and a few seconds later a drop of water spurts up at the second focus. Why?
 90. **LORAN** A radio signal was sent simultaneously from towers A and B , located several hundred miles apart on the northern California coast. A ship offshore received the signal from A 1400 microseconds before receiving the signal from B . Assuming that the signals traveled at the rate of 980 ft/microsecond, what can be said about the location of the ship relative to the two towers?

91. On a level plane, at the same instant, you hear the sound of a rifle and that of the bullet hitting the target. What can be said about your location relative to the rifle and target?
92. **Archimedes spirals** The graph of an equation of the form $r = a\theta$, where a is a nonzero constant, is called an *Archimedes spiral*. Is there anything special about the widths between the successive turns of such a spiral?
93. a. Show that the equations $x = r \cos \theta$, $y = r \sin \theta$ transform the polar equation

$$r = \frac{k}{1 + e \cos \theta}$$

into the Cartesian equation

$$(1 - e^2)x^2 + y^2 + 2kex - k^2 = 0.$$

- b. Then apply the criteria of Section 10.3 to show that

$$e = 0 \Rightarrow \text{circle.}$$

$$0 < e < 1 \Rightarrow \text{ellipse.}$$

$$e = 1 \Rightarrow \text{parabola.}$$

$$e > 1 \Rightarrow \text{hyperbola.}$$

94. **A satellite orbit** A satellite is in an orbit that passes over the North and South Poles of the earth. When it is over the South Pole it is at the highest point of its orbit, 1000 miles above the earth's surface. Above the North Pole it is at the lowest point of its orbit, 300 miles above the earth's surface.
- a. Assuming that the orbit is an ellipse with one focus at the center of the earth, find its eccentricity. (Take the diameter of the earth to be 8000 miles.)
- b. Using the north-south axis of the earth as the x -axis and the center of the earth as origin, find a polar equation for the orbit.

Chapter 10

Questions to Guide Your Review

1. What is a parabola? What are the Cartesian equations for parabolas whose vertices lie at the origin and whose foci lie on the coordinate axes? How can you find the focus and directrix of such a parabola from its equation?
2. What is an ellipse? What are the Cartesian equations for ellipses centered at the origin with foci on one of the coordinate axes? How can you find the foci, vertices, and directrices of such an ellipse from its equation?
3. What is a hyperbola? What are the Cartesian equations for hyperbolas centered at the origin with foci on one of the coordinate axes? How can you find the foci, vertices, and directrices of such an ellipse from its equation?
4. What is the eccentricity of a conic section? How can you classify conic sections by eccentricity? How are an ellipse's shape and eccentricity related?
5. Explain the equation $PF = e \cdot PD$.
6. What is a quadratic curve in the xy -plane? Give examples of degenerate and nondegenerate quadratic curves.
7. How can you find a Cartesian coordinate system in which the new equation for a conic section in the plane has no xy -term? Give an example.
8. How can you tell what kind of graph to expect from a quadratic equation in x and y ? Give examples.
9. What are some typical parametrizations for conic sections?
10. What is a cycloid? What are typical parametric equations for cycloids? What physical properties account for the importance of cycloids?
11. What are polar coordinates? What equations relate polar coordinates to Cartesian coordinates? Why might you want to change from one coordinate system to the other?
12. What consequence does the lack of uniqueness of polar coordinates have for graphing? Give an example.
13. How do you graph equations in polar coordinates? Include in your discussion symmetry, slope, behavior at the origin, and the use of Cartesian graphs. Give examples.
14. How do you find the area of a region $0 \leq r_1(\theta) \leq r \leq r_2(\theta)$, $\alpha \leq \theta \leq \beta$, in the polar coordinate plane? Give examples.
15. Under what conditions can you find the length of a curve $r = f(\theta)$, $\alpha \leq \theta \leq \beta$, in the polar coordinate plane? Give an example of a typical calculation.
16. Under what conditions can you find the area of the surface generated by revolving a curve $r = f(\theta)$, $\alpha \leq \theta \leq \beta$, about the x -axis? The y -axis? Give examples of typical calculations.
17. What are the standard equations for lines and conic sections in polar coordinates? Give examples.

Chapter 10 Technology Application Projects

Mathematica/Maple Module

Radar Tracking of a Moving Object

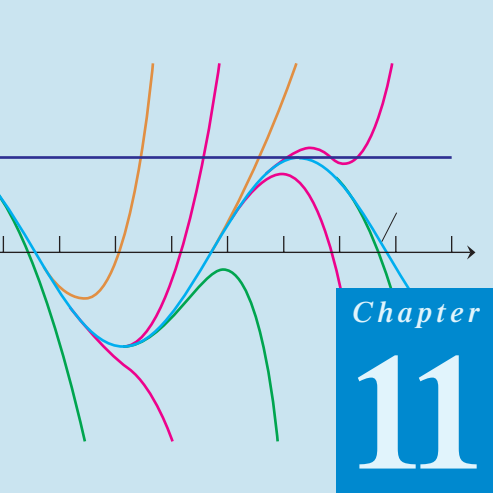
Part I: Convert from polar to Cartesian coordinates.

Mathematica/Maple Module

Parametric and Polar Equations with a Figure Skater

Part I: Visualize position, velocity, and acceleration to analyze motion defined by parametric equations.

Part II: Find and analyze the equations of motion for a figure skater tracing a polar plot.

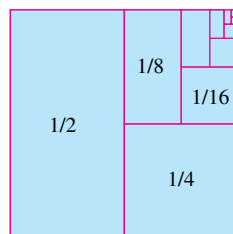


INFINITE SEQUENCES AND SERIES

OVERVIEW While everyone knows how to add together two numbers, or even several, how to add together infinitely many numbers is not so clear. In this chapter we study such questions, the subject of the theory of infinite series. Infinite series sometimes have a finite sum, as in

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = 1.$$

This sum is represented geometrically by the areas of the repeatedly halved unit square shown here. The areas of the small rectangles add together to give the area of the unit square, which they fill. Adding together more and more terms gets us closer and closer to the total.



Other infinite series do not have a finite sum, as with

$$1 + 2 + 3 + 4 + 5 + \cdots$$

The sum of the first few terms gets larger and larger as we add more and more terms. Taking enough terms makes these sums larger than any prechosen constant.

With some infinite series, such as the harmonic series

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \cdots$$

it is not obvious whether a finite sum exists. It is unclear whether adding more and more terms gets us closer to some sum, or gives sums that grow without bound.

As we develop the theory of infinite sequences and series, an important application gives a method of representing a differentiable function $f(x)$ as an infinite sum of powers of x . With this method we can extend our knowledge of how to evaluate, differentiate, and integrate polynomials to a class of functions much more general than polynomials. We also investigate a method of representing a function as an infinite sum of sine and cosine functions. This method will yield a powerful tool to study functions.

11.1

Sequences

HISTORICAL ESSAY

Sequences and Series

A sequence is a list of numbers

$$a_1, a_2, a_3, \dots, a_n, \dots$$

in a given order. Each of a_1, a_2, a_3 and so on represents a number. These are the **terms** of the sequence. For example the sequence

$$2, 4, 6, 8, 10, 12, \dots, 2n, \dots$$

has first term $a_1 = 2$, second term $a_2 = 4$ and n th term $a_n = 2n$. The integer n is called the **index** of a_n , and indicates where a_n occurs in the list. We can think of the sequence

$$a_1, a_2, a_3, \dots, a_n, \dots$$

as a function that sends 1 to a_1 , 2 to a_2 , 3 to a_3 , and in general sends the positive integer n to the n th term a_n . This leads to the formal definition of a sequence.

DEFINITION Infinite Sequence

An **infinite sequence** of numbers is a function whose domain is the set of positive integers.

The function associated to the sequence

$$2, 4, 6, 8, 10, 12, \dots, 2n, \dots$$

sends 1 to $a_1 = 2$, 2 to $a_2 = 4$, and so on. The general behavior of this sequence is described by the formula

$$a_n = 2n.$$

We can equally well make the domain the integers larger than a given number n_0 , and we allow sequences of this type also.

The sequence

$$12, 14, 16, 18, 20, 22, \dots$$

is described by the formula $a_n = 10 + 2n$. It can also be described by the simpler formula $b_n = 2n$, where the index n starts at 6 and increases. To allow such simpler formulas, we let the first index of the sequence be any integer. In the sequence above, $\{a_n\}$ starts with a_1 while $\{b_n\}$ starts with b_6 . Order is important. The sequence 1, 2, 3, 4... is not the same as the sequence 2, 1, 3, 4...

Sequences can be described by writing rules that specify their terms, such as

$$a_n = \sqrt{n},$$

$$b_n = (-1)^{n+1} \frac{1}{n},$$

$$c_n = \frac{n-1}{n},$$

$$d_n = (-1)^{n+1}$$

or by listing terms,

$$\begin{aligned}\{a_n\} &= \{\sqrt{1}, \sqrt{2}, \sqrt{3}, \dots, \sqrt{n}, \dots\} \\ \{b_n\} &= \left\{1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots, (-1)^{n+1} \frac{1}{n}, \dots\right\} \\ \{c_n\} &= \left\{0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots, \frac{n-1}{n}, \dots\right\} \\ \{d_n\} &= \{1, -1, 1, -1, 1, -1, \dots, (-1)^{n+1}, \dots\}.\end{aligned}$$

We also sometimes write

$$\{a_n\} = \{\sqrt{n}\}_{n=1}^{\infty}.$$

Figure 11.1 shows two ways to represent sequences graphically. The first marks the first few points from $a_1, a_2, a_3, \dots, a_n, \dots$ on the real axis. The second method shows the graph of the function defining the sequence. The function is defined only on integer inputs, and the graph consists of some points in the xy -plane, located at $(1, a_1), (2, a_2), \dots, (n, a_n), \dots$.

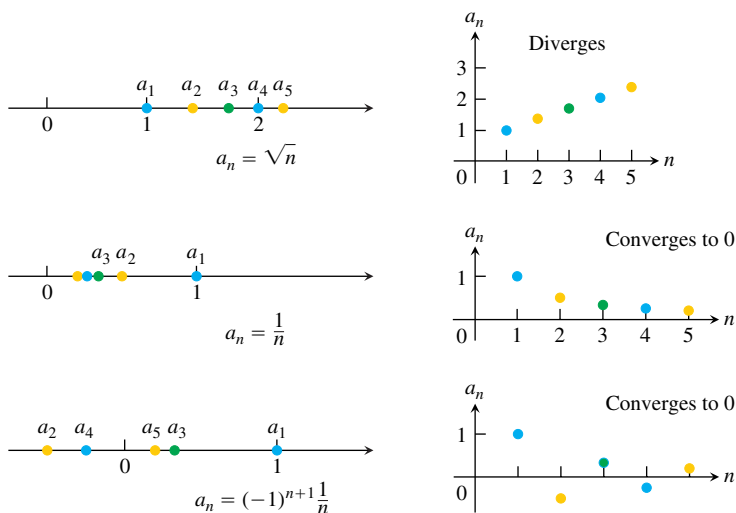


FIGURE 11.1 Sequences can be represented as points on the real line or as points in the plane where the horizontal axis n is the index number of the term and the vertical axis a_n is its value.

Convergence and Divergence

Sometimes the numbers in a sequence approach a single value as the index n increases. This happens in the sequence

$$\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n}, \dots\right\}$$

whose terms approach 0 as n gets large, and in the sequence

$$\left\{0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots, 1 - \frac{1}{n}, \dots\right\}$$

whose terms approach 1. On the other hand, sequences like

$$\{\sqrt{1}, \sqrt{2}, \sqrt{3}, \dots, \sqrt{n}, \dots\}$$

have terms that get larger than any number as n increases, and sequences like

$$\{1, -1, 1, -1, 1, -1, \dots, (-1)^{n+1}, \dots\}$$

bounce back and forth between 1 and -1 , never converging to a single value. The following definition captures the meaning of having a sequence converge to a limiting value. It says that if we go far enough out in the sequence, by taking the index n to be larger than some value N , the difference between a_n and the limit of the sequence becomes less than any preselected number $\epsilon > 0$.

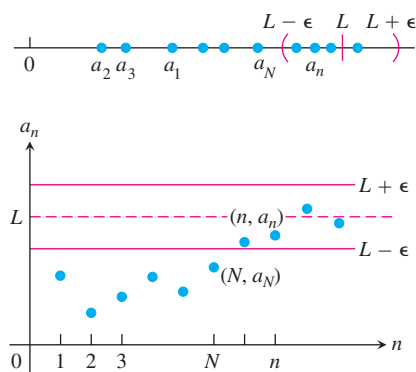


FIGURE 11.2 $a_n \rightarrow L$ if $y = L$ is a horizontal asymptote of the sequence of points $\{(n, a_n)\}$. In this figure, all the a_n 's after a_N lie within ϵ of L .

DEFINITIONS Converges, Diverges, Limit

The sequence $\{a_n\}$ **converges** to the number L if to every positive number ϵ there corresponds an integer N such that for all n ,

$$n > N \quad \Rightarrow \quad |a_n - L| < \epsilon.$$

If no such number L exists, we say that $\{a_n\}$ **diverges**.

If $\{a_n\}$ converges to L , we write $\lim_{n \rightarrow \infty} a_n = L$, or simply $a_n \rightarrow L$, and call L the **limit** of the sequence (Figure 11.2).

HISTORICAL BIOGRAPHY

Nicole Oresme
(ca. 1320–1382)

The definition is very similar to the definition of the limit of a function $f(x)$ as x tends to ∞ ($\lim_{x \rightarrow \infty} f(x)$ in Section 2.4). We will exploit this connection to calculate limits of sequences.

EXAMPLE 1 Applying the Definition

Show that

$$(a) \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad (b) \lim_{n \rightarrow \infty} k = k \quad (\text{any constant } k)$$

Solution

(a) Let $\epsilon > 0$ be given. We must show that there exists an integer N such that for all n ,

$$n > N \quad \Rightarrow \quad \left| \frac{1}{n} - 0 \right| < \epsilon.$$

This implication will hold if $(1/n) < \epsilon$ or $n > 1/\epsilon$. If N is any integer greater than $1/\epsilon$, the implication will hold for all $n > N$. This proves that $\lim_{n \rightarrow \infty} (1/n) = 0$.

(b) Let $\epsilon > 0$ be given. We must show that there exists an integer N such that for all n ,

$$n > N \quad \Rightarrow \quad |k - k| < \epsilon.$$

Since $k - k = 0$, we can use any positive integer for N and the implication will hold. This proves that $\lim_{n \rightarrow \infty} k = k$ for any constant k . ■

EXAMPLE 2 A Divergent Sequence

Show that the sequence $\{1, -1, 1, -1, 1, -1, \dots, (-1)^{n+1}, \dots\}$ diverges.

Solution Suppose the sequence converges to some number L . By choosing $\epsilon = 1/2$ in the definition of the limit, all terms a_n of the sequence with index n larger than some N must lie within $\epsilon = 1/2$ of L . Since the number 1 appears repeatedly as every other term of the sequence, we must have that the number 1 lies within the distance $\epsilon = 1/2$ of L . It follows that $|L - 1| < 1/2$, or equivalently, $1/2 < L < 3/2$. Likewise, the number -1 appears repeatedly in the sequence with arbitrarily high index. So we must also have that $|L - (-1)| < 1/2$, or equivalently, $-3/2 < L < -1/2$. But the number L cannot lie in both of the intervals $(1/2, 3/2)$ and $(-3/2, -1/2)$ because they have no overlap. Therefore, no such limit L exists and so the sequence diverges.

Note that the same argument works for any positive number ϵ smaller than 1, not just $1/2$. ■

The sequence $\{\sqrt{n}\}$ also diverges, but for a different reason. As n increases, its terms become larger than any fixed number. We describe the behavior of this sequence by writing

$$\lim_{n \rightarrow \infty} \sqrt{n} = \infty.$$

In writing infinity as the limit of a sequence, we are not saying that the differences between the terms a_n and ∞ become small as n increases. Nor are we asserting that there is some number infinity that the sequence approaches. We are merely using a notation that captures the idea that a_n eventually gets and stays larger than any fixed number as n gets large.

DEFINITION Diverges to Infinity

The sequence $\{a_n\}$ **diverges to infinity** if for every number M there is an integer N such that for all n larger than N , $a_n > M$. If this condition holds we write

$$\lim_{n \rightarrow \infty} a_n = \infty \quad \text{or} \quad a_n \rightarrow \infty.$$

Similarly if for every number m there is an integer N such that for all $n > N$ we have $a_n < m$, then we say $\{a_n\}$ **diverges to negative infinity** and write

$$\lim_{n \rightarrow \infty} a_n = -\infty \quad \text{or} \quad a_n \rightarrow -\infty.$$

A sequence may diverge without diverging to infinity or negative infinity. We saw this in Example 2, and the sequences $\{1, -2, 3, -4, 5, -6, 7, -8, \dots\}$ and $\{1, 0, 2, 0, 3, 0, \dots\}$ are also examples of such divergence.

Calculating Limits of Sequences

If we always had to use the formal definition of the limit of a sequence, calculating with ϵ 's and N 's, then computing limits of sequences would be a formidable task. Fortunately we can derive a few basic examples, and then use these to quickly analyze the limits of many more sequences. We will need to understand how to combine and compare sequences. Since sequences are functions with domain restricted to the positive integers, it is not too surprising that the theorems on limits of functions given in Chapter 2 have versions for sequences.

THEOREM 1

Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers and let A and B be real numbers. The following rules hold if $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$.

1. *Sum Rule:* $\lim_{n \rightarrow \infty} (a_n + b_n) = A + B$
2. *Difference Rule:* $\lim_{n \rightarrow \infty} (a_n - b_n) = A - B$
3. *Product Rule:* $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = A \cdot B$
4. *Constant Multiple Rule:* $\lim_{n \rightarrow \infty} (k \cdot b_n) = k \cdot B$ (Any number k)
5. *Quotient Rule:* $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B}$ if $B \neq 0$

The proof is similar to that of Theorem 1 of Section 2.2, and is omitted.

EXAMPLE 3 Applying Theorem 1

By combining Theorem 1 with the limits of Example 1, we have:

- (a) $\lim_{n \rightarrow \infty} \left(-\frac{1}{n}\right) = -1 \cdot \lim_{n \rightarrow \infty} \frac{1}{n} = -1 \cdot 0 = 0$ Constant Multiple Rule and Example 1a
- (b) $\lim_{n \rightarrow \infty} \left(\frac{n-1}{n}\right) = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) = \lim_{n \rightarrow \infty} 1 - \lim_{n \rightarrow \infty} \frac{1}{n} = 1 - 0 = 1$ Difference Rule and Example 1a
- (c) $\lim_{n \rightarrow \infty} \frac{5}{n^2} = 5 \cdot \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \lim_{n \rightarrow \infty} \frac{1}{n} = 5 \cdot 0 \cdot 0 = 0$ Product Rule
- (d) $\lim_{n \rightarrow \infty} \frac{4 - 7n^6}{n^6 + 3} = \lim_{n \rightarrow \infty} \frac{(4/n^6) - 7}{1 + (3/n^6)} = \frac{0 - 7}{1 + 0} = -7.$ Sum and Quotient Rules ■

Be cautious in applying Theorem 1. It does not say, for example, that each of the sequences $\{a_n\}$ and $\{b_n\}$ have limits if their sum $\{a_n + b_n\}$ has a limit. For instance, $\{a_n\} = \{1, 2, 3, \dots\}$ and $\{b_n\} = \{-1, -2, -3, \dots\}$ both diverge, but their sum $\{a_n + b_n\} = \{0, 0, 0, \dots\}$ clearly converges to 0.

One consequence of Theorem 1 is that every nonzero multiple of a divergent sequence $\{a_n\}$ diverges. For suppose, to the contrary, that $\{ca_n\}$ converges for some number $c \neq 0$. Then, by taking $k = 1/c$ in the Constant Multiple Rule in Theorem 1, we see that the sequence

$$\left\{\frac{1}{c} \cdot ca_n\right\} = \{a_n\}$$

converges. Thus, $\{ca_n\}$ cannot converge unless $\{a_n\}$ also converges. If $\{a_n\}$ does not converge, then $\{ca_n\}$ does not converge.

The next theorem is the sequence version of the Sandwich Theorem in Section 2.2. You are asked to prove the theorem in Exercise 95.

THEOREM 2 The Sandwich Theorem for Sequences

Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ be sequences of real numbers. If $a_n \leq b_n \leq c_n$ holds for all n beyond some index N , and if $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$ also.

An immediate consequence of Theorem 2 is that, if $|b_n| \leq c_n$ and $c_n \rightarrow 0$, then $b_n \rightarrow 0$ because $-c_n \leq b_n \leq c_n$. We use this fact in the next example.

EXAMPLE 4 Applying the Sandwich Theorem

Since $1/n \rightarrow 0$, we know that

- (a) $\frac{\cos n}{n} \rightarrow 0$ because $-\frac{1}{n} \leq \frac{\cos n}{n} \leq \frac{1}{n}$;
 (b) $\frac{1}{2^n} \rightarrow 0$ because $0 \leq \frac{1}{2^n} \leq \frac{1}{n}$;
 (c) $(-1)^n \frac{1}{n} \rightarrow 0$ because $-\frac{1}{n} \leq (-1)^n \frac{1}{n} \leq \frac{1}{n}$. ■

The application of Theorems 1 and 2 is broadened by a theorem stating that applying a continuous function to a convergent sequence produces a convergent sequence. We state the theorem without proof (Exercise 96).

THEOREM 3 The Continuous Function Theorem for Sequences

Let $\{a_n\}$ be a sequence of real numbers. If $a_n \rightarrow L$ and if f is a function that is continuous at L and defined at all a_n , then $f(a_n) \rightarrow f(L)$.

EXAMPLE 5 Applying Theorem 3

Show that $\sqrt{(n+1)/n} \rightarrow 1$.

Solution We know that $(n+1)/n \rightarrow 1$. Taking $f(x) = \sqrt{x}$ and $L = 1$ in Theorem 3 gives $\sqrt{(n+1)/n} \rightarrow \sqrt{1} = 1$. ■

EXAMPLE 6 The Sequence $\{2^{1/n}\}$

The sequence $\{1/n\}$ converges to 0. By taking $a_n = 1/n$, $f(x) = 2^x$, and $L = 0$ in Theorem 3, we see that $2^{1/n} = f(1/n) \rightarrow f(L) = 2^0 = 1$. The sequence $\{2^{1/n}\}$ converges to 1 (Figure 11.3). ■

Using l'Hôpital's Rule

The next theorem enables us to use l'Hôpital's Rule to find the limits of some sequences. It formalizes the connection between $\lim_{n \rightarrow \infty} a_n$ and $\lim_{x \rightarrow \infty} f(x)$.

THEOREM 4

Suppose that $f(x)$ is a function defined for all $x \geq n_0$ and that $\{a_n\}$ is a sequence of real numbers such that $a_n = f(n)$ for $n \geq n_0$. Then

$$\lim_{x \rightarrow \infty} f(x) = L \quad \Rightarrow \quad \lim_{n \rightarrow \infty} a_n = L.$$

Proof Suppose that $\lim_{x \rightarrow \infty} f(x) = L$. Then for each positive number ϵ there is a number M such that for all x ,

$$x > M \quad \Rightarrow \quad |f(x) - L| < \epsilon.$$

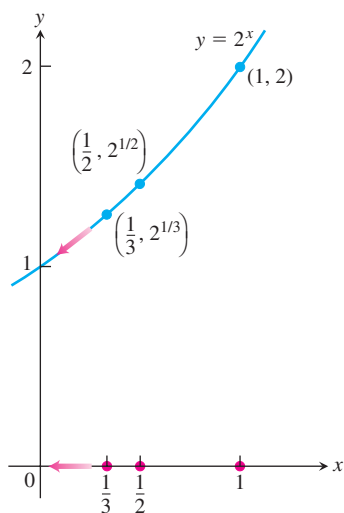


FIGURE 11.3 As $n \rightarrow \infty$, $1/n \rightarrow 0$ and $2^{1/n} \rightarrow 2^0$ (Example 6).

Let N be an integer greater than M and greater than or equal to n_0 . Then

$$n > N \quad \Rightarrow \quad a_n = f(n) \quad \text{and} \quad |a_n - L| = |f(n) - L| < \epsilon. \quad \blacksquare$$

EXAMPLE 7 Applying L'Hôpital's Rule

Show that

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0.$$

Solution The function $(\ln x)/x$ is defined for all $x \geq 1$ and agrees with the given sequence at positive integers. Therefore, by Theorem 5, $\lim_{n \rightarrow \infty} (\ln n)/n$ will equal $\lim_{x \rightarrow \infty} (\ln x)/x$ if the latter exists. A single application of l'Hôpital's Rule shows that

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = \frac{0}{1} = 0.$$

We conclude that $\lim_{n \rightarrow \infty} (\ln n)/n = 0$. ■

When we use l'Hôpital's Rule to find the limit of a sequence, we often treat n as a continuous real variable and differentiate directly with respect to n . This saves us from having to rewrite the formula for a_n as we did in Example 7.

EXAMPLE 8 Applying L'Hôpital's Rule

Find

$$\lim_{n \rightarrow \infty} \frac{2^n}{5n}.$$

Solution By l'Hôpital's Rule (differentiating with respect to n),

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{2^n}{5n} &= \lim_{n \rightarrow \infty} \frac{2^n \cdot \ln 2}{5} \\ &= \infty. \end{aligned} \quad \blacksquare$$

EXAMPLE 9 Applying L'Hôpital's Rule to Determine Convergence

Does the sequence whose n th term is

$$a_n = \left(\frac{n+1}{n-1} \right)^n$$

converge? If so, find $\lim_{n \rightarrow \infty} a_n$.

Solution The limit leads to the indeterminate form 1^∞ . We can apply l'Hôpital's Rule if we first change the form to $\infty \cdot 0$ by taking the natural logarithm of a_n :

$$\begin{aligned} \ln a_n &= \ln \left(\frac{n+1}{n-1} \right)^n \\ &= n \ln \left(\frac{n+1}{n-1} \right). \end{aligned}$$

Then,

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \ln a_n &= \lim_{n \rightarrow \infty} n \ln \left(\frac{n+1}{n-1} \right) && \infty \cdot 0 \\
 &= \lim_{n \rightarrow \infty} \frac{\ln \left(\frac{n+1}{n-1} \right)}{1/n} && \frac{0}{0} \\
 &= \lim_{n \rightarrow \infty} \frac{-2/(n^2-1)}{-1/n^2} && \text{L'Hôpital's Rule} \\
 &= \lim_{n \rightarrow \infty} \frac{2n^2}{n^2-1} = 2.
 \end{aligned}$$

Since $\ln a_n \rightarrow 2$ and $f(x) = e^x$ is continuous, Theorem 4 tells us that

$$a_n = e^{\ln a_n} \rightarrow e^2.$$

The sequence $\{a_n\}$ converges to e^2 . ■

Commonly Occurring Limits

The next theorem gives some limits that arise frequently.

THEOREM 5

The following six sequences converge to the limits listed below:

1. $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$
2. $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$
3. $\lim_{n \rightarrow \infty} x^{1/n} = 1 \quad (x > 0)$
4. $\lim_{n \rightarrow \infty} x^n = 0 \quad (|x| < 1)$
5. $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n = e^x \quad (\text{any } x)$
6. $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0 \quad (\text{any } x)$

In Formulas (3) through (6), x remains fixed as $n \rightarrow \infty$.

Factorial Notation

The notation $n!$ (“ n factorial”) means the product $1 \cdot 2 \cdot 3 \cdots n$ of the integers from 1 to n . Notice that $(n+1)! = (n+1) \cdot n!$. Thus, $4! = 1 \cdot 2 \cdot 3 \cdot 4 = 24$ and $5! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 5 \cdot 4! = 120$. We define $0!$ to be 1. Factorials grow even faster than exponentials, as the table suggests.

n	e^n (rounded)	$n!$
1	3	1
5	148	120
10	22,026	3,628,800
20	4.9×10^8	2.4×10^{18}

Proof The first limit was computed in Example 7. The next two can be proved by taking logarithms and applying Theorem 4 (Exercises 93 and 94). The remaining proofs are given in Appendix 3. ■

EXAMPLE 10 Applying Theorem 5

- (a) $\frac{\ln(n^2)}{n} = \frac{2 \ln n}{n} \rightarrow 2 \cdot 0 = 0$ Formula 1
- (b) $\sqrt[n]{n^2} = n^{2/n} = (n^{1/n})^2 \rightarrow (1)^2 = 1$ Formula 2
- (c) $\sqrt[n]{3n} = 3^{1/n} (n^{1/n}) \rightarrow 1 \cdot 1 = 1$ Formula 3 with $x = 3$ and Formula 2

- (d) $\left(-\frac{1}{2}\right)^n \rightarrow 0$ Formula 4 with $x = -\frac{1}{2}$
- (e) $\left(\frac{n-2}{n}\right)^n = \left(1 + \frac{-2}{n}\right)^n \rightarrow e^{-2}$ Formula 5 with $x = -2$
- (f) $\frac{100^n}{n!} \rightarrow 0$ Formula 6 with $x = 100$ ■

Recursive Definitions

So far, we have calculated each a_n directly from the value of n . But sequences are often defined **recursively** by giving

1. The value(s) of the initial term or terms, and
2. A rule, called a **recursion formula**, for calculating any later term from terms that precede it.

EXAMPLE 11 Sequences Constructed Recursively

- (a) The statements $a_1 = 1$ and $a_n = a_{n-1} + 1$ define the sequence $1, 2, 3, \dots, n, \dots$ of positive integers. With $a_1 = 1$, we have $a_2 = a_1 + 1 = 2$, $a_3 = a_2 + 1 = 3$, and so on.
- (b) The statements $a_1 = 1$ and $a_n = n \cdot a_{n-1}$ define the sequence $1, 2, 6, 24, \dots, n!, \dots$ of factorials. With $a_1 = 1$, we have $a_2 = 2 \cdot a_1 = 2$, $a_3 = 3 \cdot a_2 = 6$, $a_4 = 4 \cdot a_3 = 24$, and so on.
- (c) The statements $a_1 = 1$, $a_2 = 1$, and $a_{n+1} = a_n + a_{n-1}$ define the sequence $1, 1, 2, 3, 5, \dots$ of **Fibonacci numbers**. With $a_1 = 1$ and $a_2 = 1$, we have $a_3 = 1 + 1 = 2$, $a_4 = 2 + 1 = 3$, $a_5 = 3 + 2 = 5$, and so on.
- (d) As we can see by applying Newton's method, the statements $x_0 = 1$ and $x_{n+1} = x_n - [(\sin x_n - x_n^2)/(\cos x_n - 2x_n)]$ define a sequence that converges to a solution of the equation $\sin x - x^2 = 0$. ■

Bounded Nondecreasing Sequences

The terms of a general sequence can bounce around, sometimes getting larger, sometimes smaller. An important special kind of sequence is one for which each term is at least as large as its predecessor.

DEFINITION Nondecreasing Sequence

A sequence $\{a_n\}$ with the property that $a_n \leq a_{n+1}$ for all n is called a **nondecreasing sequence**.

EXAMPLE 12 Nondecreasing Sequences

- (a) The sequence $1, 2, 3, \dots, n, \dots$ of natural numbers
- (b) The sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots, \frac{n}{n+1}, \dots$
- (c) The constant sequence $\{3\}$ ■

There are two kinds of nondecreasing sequences—those whose terms increase beyond any finite bound and those whose terms do not.

DEFINITIONS Bounded, Upper Bound, Least Upper Bound

A sequence $\{a_n\}$ is **bounded from above** if there exists a number M such that $a_n \leq M$ for all n . The number M is an **upper bound** for $\{a_n\}$. If M is an upper bound for $\{a_n\}$ but no number less than M is an upper bound for $\{a_n\}$, then M is the **least upper bound** for $\{a_n\}$.

EXAMPLE 13 Applying the Definition for Boundedness

- (a) The sequence $1, 2, 3, \dots, n, \dots$ has no upper bound.
- (b) The sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots, \frac{n}{n+1}, \dots$ is bounded above by $M = 1$.

No number less than 1 is an upper bound for the sequence, so 1 is the least upper bound (Exercise 113). ■

A nondecreasing sequence that is bounded from above always has a least upper bound. This is the completeness property of the real numbers, discussed in Appendix 4. We will prove that if L is the least upper bound then the sequence converges to L .

Suppose we plot the points $(1, a_1), (2, a_2), \dots, (n, a_n), \dots$ in the xy -plane. If M is an upper bound of the sequence, all these points will lie on or below the line $y = M$ (Figure 11.4). The line $y = L$ is the lowest such line. None of the points (n, a_n) lies above $y = L$, but some do lie above any lower line $y = L - \epsilon$, if ϵ is a positive number. The sequence converges to L because

- (a) $a_n \leq L$ for all values of n and
- (b) given any $\epsilon > 0$, there exists at least one integer N for which $a_N > L - \epsilon$.

The fact that $\{a_n\}$ is nondecreasing tells us further that

$$a_n \geq a_N > L - \epsilon \quad \text{for all } n \geq N.$$

Thus, all the numbers a_n beyond the N th number lie within ϵ of L . This is precisely the condition for L to be the limit of the sequence $\{a_n\}$.

The facts for nondecreasing sequences are summarized in the following theorem. A similar result holds for nonincreasing sequences (Exercise 107).

THEOREM 6 The Nondecreasing Sequence Theorem

A nondecreasing sequence of real numbers converges if and only if it is bounded from above. If a nondecreasing sequence converges, it converges to its least upper bound.

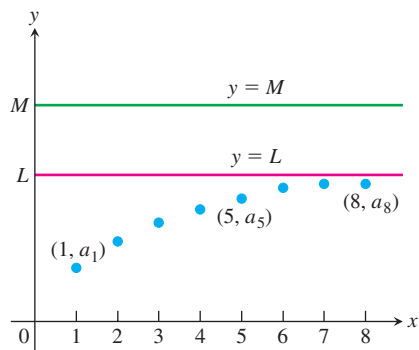


FIGURE 11.4 If the terms of a nondecreasing sequence have an upper bound M , they have a limit $L \leq M$.

Theorem 6 implies that a nondecreasing sequence converges when it is bounded from above. It diverges to infinity if it is not bounded from above.

EXERCISES 11.1

Finding Terms of a Sequence

Each of Exercises 1–6 gives a formula for the n th term a_n of a sequence $\{a_n\}$. Find the values of a_1 , a_2 , a_3 , and a_4 .

1. $a_n = \frac{1-n}{n^2}$

2. $a_n = \frac{1}{n!}$

3. $a_n = \frac{(-1)^{n+1}}{2n-1}$

4. $a_n = 2 + (-1)^n$

5. $a_n = \frac{2^n}{2^{n+1}}$

6. $a_n = \frac{2^n - 1}{2^n}$

Each of Exercises 7–12 gives the first term or two of a sequence along with a recursion formula for the remaining terms. Write out the first ten terms of the sequence.

7. $a_1 = 1, \quad a_{n+1} = a_n + (1/2^n)$

8. $a_1 = 1, \quad a_{n+1} = a_n/(n+1)$

9. $a_1 = 2, \quad a_{n+1} = (-1)^{n+1}a_n/2$

10. $a_1 = -2, \quad a_{n+1} = na_n/(n+1)$

11. $a_1 = a_2 = 1, \quad a_{n+2} = a_{n+1} + a_n$

12. $a_1 = 2, \quad a_2 = -1, \quad a_{n+2} = a_{n+1}/a_n$

Finding a Sequence's Formula

In Exercises 13–22, find a formula for the n th term of the sequence.

13. The sequence 1, -1, 1, -1, 1, ... 1's with alternating signs

14. The sequence -1, 1, -1, 1, -1, ... 1's with alternating signs

15. The sequence 1, -4, 9, -16, 25, ... Squares of the positive integers; with alternating signs

16. The sequence $1, -\frac{1}{4}, \frac{1}{9}, -\frac{1}{16}, \frac{1}{25}, \dots$ Reciprocals of squares of the positive integers, with alternating signs

17. The sequence 0, 3, 8, 15, 24, ... Squares of the positive integers diminished by 1

18. The sequence -3, -2, -1, 0, 1, ... Integers beginning with -3

19. The sequence 1, 5, 9, 13, 17, ... Every other odd positive integer

20. The sequence 2, 6, 10, 14, 18, ... Every other even positive integer

21. The sequence 1, 0, 1, 0, 1, ... Alternating 1's and 0's

22. The sequence 0, 1, 1, 2, 2, 3, 3, 4, ... Each positive integer repeated

Finding Limits

Which of the sequences $\{a_n\}$ in Exercises 23–84 converge, and which diverge? Find the limit of each convergent sequence.

23. $a_n = 2 + (0.1)^n$

25. $a_n = \frac{1-2n}{1+2n}$

27. $a_n = \frac{1-5n^4}{n^4+8n^3}$

29. $a_n = \frac{n^2-2n+1}{n-1}$

31. $a_n = 1 + (-1)^n$

33. $a_n = \left(\frac{n+1}{2n}\right)\left(1-\frac{1}{n}\right)$

35. $a_n = \frac{(-1)^{n+1}}{2n-1}$

37. $a_n = \sqrt{\frac{2n}{n+1}}$

39. $a_n = \sin\left(\frac{\pi}{2} + \frac{1}{n}\right)$

41. $a_n = \frac{\sin n}{n}$

43. $a_n = \frac{n}{2^n}$

45. $a_n = \frac{\ln(n+1)}{\sqrt{n}}$

47. $a_n = 8^{1/n}$

49. $a_n = \left(1 + \frac{7}{n}\right)^n$

51. $a_n = \sqrt[n]{10n}$

53. $a_n = \left(\frac{3}{n}\right)^{1/n}$

55. $a_n = \frac{\ln n}{n^{1/n}}$

57. $a_n = \sqrt[n]{4^n n}$

59. $a_n = \frac{n!}{n^n}$ (Hint: Compare with $1/n$.)

24. $a_n = \frac{n + (-1)^n}{n}$

26. $a_n = \frac{2n+1}{1-3\sqrt{n}}$

28. $a_n = \frac{n+3}{n^2+5n+6}$

30. $a_n = \frac{1-n^3}{70-4n^2}$

32. $a_n = (-1)^n \left(1 - \frac{1}{n}\right)$

34. $a_n = \left(2 - \frac{1}{2^n}\right)\left(3 + \frac{1}{2^n}\right)$

36. $a_n = \left(-\frac{1}{2}\right)^n$

38. $a_n = \frac{1}{(0.9)^n}$

40. $a_n = n\pi \cos(n\pi)$

42. $a_n = \frac{\sin^2 n}{2^n}$

44. $a_n = \frac{3^n}{n^3}$

46. $a_n = \frac{\ln n}{\ln 2n}$

48. $a_n = (0.03)^{1/n}$

50. $a_n = \left(1 - \frac{1}{n}\right)^n$

52. $a_n = \sqrt[n]{n^2}$

54. $a_n = (n+4)^{1/(n+4)}$

56. $a_n = \ln n - \ln(n+1)$

58. $a_n = \sqrt[n]{3^{2n+1}}$

60. $a_n = \frac{(-4)^n}{n!}$
62. $a_n = \frac{n!}{2^n \cdot 3^n}$
64. $a_n = \ln\left(1 + \frac{1}{n}\right)^n$
66. $a_n = \left(\frac{n}{n+1}\right)^n$
68. $a_n = \left(1 - \frac{1}{n^2}\right)^n$
70. $a_n = \frac{(10/11)^n}{(9/10)^n + (11/12)^n}$
72. $a_n = \sinh(\ln n)$
74. $a_n = n\left(1 - \cos \frac{1}{n}\right)$
76. $a_n = \frac{1}{\sqrt{n}} \tan^{-1} n$
78. $a_n = \sqrt[n]{n^2 + n}$
80. $a_n = \frac{(\ln n)^5}{\sqrt{n}}$
82. $a_n = \frac{1}{\sqrt{n^2 - 1} - \sqrt{n^2 + n}}$
83. $a_n = \frac{1}{n} \int_1^n \frac{1}{x} dx$
61. $a_n = \frac{n!}{10^{6n}}$
63. $a_n = \left(\frac{1}{n}\right)^{1/(\ln n)}$
65. $a_n = \left(\frac{3n+1}{3n-1}\right)^n$
67. $a_n = \left(\frac{x^n}{2n+1}\right)^{1/n}, \quad x > 0$
69. $a_n = \frac{3^n \cdot 6^n}{2^{-n} \cdot n!}$
71. $a_n = \tanh n$
73. $a_n = \frac{n^2}{2n-1} \sin \frac{1}{n}$
75. $a_n = \tan^{-1} n$
77. $a_n = \left(\frac{1}{3}\right)^n + \frac{1}{\sqrt{2^n}}$
79. $a_n = \frac{(\ln n)^{200}}{n}$
81. $a_n = n - \sqrt{n^2 - n}$
84. $a_n = \int_1^n \frac{1}{x^p} dx, \quad p > 1$

Theory and Examples

85. The first term of a sequence is $x_1 = 1$. Each succeeding term is the sum of all those that come before it:

$$x_{n+1} = x_1 + x_2 + \cdots + x_n.$$

Write out enough early terms of the sequence to deduce a general formula for x_n that holds for $n \geq 2$.

86. A sequence of rational numbers is described as follows:

$$\frac{1}{1}, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \dots, \frac{a}{b}, \frac{a+2b}{a+b}, \dots$$

Here the numerators form one sequence, the denominators form a second sequence, and their ratios form a third sequence. Let x_n and y_n be, respectively, the numerator and the denominator of the n th fraction $r_n = x_n/y_n$.

- a. Verify that $x_1^2 - 2y_1^2 = -1$, $x_2^2 - 2y_2^2 = +1$ and, more generally, that if $a^2 - 2b^2 = -1$ or $+1$, then

$$(a + 2b)^2 - 2(a + b)^2 = +1 \quad \text{or} \quad -1,$$

respectively.

- b. The fractions $r_n = x_n/y_n$ approach a limit as n increases. What is that limit? (*Hint*: Use part (a) to show that $r_n^2 - 2 = \pm(1/y_n)^2$ and that y_n is not less than n .)

87. **Newton's method** The following sequences come from the recursion formula for Newton's method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Do the sequences converge? If so, to what value? In each case, begin by identifying the function f that generates the sequence.

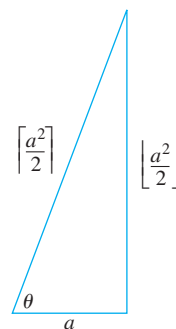
- a. $x_0 = 1, \quad x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{x_n}{2} + \frac{1}{x_n}$
- b. $x_0 = 1, \quad x_{n+1} = x_n - \frac{\tan x_n - 1}{\sec^2 x_n}$
- c. $x_0 = 1, \quad x_{n+1} = x_n - 1$
88. a. Suppose that $f(x)$ is differentiable for all x in $[0, 1]$ and that $f(0) = 0$. Define the sequence $\{a_n\}$ by the rule $a_n = nf(1/n)$. Show that $\lim_{n \rightarrow \infty} a_n = f'(0)$.

Use the result in part (a) to find the limits of the following sequences $\{a_n\}$.

- b. $a_n = n \tan^{-1} \frac{1}{n}$
- c. $a_n = n(e^{1/n} - 1)$
- d. $a_n = n \ln\left(1 + \frac{2}{n}\right)$
89. **Pythagorean triples** A triple of positive integers a , b , and c is called a **Pythagorean triple** if $a^2 + b^2 = c^2$. Let a be an odd positive integer and let

$$b = \left\lfloor \frac{a^2}{2} \right\rfloor \quad \text{and} \quad c = \left\lceil \frac{a^2}{2} \right\rceil$$

be, respectively, the integer floor and ceiling for $a^2/2$.



- a. Show that $a^2 + b^2 = c^2$. (*Hint*: Let $a = 2n + 1$ and express b and c in terms of n .)

- b. By direct calculation, or by appealing to the figure here, find

$$\lim_{a \rightarrow \infty} \frac{\left\lfloor \frac{a^2}{2} \right\rfloor}{\left\lceil \frac{a^2}{2} \right\rceil}.$$

90. The n th root of $n!$

- a. Show that $\lim_{n \rightarrow \infty} (2n\pi)^{1/(2n)} = 1$ and hence, using Stirling's approximation (Chapter 8, Additional Exercise 50a), that

$$\sqrt[n]{n!} \approx \frac{n}{e} \quad \text{for large values of } n.$$

- T** b. Test the approximation in part (a) for $n = 40, 50, 60, \dots$, as far as your calculator will allow.

- 91. a.** Assuming that $\lim_{n \rightarrow \infty} (1/n^c) = 0$ if c is any positive constant, show that

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n^c} = 0$$

if c is any positive constant.

- b. Prove that $\lim_{n \rightarrow \infty} (1/n^c) = 0$ if c is any positive constant. (*Hint:* If $\epsilon = 0.001$ and $c = 0.04$, how large should N be to ensure that $|1/n^c - 0| < \epsilon$ if $n > N$?)

- 92. The zipper theorem** Prove the “zipper theorem” for sequences: If $\{a_n\}$ and $\{b_n\}$ both converge to L , then the sequence

$$a_1, b_1, a_2, b_2, \dots, a_n, b_n, \dots$$

converges to L .

- 93.** Prove that $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$.
94. Prove that $\lim_{n \rightarrow \infty} x^{1/n} = 1$, ($x > 0$).
95. Prove Theorem 2. **96.** Prove Theorem 3.

In Exercises 97–100, determine if the sequence is nondecreasing and if it is bounded from above.

- 97.** $a_n = \frac{3n+1}{n+1}$ **98.** $a_n = \frac{(2n+3)!}{(n+1)!}$
99. $a_n = \frac{2^n 3^n}{n!}$ **100.** $a_n = 2 - \frac{2}{n} - \frac{1}{2^n}$

Which of the sequences in Exercises 101–106 converge, and which diverge? Give reasons for your answers.

- 101.** $a_n = 1 - \frac{1}{n}$ **102.** $a_n = n - \frac{1}{n}$
103. $a_n = \frac{2^n - 1}{2^n}$ **104.** $a_n = \frac{2^n - 1}{3^n}$
105. $a_n = ((-1)^n + 1) \left(\frac{n+1}{n} \right)$

- 106.** The first term of a sequence is $x_1 = \cos(1)$. The next terms are $x_2 = x_1$ or $\cos(2)$, whichever is larger; and $x_3 = x_2$ or $\cos(3)$, whichever is larger (farther to the right). In general,

$$x_{n+1} = \max \{x_n, \cos(n+1)\}.$$

- 107. Nonincreasing sequences** A sequence of numbers $\{a_n\}$ in which $a_n \geq a_{n+1}$ for every n is called a **nonincreasing sequence**. A sequence $\{a_n\}$ is **bounded from below** if there is a number M with $M \leq a_n$ for every n . Such a number M is called a **lower bound** for the sequence. Deduce from Theorem 6 that a nonincreasing sequence that is bounded from below converges and that a nonincreasing sequence that is not bounded from below diverges.

(*Continuation of Exercise 107.*) Using the conclusion of Exercise 107, determine which of the sequences in Exercises 108–112 converge and which diverge.

- 108.** $a_n = \frac{n+1}{n}$ **109.** $a_n = \frac{1 + \sqrt{2n}}{\sqrt{n}}$
110. $a_n = \frac{1 - 4^n}{2^n}$ **111.** $a_n = \frac{4^{n+1} + 3^n}{4^n}$

- 112.** $a_1 = 1, \quad a_{n+1} = 2a_n - 3$

- 113. The sequence $\{n/(n+1)\}$ has a least upper bound of 1** Show that if M is a number less than 1, then the terms of $\{n/(n+1)\}$ eventually exceed M . That is, if $M < 1$ there is an integer N such that $n/(n+1) > M$ whenever $n > N$. Since $n/(n+1) < 1$ for every n , this proves that 1 is a least upper bound for $\{n/(n+1)\}$.

- 114. Uniqueness of least upper bounds** Show that if M_1 and M_2 are least upper bounds for the sequence $\{a_n\}$, then $M_1 = M_2$. That is, a sequence cannot have two different least upper bounds.

- 115.** Is it true that a sequence $\{a_n\}$ of positive numbers must converge if it is bounded from above? Give reasons for your answer.

- 116.** Prove that if $\{a_n\}$ is a convergent sequence, then to every positive number ϵ there corresponds an integer N such that for all m and n ,

$$m > N \quad \text{and} \quad n > N \quad \Rightarrow \quad |a_m - a_n| < \epsilon.$$

- 117. Uniqueness of limits** Prove that limits of sequences are unique. That is, show that if L_1 and L_2 are numbers such that $a_n \rightarrow L_1$ and $a_n \rightarrow L_2$, then $L_1 = L_2$.

- 118. Limits and subsequences** If the terms of one sequence appear in another sequence in their given order, we call the first sequence a **subsequence** of the second. Prove that if two subsequences of a sequence $\{a_n\}$ have different limits $L_1 \neq L_2$, then $\{a_n\}$ diverges.

- 119.** For a sequence $\{a_n\}$ the terms of even index are denoted by a_{2k} and the terms of odd index by a_{2k+1} . Prove that if $a_{2k} \rightarrow L$ and $a_{2k+1} \rightarrow L$, then $a_n \rightarrow L$.

- 120.** Prove that a sequence $\{a_n\}$ converges to 0 if and only if the sequence of absolute values $\{|a_n|\}$ converges to 0.

T Calculator Explorations of Limits

In Exercises 121–124, experiment with a calculator to find a value of N that will make the inequality hold for all $n > N$. Assuming that the inequality is the one from the formal definition of the limit of a sequence, what sequence is being considered in each case and what is its limit?

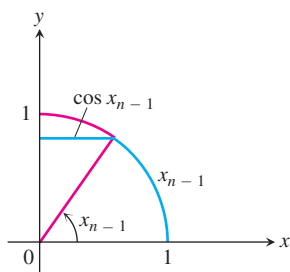
$$121. |\sqrt[n]{0.5} - 1| < 10^{-3} \quad 122. |\sqrt[n]{n} - 1| < 10^{-3}$$

$$123. (0.9)^n < 10^{-3} \quad 124. 2^n/n! < 10^{-7}$$

125. Sequences generated by Newton's method Newton's method, applied to a differentiable function $f(x)$, begins with a starting value x_0 and constructs from it a sequence of numbers $\{x_n\}$ that under favorable circumstances converges to a zero of f . The recursion formula for the sequence is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

- Show that the recursion formula for $f(x) = x^2 - a$, $a > 0$, can be written as $x_{n+1} = (x_n + a/x_n)/2$.
 - Starting with $x_0 = 1$ and $a = 3$, calculate successive terms of the sequence until the display begins to repeat. What number is being approximated? Explain.
- 126. (Continuation of Exercise 125.)** Repeat part (b) of Exercise 125 with $a = 2$ in place of $a = 3$.
- 127. A recursive definition of $\pi/2$** If you start with $x_1 = 1$ and define the subsequent terms of $\{x_n\}$ by the rule $x_n = x_{n-1} + \cos x_{n-1}$, you generate a sequence that converges rapidly to $\pi/2$. **a.** Try it. **b.** Use the accompanying figure to explain why the convergence is so rapid.



- 128.** According to a front-page article in the December 15, 1992, issue of the *Wall Street Journal*, Ford Motor Company used about $7\frac{1}{4}$ hours of labor to produce stampings for the average vehicle, down from an estimated 15 hours in 1980. The Japanese needed only about $3\frac{1}{2}$ hours.

Ford's improvement since 1980 represents an average decrease of 6% per year. If that rate continues, then n years from 1992 Ford will use about

$$S_n = 7.25(0.94)^n$$

hours of labor to produce stampings for the average vehicle. Assuming that the Japanese continue to spend $3\frac{1}{2}$ hours per vehicle,

how many more years will it take Ford to catch up? Find out two ways:

- Find the first term of the sequence $\{S_n\}$ that is less than or equal to 3.5.

T **b.** Graph $f(x) = 7.25(0.94)^x$ and use Trace to find where the graph crosses the line $y = 3.5$.

COMPUTER EXPLORATIONS

Use a CAS to perform the following steps for the sequences in Exercises 129–140.

- Calculate and then plot the first 25 terms of the sequence. Does the sequence appear to be bounded from above or below? Does it appear to converge or diverge? If it does converge, what is the limit L ?
- If the sequence converges, find an integer N such that $|a_n - L| \leq 0.01$ for $n \geq N$. How far in the sequence do you have to get for the terms to lie within 0.0001 of L ?

$$129. a_n = \sqrt[n]{n} \quad 130. a_n = \left(1 + \frac{0.5}{n}\right)^n$$

$$131. a_1 = 1, \quad a_{n+1} = a_n + \frac{1}{5^n}$$

$$132. a_1 = 1, \quad a_{n+1} = a_n + (-2)^n$$

$$133. a_n = \sin n \quad 134. a_n = n \sin \frac{1}{n}$$

$$135. a_n = \frac{\sin n}{n} \quad 136. a_n = \frac{\ln n}{n}$$

$$137. a_n = (0.9999)^n \quad 138. a_n = 123456^{1/n}$$

$$139. a_n = \frac{8^n}{n!} \quad 140. a_n = \frac{n^{41}}{19^n}$$

- 141. Compound interest, deposits, and withdrawals** If you invest an amount of money A_0 at a fixed annual interest rate r compounded m times per year, and if the constant amount b is added to the account at the end of each compounding period (or taken from the account if $b < 0$), then the amount you have after $n + 1$ compounding periods is

$$A_{n+1} = \left(1 + \frac{r}{m}\right)A_n + b. \quad (1)$$

- If $A_0 = 1000$, $r = 0.02015$, $m = 12$, and $b = 50$, calculate and plot the first 100 points (n, A_n) . How much money is in your account at the end of 5 years? Does $\{A_n\}$ converge? Is $\{A_n\}$ bounded?
- Repeat part (a) with $A_0 = 5000$, $r = 0.0589$, $m = 12$, and $b = -50$.
- If you invest 5000 dollars in a certificate of deposit (CD) that pays 4.5% annually, compounded quarterly, and you make no further investments in the CD, approximately how many years will it take before you have 20,000 dollars? What if the CD earns 6.25%?

- d. It can be shown that for any $k \geq 0$, the sequence defined recursively by Equation (1) satisfies the relation

$$A_k = \left(1 + \frac{r}{m}\right)^k \left(A_0 + \frac{mb}{r}\right) - \frac{mb}{r}. \quad (2)$$

For the values of the constants A_0 , r , m , and b given in part (a), validate this assertion by comparing the values of the first 50 terms of both sequences. Then show by direct substitution that the terms in Equation (2) satisfy the recursion formula in Equation (1).

142. Logistic difference equation The recursive relation

$$a_{n+1} = ra_n(1 - a_n)$$

is called the *logistic difference equation*, and when the initial value a_0 is given the equation defines the *logistic sequence* $\{a_n\}$. Throughout this exercise we choose a_0 in the interval $0 < a_0 < 1$, say $a_0 = 0.3$.

- a. Choose $r = 3/4$. Calculate and plot the points (n, a_n) for the first 100 terms in the sequence. Does it appear to converge? What do you guess is the limit? Does the limit seem to depend on your choice of a_0 ?
- b. Choose several values of r in the interval $1 < r < 3$ and repeat the procedures in part (a). Be sure to choose some points near the endpoints of the interval. Describe the behavior of the sequences you observe in your plots.
- c. Now examine the behavior of the sequence for values of r near the endpoints of the interval $3 < r < 3.45$. The transition value $r = 3$ is called a **bifurcation value** and the new behavior of the sequence in the interval is called an **attracting 2-cycle**. Explain why this reasonably describes the behavior.
- d. Next explore the behavior for r values near the endpoints of each of the intervals $3.45 < r < 3.54$ and $3.54 < r < 3.55$. Plot the first 200 terms of the sequences. Describe in your own words the behavior observed in your plots for each interval. Among how many values does the sequence appear to oscillate for each interval? The values $r = 3.45$ and $r = 3.54$ (rounded to two decimal places) are also called bifurcation values because the behavior of the sequence changes as r crosses over those values.
- e. The situation gets even more interesting. There is actually an increasing sequence of bifurcation values $3 < 3.45 < 3.54 < \dots < c_n < c_{n+1} \dots$ such that for $c_n < r < c_{n+1}$ the logistic sequence $\{a_n\}$ eventually oscillates steadily among 2^n values, called an **attracting 2^n -cycle**. Moreover, the bifurcation sequence $\{c_n\}$ is bounded above by 3.57 (so it converges). If you choose a value of $r < 3.57$ you will observe a 2^n -cycle of some sort. Choose $r = 3.5695$ and plot 300 points.
- f. Let us see what happens when $r > 3.57$. Choose $r = 3.65$ and calculate and plot the first 300 terms of $\{a_n\}$. Observe how the terms wander around in an unpredictable, chaotic fashion. You cannot predict the value of a_{n+1} from previous values of the sequence.
- g. For $r = 3.65$ choose two starting values of a_0 that are close together, say, $a_0 = 0.3$ and $a_0 = 0.301$. Calculate and plot the first 300 values of the sequences determined by each starting value. Compare the behaviors observed in your plots. How far out do you go before the corresponding terms of your two sequences appear to depart from each other? Repeat the exploration for $r = 3.75$. Can you see how the plots look different depending on your choice of a_0 ? We say that the logistic sequence is *sensitive to the initial condition* a_0 .

11.2

Infinite Series

An *infinite series* is the sum of an infinite sequence of numbers

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

The goal of this section is to understand the meaning of such an infinite sum and to develop methods to calculate it. Since there are infinitely many terms to add in an infinite series, we cannot just keep adding to see what comes out. Instead we look at what we get by summing the first n terms of the sequence and stopping. The sum of the first n terms

$$s_n = a_1 + a_2 + a_3 + \cdots + a_n$$

is an ordinary finite sum and can be calculated by normal addition. It is called the *n*th partial sum. As *n* gets larger, we expect the partial sums to get closer and closer to a limiting value in the same sense that the terms of a sequence approach a limit, as discussed in Section 11.1.

For example, to assign meaning to an expression like

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$$

We add the terms one at a time from the beginning and look for a pattern in how these partial sums grow.

Partial sum		Suggestive expression for partial sum	Value
First:	$s_1 = 1$	$2 - 1$	1
Second:	$s_2 = 1 + \frac{1}{2}$	$2 - \frac{1}{2}$	$\frac{3}{2}$
Third:	$s_3 = 1 + \frac{1}{2} + \frac{1}{4}$	$2 - \frac{1}{4}$	$\frac{7}{4}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>n</i> th:	$s_n = 1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-1}}$	$2 - \frac{1}{2^{n-1}}$	$\frac{2^n - 1}{2^{n-1}}$

Indeed there is a pattern. The partial sums form a sequence whose *n*th term is

$$s_n = 2 - \frac{1}{2^{n-1}}.$$

This sequence of partial sums converges to 2 because $\lim_{n \rightarrow \infty} (1/2^n) = 0$. We say

“the sum of the infinite series $1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-1}} + \cdots$ is 2.”

Is the sum of any finite number of terms in this series equal to 2? No. Can we actually add an infinite number of terms one by one? No. But we can still define their sum by defining it to be the limit of the sequence of partial sums as $n \rightarrow \infty$, in this case 2 (Figure 11.5). Our knowledge of sequences and limits enables us to break away from the confines of finite sums.

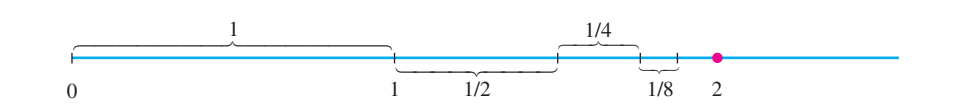


FIGURE 11.5 As the lengths $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$ are added one by one, the sum approaches 2.

HISTORICAL BIOGRAPHY

Blaise Pascal
(1623–1662)

DEFINITIONS Infinite Series, n th Term, Partial Sum, Converges, Sum

Given a sequence of numbers $\{a_n\}$, an expression of the form

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

is an **infinite series**. The number a_n is the **n th term** of the series. The sequence $\{s_n\}$ defined by

$$\begin{aligned} s_1 &= a_1 \\ s_2 &= a_1 + a_2 \\ &\vdots \\ s_n &= a_1 + a_2 + \cdots + a_n = \sum_{k=1}^n a_k \\ &\vdots \end{aligned}$$

is the **sequence of partial sums** of the series, the number s_n being the **n th partial sum**. If the sequence of partial sums converges to a limit L , we say that the series **converges** and that its **sum** is L . In this case, we also write

$$a_1 + a_2 + \cdots + a_n + \cdots = \sum_{n=1}^{\infty} a_n = L.$$

If the sequence of partial sums of the series does not converge, we say that the series **diverges**.

When we begin to study a given series $a_1 + a_2 + \cdots + a_n + \cdots$, we might not know whether it converges or diverges. In either case, it is convenient to use sigma notation to write the series as

$$\sum_{n=1}^{\infty} a_n, \quad \sum_{k=1}^{\infty} a_k, \quad \text{or} \quad \sum a_n$$

A useful shorthand
when summation
from 1 to ∞ is
understood

Geometric Series

Geometric series are series of the form

$$a + ar + ar^2 + \cdots + ar^{n-1} + \cdots = \sum_{n=1}^{\infty} ar^{n-1}$$

in which a and r are fixed real numbers and $a \neq 0$. The series can also be written as $\sum_{n=0}^{\infty} ar^n$. The **ratio** r can be positive, as in

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \left(\frac{1}{2}\right)^{n-1} + \cdots,$$

or negative, as in

$$1 - \frac{1}{3} + \frac{1}{9} - \cdots + \left(-\frac{1}{3}\right)^{n-1} + \cdots.$$

If $r = 1$, the n th partial sum of the geometric series is

$$s_n = a + a(1) + a(1)^2 + \cdots + a(1)^{n-1} = na,$$

and the series diverges because $\lim_{n \rightarrow \infty} s_n = \pm\infty$, depending on the sign of a . If $r = -1$, the series diverges because the n th partial sums alternate between a and 0. If $|r| \neq 1$, we can determine the convergence or divergence of the series in the following way:

$$\begin{aligned} s_n &= a + ar + ar^2 + \cdots + ar^{n-1} \\ rs_n &= ar + ar^2 + \cdots + ar^{n-1} + ar^n && \text{Multiply } s_n \text{ by } r. \\ s_n - rs_n &= a - ar^n && \text{Subtract } rs_n \text{ from } s_n. \text{ Most of} \\ &&& \text{the terms on the right cancel.} \\ s_n(1 - r) &= a(1 - r^n) && \text{Factor.} \\ s_n &= \frac{a(1 - r^n)}{1 - r}, \quad (r \neq 1). && \text{We can solve for } s_n \text{ if } r \neq 1. \end{aligned}$$

If $|r| < 1$, then $r^n \rightarrow 0$ as $n \rightarrow \infty$ (as in Section 11.1) and $s_n \rightarrow a/(1 - r)$. If $|r| > 1$, then $|r^n| \rightarrow \infty$ and the series diverges.

If $|r| < 1$, the geometric series $a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$ converges to $a/(1 - r)$:

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1 - r}, \quad |r| < 1.$$

If $|r| \geq 1$, the series diverges.

We have determined when a geometric series converges or diverges, and to what value. Often we can determine that a series converges without knowing the value to which it converges, as we will see in the next several sections. The formula $a/(1 - r)$ for the sum of a geometric series applies *only* when the summation index begins with $n = 1$ in the expression $\sum_{n=1}^{\infty} ar^{n-1}$ (or with the index $n = 0$ if we write the series as $\sum_{n=0}^{\infty} ar^n$).

EXAMPLE 1 Index Starts with $n = 1$

The geometric series with $a = 1/9$ and $r = 1/3$ is

$$\frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \cdots = \sum_{n=1}^{\infty} \frac{1}{9} \left(\frac{1}{3}\right)^{n-1} = \frac{1/9}{1 - (1/3)} = \frac{1}{6}. \quad \blacksquare$$

EXAMPLE 2 Index Starts with $n = 0$

The series

$$\sum_{n=0}^{\infty} \frac{(-1)^n 5}{4^n} = 5 - \frac{5}{4} + \frac{5}{16} - \frac{5}{64} + \cdots$$

is a geometric series with $a = 5$ and $r = -1/4$. It converges to

$$\frac{a}{1 - r} = \frac{5}{1 + (1/4)} = 4. \quad \blacksquare$$

EXAMPLE 3 A Bouncing Ball

You drop a ball from a meters above a flat surface. Each time the ball hits the surface after falling a distance h , it rebounds a distance rh , where r is positive but less than 1. Find the total distance the ball travels up and down (Figure 11.6).

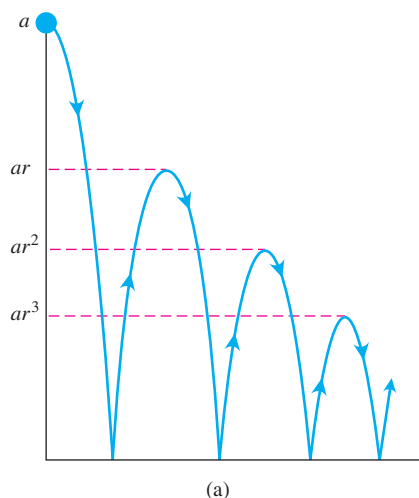


FIGURE 11.6 (a) Example 3 shows how to use a geometric series to calculate the total vertical distance traveled by a bouncing ball if the height of each rebound is reduced by the factor r . (b) A stroboscopic photo of a bouncing ball.

Solution The total distance is

$$s = a + \underbrace{2ar + 2ar^2 + 2ar^3 + \cdots}_{\text{This sum is } 2ar/(1-r)} = a + \frac{2ar}{1-r} = a \frac{1+r}{1-r}.$$

If $a = 6$ m and $r = 2/3$, for instance, the distance is

$$s = 6 \frac{1 + (2/3)}{1 - (2/3)} = 6 \left(\frac{5/3}{1/3} \right) = 30 \text{ m.}$$

EXAMPLE 4 Repeating Decimals

Express the repeating decimal $5.232323 \dots$ as the ratio of two integers.

Solution

$$\begin{aligned} 5.232323 \dots &= 5 + \frac{23}{100} + \frac{23}{(100)^2} + \frac{23}{(100)^3} + \cdots \\ &= 5 + \frac{23}{100} \underbrace{\left(1 + \frac{1}{100} + \left(\frac{1}{100} \right)^2 + \cdots \right)}_{1/(1-0.01)} \quad \begin{array}{l} a = 1, \\ r = 1/100 \end{array} \\ &= 5 + \frac{23}{100} \left(\frac{1}{0.99} \right) = 5 + \frac{23}{99} = \frac{518}{99} \end{aligned}$$

Unfortunately, formulas like the one for the sum of a convergent geometric series are rare and we usually have to settle for an estimate of a series' sum (more about this later). The next example, however, is another case in which we can find the sum exactly.

EXAMPLE 5 A Nongeometric but Telescoping Series

Find the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

Solution We look for a pattern in the sequence of partial sums that might lead to a formula for s_k . The key observation is the partial fraction decomposition

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1},$$

so

$$\sum_{n=1}^k \frac{1}{n(n+1)} = \sum_{n=1}^k \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

and

$$s_k = \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \cdots + \left(\frac{1}{k} - \frac{1}{k+1} \right).$$

Removing parentheses and canceling adjacent terms of opposite sign collapses the sum to

$$s_k = 1 - \frac{1}{k+1}.$$

We now see that $s_k \rightarrow 1$ as $k \rightarrow \infty$. The series converges, and its sum is 1:

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1. \quad \blacksquare$$

Divergent Series

One reason that a series may fail to converge is that its terms don't become small.

EXAMPLE 6 Partial Sums Outgrow Any Number

(a) The series

$$\sum_{n=1}^{\infty} n^2 = 1 + 4 + 9 + \cdots + n^2 + \cdots$$

diverges because the partial sums grow beyond every number L . After $n = 1$, the partial sum $s_n = 1 + 4 + 9 + \cdots + n^2$ is greater than n^2 .

(b) The series

$$\sum_{n=1}^{\infty} \frac{n+1}{n} = \frac{2}{1} + \frac{3}{2} + \frac{4}{3} + \cdots + \frac{n+1}{n} + \cdots$$

diverges because the partial sums eventually outgrow every preassigned number. Each term is greater than 1, so the sum of n terms is greater than n . ■

The n th-Term Test for Divergence

Observe that $\lim_{n \rightarrow \infty} a_n$ must equal zero if the series $\sum_{n=1}^{\infty} a_n$ converges. To see why, let S represent the series' sum and $s_n = a_1 + a_2 + \cdots + a_n$ the n th partial sum. When n is large, both s_n and s_{n-1} are close to S , so their difference, a_n , is close to zero. More formally,

$$a_n = s_n - s_{n-1} \rightarrow S - S = 0. \quad \text{Difference Rule for sequences}$$

This establishes the following theorem.

Caution

Theorem 7 *does not say* that $\sum_{n=1}^{\infty} a_n$ converges if $a_n \rightarrow 0$. It is possible for a series to diverge when $a_n \rightarrow 0$.

THEOREM 7

If $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \rightarrow 0$.

Theorem 7 leads to a test for detecting the kind of divergence that occurred in Example 6.

The n th-Term Test for Divergence

$\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n \rightarrow \infty} a_n$ fails to exist or is different from zero.

EXAMPLE 7 Applying the n th-Term Test

- (a) $\sum_{n=1}^{\infty} n^2$ diverges because $n^2 \rightarrow \infty$
- (b) $\sum_{n=1}^{\infty} \frac{n+1}{n}$ diverges because $\frac{n+1}{n} \rightarrow 1$
- (c) $\sum_{n=1}^{\infty} (-1)^{n+1}$ diverges because $\lim_{n \rightarrow \infty} (-1)^{n+1}$ does not exist
- (d) $\sum_{n=1}^{\infty} \frac{-n}{2n+5}$ diverges because $\lim_{n \rightarrow \infty} \frac{-n}{2n+5} = -\frac{1}{2} \neq 0$. ■

EXAMPLE 8 $a_n \rightarrow 0$ but the Series Diverges

The series

$$1 + \underbrace{\frac{1}{2} + \frac{1}{2}}_{2 \text{ terms}} + \underbrace{\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}}_{4 \text{ terms}} + \cdots + \underbrace{\frac{1}{2^n} + \frac{1}{2^n} + \cdots + \frac{1}{2^n}}_{2^n \text{ terms}} + \cdots$$

diverges because the terms are grouped into clusters that add to 1, so the partial sums increase without bound. However, the terms of the series form a sequence that converges to 0. Example 1 of Section 11.3 shows that the harmonic series also behaves in this manner. ■

Combining Series

Whenever we have two convergent series, we can add them term by term, subtract them term by term, or multiply them by constants to make new convergent series.

THEOREM 8

If $\sum a_n = A$ and $\sum b_n = B$ are convergent series, then

1. *Sum Rule:* $\sum (a_n + b_n) = \sum a_n + \sum b_n = A + B$
2. *Difference Rule:* $\sum (a_n - b_n) = \sum a_n - \sum b_n = A - B$
3. *Constant Multiple Rule:* $\sum ka_n = k \sum a_n = kA$ (Any number k).

Proof The three rules for series follow from the analogous rules for sequences in Theorem 1, Section 11.1. To prove the Sum Rule for series, let

$$A_n = a_1 + a_2 + \cdots + a_n, \quad B_n = b_1 + b_2 + \cdots + b_n.$$

Then the partial sums of $\sum (a_n + b_n)$ are

$$\begin{aligned} s_n &= (a_1 + b_1) + (a_2 + b_2) + \cdots + (a_n + b_n) \\ &= (a_1 + \cdots + a_n) + (b_1 + \cdots + b_n) \\ &= A_n + B_n. \end{aligned}$$

Since $A_n \rightarrow A$ and $B_n \rightarrow B$, we have $s_n \rightarrow A + B$ by the Sum Rule for sequences. The proof of the Difference Rule is similar.

To prove the Constant Multiple Rule for series, observe that the partial sums of $\sum ka_n$ form the sequence

$$s_n = ka_1 + ka_2 + \cdots + ka_n = k(a_1 + a_2 + \cdots + a_n) = kA_n,$$

which converges to kA by the Constant Multiple Rule for sequences. ■

As corollaries of Theorem 8, we have

1. Every nonzero constant multiple of a divergent series diverges.
2. If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum(a_n + b_n)$ and $\sum(a_n - b_n)$ both diverge.

We omit the proofs.

CAUTION Remember that $\sum(a_n + b_n)$ can converge when $\sum a_n$ and $\sum b_n$ both diverge. For example, $\sum a_n = 1 + 1 + 1 + \cdots$ and $\sum b_n = (-1) + (-1) + (-1) + \cdots$ diverge, whereas $\sum(a_n + b_n) = 0 + 0 + 0 + \cdots$ converges to 0.

EXAMPLE 9 Find the sums of the following series.

$$\begin{aligned}
 \text{(a)} \quad \sum_{n=1}^{\infty} \frac{3^{n-1} - 1}{6^{n-1}} &= \sum_{n=1}^{\infty} \left(\frac{1}{2^{n-1}} - \frac{1}{6^{n-1}} \right) \\
 &= \sum_{n=1}^{\infty} \frac{1}{2^{n-1}} - \sum_{n=1}^{\infty} \frac{1}{6^{n-1}} && \text{Difference Rule} \\
 &= \frac{1}{1 - (1/2)} - \frac{1}{1 - (1/6)} && \text{Geometric series with } a = 1 \text{ and } r = 1/2, 1/6 \\
 &= 2 - \frac{6}{5} \\
 &= \frac{4}{5}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad \sum_{n=0}^{\infty} \frac{4}{2^n} &= 4 \sum_{n=0}^{\infty} \frac{1}{2^n} && \text{Constant Multiple Rule} \\
 &= 4 \left(\frac{1}{1 - (1/2)} \right) && \text{Geometric series with } a = 1, r = 1/2 \\
 &= 8
 \end{aligned}$$

Adding or Deleting Terms

We can add a finite number of terms to a series or delete a finite number of terms without altering the series' convergence or divergence, although in the case of convergence this will usually change the sum. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$ and

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \cdots + a_{k-1} + \sum_{n=k}^{\infty} a_n.$$

Conversely, if $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$, then $\sum_{n=1}^{\infty} a_n$ converges. Thus,

$$\sum_{n=1}^{\infty} \frac{1}{5^n} = \frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \sum_{n=4}^{\infty} \frac{1}{5^n}$$

and

$$\sum_{n=4}^{\infty} \frac{1}{5^n} = \left(\sum_{n=1}^{\infty} \frac{1}{5^n} \right) - \frac{1}{5} - \frac{1}{25} - \frac{1}{125}.$$

HISTORICAL BIOGRAPHY

Richard Dedekind
(1831–1916)

Reindexing

As long as we preserve the order of its terms, we can reindex any series without altering its convergence. To raise the starting value of the index h units, replace the n in the formula for a_n by $n - h$:

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1+h}^{\infty} a_{n-h} = a_1 + a_2 + a_3 + \cdots.$$

To lower the starting value of the index h units, replace the n in the formula for a_n by $n + h$:

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1-h}^{\infty} a_{n+h} = a_1 + a_2 + a_3 + \cdots.$$

It works like a horizontal shift. We saw this in starting a geometric series with the index $n = 0$ instead of the index $n = 1$, but we can use any other starting index value as well. We usually give preference to indexings that lead to simple expressions.

EXAMPLE 10 Reindexing a Geometric Series

We can write the geometric series

$$\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 1 + \frac{1}{2} + \frac{1}{4} + \cdots$$

as

$$\sum_{n=0}^{\infty} \frac{1}{2^n}, \quad \sum_{n=5}^{\infty} \frac{1}{2^{n-5}}, \quad \text{or even} \quad \sum_{n=-4}^{\infty} \frac{1}{2^{n+4}}.$$

The partial sums remain the same no matter what indexing we choose. ■

EXERCISES 11.2

Finding n th Partial Sums

In Exercises 1–6, find a formula for the n th partial sum of each series and use it to find the series' sum if the series converges.

1. $2 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \cdots + \frac{2}{3^{n-1}} + \cdots$

2. $\frac{9}{100} + \frac{9}{100^2} + \frac{9}{100^3} + \cdots + \frac{9}{100^n} + \cdots$

3. $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots + (-1)^{n-1} \frac{1}{2^{n-1}} + \cdots$

4. $1 - 2 + 4 - 8 + \cdots + (-1)^{n-1} 2^{n-1} + \cdots$

5. $\frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \cdots + \frac{1}{(n+1)(n+2)} + \cdots$

6. $\frac{5}{1 \cdot 2} + \frac{5}{2 \cdot 3} + \frac{5}{3 \cdot 4} + \cdots + \frac{5}{n(n+1)} + \cdots$

Series with Geometric Terms

In Exercises 7–14, write out the first few terms of each series to show how the series starts. Then find the sum of the series.

7. $\sum_{n=0}^{\infty} \frac{(-1)^n}{4^n}$

8. $\sum_{n=2}^{\infty} \frac{1}{4^n}$

$$\begin{array}{ll}
 9. \sum_{n=1}^{\infty} \frac{7}{4^n} & 10. \sum_{n=0}^{\infty} (-1)^n \frac{5}{4^n} \\
 11. \sum_{n=0}^{\infty} \left(\frac{5}{2^n} + \frac{1}{3^n} \right) & 12. \sum_{n=0}^{\infty} \left(\frac{5}{2^n} - \frac{1}{3^n} \right) \\
 13. \sum_{n=0}^{\infty} \left(\frac{1}{2^n} + \frac{(-1)^n}{5^n} \right) & 14. \sum_{n=0}^{\infty} \left(\frac{2^{n+1}}{5^n} \right)
 \end{array}$$

Telescoping Series

Use partial fractions to find the sum of each series in Exercises 15–22.

$$\begin{array}{ll}
 15. \sum_{n=1}^{\infty} \frac{4}{(4n-3)(4n+1)} & 16. \sum_{n=1}^{\infty} \frac{6}{(2n-1)(2n+1)} \\
 17. \sum_{n=1}^{\infty} \frac{40n}{(2n-1)^2(2n+1)^2} & 18. \sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+1)^2} \\
 19. \sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right) & 20. \sum_{n=1}^{\infty} \left(\frac{1}{2^{1/n}} - \frac{1}{2^{1/(n+1)}} \right) \\
 21. \sum_{n=1}^{\infty} \left(\frac{1}{\ln(n+2)} - \frac{1}{\ln(n+1)} \right) & \\
 22. \sum_{n=1}^{\infty} (\tan^{-1}(n) - \tan^{-1}(n+1)) &
 \end{array}$$

Convergence or Divergence

Which series in Exercises 23–40 converge, and which diverge? Give reasons for your answers. If a series converges, find its sum.

$$\begin{array}{ll}
 23. \sum_{n=0}^{\infty} \left(\frac{1}{\sqrt{2}} \right)^n & 24. \sum_{n=0}^{\infty} (\sqrt{2})^n \\
 25. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{3}{2^n} & 26. \sum_{n=1}^{\infty} (-1)^{n+1} n \\
 27. \sum_{n=0}^{\infty} \cos n\pi & 28. \sum_{n=0}^{\infty} \frac{\cos n\pi}{5^n} \\
 29. \sum_{n=0}^{\infty} e^{-2n} & 30. \sum_{n=1}^{\infty} \ln \frac{1}{n} \\
 31. \sum_{n=1}^{\infty} \frac{2}{10^n} & 32. \sum_{n=0}^{\infty} \frac{1}{x^n}, \quad |x| > 1 \\
 33. \sum_{n=0}^{\infty} \frac{2^n - 1}{3^n} & 34. \sum_{n=1}^{\infty} \left(1 - \frac{1}{n} \right)^n \\
 35. \sum_{n=0}^{\infty} \frac{n!}{1000^n} & 36. \sum_{n=1}^{\infty} \frac{n^n}{n!} \\
 37. \sum_{n=1}^{\infty} \ln \left(\frac{n}{n+1} \right) & 38. \sum_{n=1}^{\infty} \ln \left(\frac{n}{2n+1} \right) \\
 39. \sum_{n=0}^{\infty} \left(\frac{e}{\pi} \right)^n & 40. \sum_{n=0}^{\infty} \frac{e^{n\pi}}{\pi^{ne}}
 \end{array}$$

Geometric Series

In each of the geometric series in Exercises 41–44, write out the first few terms of the series to find a and r , and find the sum of the series.

Then express the inequality $|r| < 1$ in terms of x and find the values of x for which the inequality holds and the series converges.

$$\begin{array}{ll}
 41. \sum_{n=0}^{\infty} (-1)^n x^n & 42. \sum_{n=0}^{\infty} (-1)^n x^{2n} \\
 43. \sum_{n=0}^{\infty} 3 \left(\frac{x-1}{2} \right)^n & 44. \sum_{n=0}^{\infty} \frac{(-1)^n}{2} \left(\frac{1}{3 + \sin x} \right)^n
 \end{array}$$

In Exercises 45–50, find the values of x for which the given geometric series converges. Also, find the sum of the series (as a function of x) for those values of x .

$$\begin{array}{ll}
 45. \sum_{n=0}^{\infty} 2^n x^n & 46. \sum_{n=0}^{\infty} (-1)^n x^{-2n} \\
 47. \sum_{n=0}^{\infty} (-1)^n (x+1)^n & 48. \sum_{n=0}^{\infty} \left(-\frac{1}{2} \right)^n (x-3)^n \\
 49. \sum_{n=0}^{\infty} \sin^n x & 50. \sum_{n=0}^{\infty} (\ln x)^n
 \end{array}$$

Repeating Decimals

Express each of the numbers in Exercises 51–58 as the ratio of two integers.

$$\begin{array}{ll}
 51. 0.\overline{23} = 0.23\,23\,23\ldots & \\
 52. 0.\overline{234} = 0.234\,234\,234\ldots & \\
 53. 0.\overline{7} = 0.7777\ldots & \\
 54. 0.\overline{d} = 0.d\,d\,d\,d\ldots, \text{ where } d \text{ is a digit} & \\
 55. 0.\overline{06} = 0.06666\ldots & \\
 56. 1.\overline{414} = 1.414\,414\,414\ldots & \\
 57. 1.24\overline{123} = 1.24\,123\,123\,123\ldots & \\
 58. 3.\overline{142857} = 3.142857\,142857\ldots &
 \end{array}$$

Theory and Examples

59. The series in Exercise 5 can also be written as

$$\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)} \quad \text{and} \quad \sum_{n=-1}^{\infty} \frac{1}{(n+3)(n+4)}.$$

Write it as a sum beginning with (a) $n = -2$, (b) $n = 0$, (c) $n = 5$.

60. The series in Exercise 6 can also be written as

$$\sum_{n=1}^{\infty} \frac{5}{n(n+1)} \quad \text{and} \quad \sum_{n=0}^{\infty} \frac{5}{(n+1)(n+2)}.$$

Write it as a sum beginning with (a) $n = -1$, (b) $n = 3$, (c) $n = 20$.

61. Make up an infinite series of nonzero terms whose sum is
a. 1 b. -3 c. 0.

62. (Continuation of Exercise 61.) Can you make an infinite series of nonzero terms that converges to any number you want? Explain.

63. Show by example that $\sum (a_n/b_n)$ may diverge even though $\sum a_n$ and $\sum b_n$ converge and no b_n equals 0.

64. Find convergent geometric series $A = \sum a_n$ and $B = \sum b_n$ that illustrate the fact that $\sum a_n b_n$ may converge without being equal to AB .
65. Show by example that $\sum (a_n/b_n)$ may converge to something other than A/B even when $A = \sum a_n$, $B = \sum b_n \neq 0$, and no b_n equals 0.
66. If $\sum a_n$ converges and $a_n > 0$ for all n , can anything be said about $\sum (1/a_n)$? Give reasons for your answer.
67. What happens if you add a finite number of terms to a divergent series or delete a finite number of terms from a divergent series? Give reasons for your answer.
68. If $\sum a_n$ converges and $\sum b_n$ diverges, can anything be said about their term-by-term sum $\sum (a_n + b_n)$? Give reasons for your answer.
69. Make up a geometric series $\sum ar^{n-1}$ that converges to the number 5 if
- $a = 2$
 - $a = 13/2$.

70. Find the value of b for which

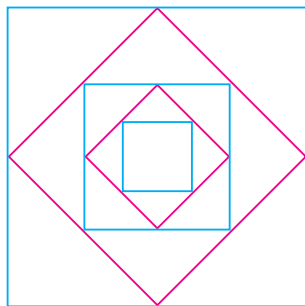
$$1 + e^b + e^{2b} + e^{3b} + \cdots = 9.$$

71. For what values of r does the infinite series

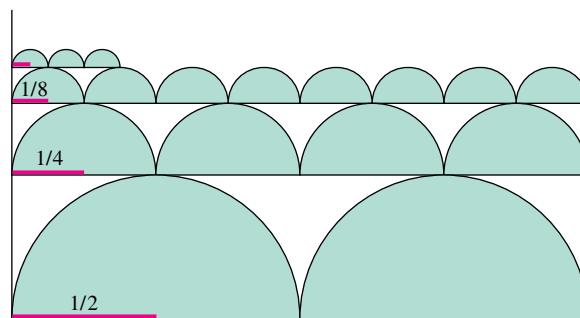
$$1 + 2r + r^2 + 2r^3 + r^4 + 2r^5 + r^6 + \cdots$$

converge? Find the sum of the series when it converges.

72. Show that the error $(L - s_n)$ obtained by replacing a convergent geometric series with one of its partial sums s_n is $ar^n/(1 - r)$.
73. A ball is dropped from a height of 4 m. Each time it strikes the pavement after falling from a height of h meters it rebounds to a height of $0.75h$ meters. Find the total distance the ball travels up and down.
74. (Continuation of Exercise 73.) Find the total number of seconds the ball in Exercise 73 is traveling. (Hint: The formula $s = 4.9t^2$ gives $t = \sqrt{s/4.9}$.)
75. The accompanying figure shows the first five of a sequence of squares. The outermost square has an area of 4 m^2 . Each of the other squares is obtained by joining the midpoints of the sides of the squares before it. Find the sum of the areas of all the squares.

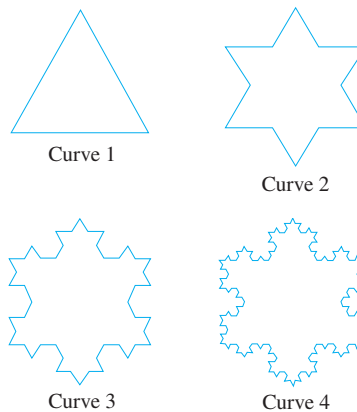


76. The accompanying figure shows the first three rows and part of the fourth row of a sequence of rows of semicircles. There are 2^n semicircles in the n th row, each of radius $1/2^n$. Find the sum of the areas of all the semicircles.

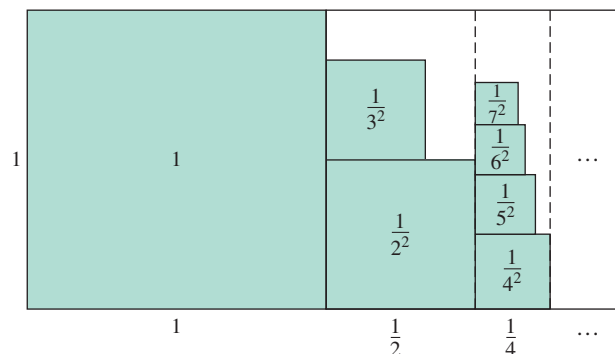


77. **Helga von Koch's snowflake curve** Helga von Koch's snowflake is a curve of infinite length that encloses a region of finite area. To see why this is so, suppose the curve is generated by starting with an equilateral triangle whose sides have length 1.

- Find the length L_n of the n th curve C_n and show that $\lim_{n \rightarrow \infty} L_n = \infty$.
- Find the area A_n of the region enclosed by C_n and calculate $\lim_{n \rightarrow \infty} A_n$.



78. The accompanying figure provides an informal proof that $\sum_{n=1}^{\infty} (1/n^2)$ is less than 2. Explain what is going on. (Source: "Convergence with Pictures" by P. J. Rippon, *American Mathematical Monthly*, Vol. 93, No. 6, 1986, pp. 476–478.)



11.3

The Integral Test

Given a series $\sum a_n$, we have two questions:

1. Does the series converge?
2. If it converges, what is its sum?

Much of the rest of this chapter is devoted to the first question, and in this section we answer that question by making a connection to the convergence of the improper integral $\int_1^\infty f(x) dx$. However, as a practical matter the second question is also important, and we will return to it later.

In this section and the next two, we study series that do not have negative terms. The reason for this restriction is that the partial sums of these series form nondecreasing sequences, and nondecreasing sequences that are bounded from above always converge (Theorem 6, Section 11.1). To show that a series of nonnegative terms converges, we need only show that its partial sums are bounded from above.

It may at first seem to be a drawback that this approach establishes the fact of convergence without producing the sum of the series in question. Surely it would be better to compute sums of series directly from formulas for their partial sums. But in most cases such formulas are not available, and in their absence we have to turn instead to the two-step procedure of first establishing convergence and then approximating the sum.

Nondecreasing Partial Sums

Suppose that $\sum_{n=1}^\infty a_n$ is an infinite series with $a_n \geq 0$ for all n . Then each partial sum is greater than or equal to its predecessor because $s_{n+1} = s_n + a_n$:

$$s_1 \leq s_2 \leq s_3 \leq \cdots \leq s_n \leq s_{n+1} \leq \cdots$$

Since the partial sums form a nondecreasing sequence, the Nondecreasing Sequence Theorem (Theorem 6, Section 11.1) tells us that the series will converge if and only if the partial sums are bounded from above.

Corollary of Theorem 6

A series $\sum_{n=1}^\infty a_n$ of nonnegative terms converges if and only if its partial sums are bounded from above.

EXAMPLE 1 The Harmonic Series

The series

$$\sum_{n=1}^\infty \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \cdots$$

is called the **harmonic series**. The harmonic series is divergent, but this doesn't follow from the n th-Term Test. The n th term $1/n$ does go to zero, but the series still diverges. The reason it diverges is because there is no upper bound for its partial sums. To see why, group the terms of the series in the following way:

$$1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{> \frac{2}{4} = \frac{1}{2}} + \underbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)}_{> \frac{4}{8} = \frac{1}{2}} + \underbrace{\left(\frac{1}{9} + \frac{1}{10} + \cdots + \frac{1}{16}\right)}_{> \frac{8}{16} = \frac{1}{2}} + \cdots$$

HISTORICAL BIOGRAPHY

Nicole Oresme
(1320–1382)

The sum of the first two terms is 1.5. The sum of the next two terms is $1/3 + 1/4$, which is greater than $1/4 + 1/4 = 1/2$. The sum of the next four terms is $1/5 + 1/6 + 1/7 + 1/8$, which is greater than $1/8 + 1/8 + 1/8 + 1/8 = 1/2$. The sum of the next eight terms is $1/9 + 1/10 + 1/11 + 1/12 + 1/13 + 1/14 + 1/15 + 1/16$, which is greater than $8/16 = 1/2$. The sum of the next 16 terms is greater than $16/32 = 1/2$, and so on. In general, the sum of 2^n terms ending with $1/2^{n+1}$ is greater than $2^n/2^{n+1} = 1/2$. The sequence of partial sums is not bounded from above: If $n = 2^k$, the partial sum s_n is greater than $k/2$. The harmonic series diverges. ■

The Integral Test

We introduce the Integral Test with a series that is related to the harmonic series, but whose n th term is $1/n^2$ instead of $1/n$.

EXAMPLE 2 Does the following series converge?

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots + \frac{1}{n^2} + \cdots$$

Solution We determine the convergence of $\sum_{n=1}^{\infty} (1/n^2)$ by comparing it with $\int_1^{\infty} (1/x^2) dx$. To carry out the comparison, we think of the terms of the series as values of the function $f(x) = 1/x^2$ and interpret these values as the areas of rectangles under the curve $y = 1/x^2$.

As Figure 11.7 shows,

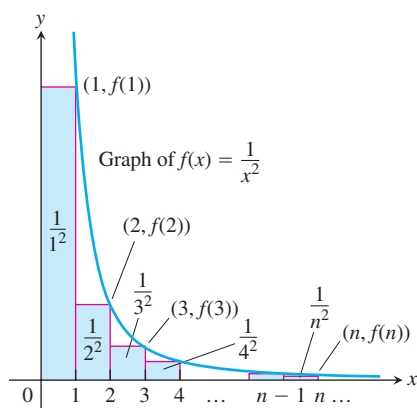


FIGURE 11.7 The sum of the areas of the rectangles under the graph of $f(x) = 1/x^2$ is less than the area under the graph (Example 2).

$$\begin{aligned} s_n &= \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} \\ &= f(1) + f(2) + f(3) + \cdots + f(n) \\ &< f(1) + \int_1^n \frac{1}{x^2} dx \\ &< 1 + \int_1^{\infty} \frac{1}{x^2} dx \\ &< 1 + 1 = 2. \end{aligned}$$

As in Section 8.8, Example 3,
 $\int_1^{\infty} (1/x^2) dx = 1$.

Thus the partial sums of $\sum_{n=1}^{\infty} 1/n^2$ are bounded from above (by 2) and the series converges. The sum of the series is known to be $\pi^2/6 \approx 1.64493$. (See Exercise 16 in Section 11.11.) ■

Caution

The series and integral need not have the same value in the convergent case. As we noted in Example 2, $\sum_{n=1}^{\infty} (1/n^2) = \pi^2/6$ while $\int_1^{\infty} (1/x^2) dx = 1$.

THEOREM 9 The Integral Test

Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a continuous, positive, decreasing function of x for all $x \geq N$ (N a positive integer). Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(x) dx$ both converge or both diverge.

Proof We establish the test for the case $N = 1$. The proof for general N is similar.

We start with the assumption that f is a decreasing function with $f(n) = a_n$ for every n . This leads us to observe that the rectangles in Figure 11.8a, which have areas

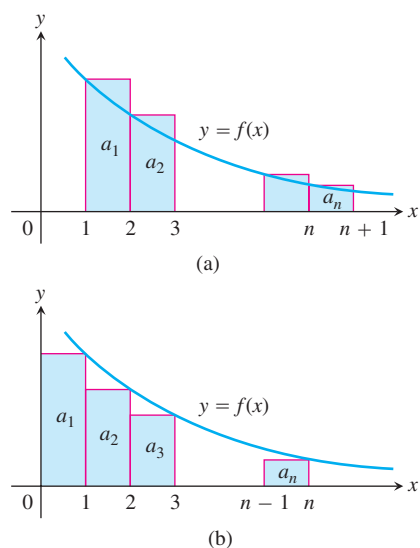


FIGURE 11.8 Subject to the conditions of the Integral Test, the series $\sum_{n=1}^{\infty} a_n$ and the integral $\int_1^{\infty} f(x) dx$ both converge or both diverge.

$a_1, a_2, \dots, a_n$, collectively enclose more area than that under the curve $y = f(x)$ from $x = 1$ to $x = n + 1$. That is,

$$\int_1^{n+1} f(x) dx \leq a_1 + a_2 + \cdots + a_n.$$

In Figure 11.8b the rectangles have been faced to the left instead of to the right. If we momentarily disregard the first rectangle, of area a_1 , we see that

$$a_2 + a_3 + \cdots + a_n \leq \int_1^n f(x) dx.$$

If we include a_1 , we have

$$a_1 + a_2 + \cdots + a_n \leq a_1 + \int_1^n f(x) dx.$$

Combining these results gives

$$\int_1^{n+1} f(x) dx \leq a_1 + a_2 + \cdots + a_n \leq a_1 + \int_1^n f(x) dx.$$

These inequalities hold for each n , and continue to hold as $n \rightarrow \infty$.

If $\int_1^{\infty} f(x) dx$ is finite, the right-hand inequality shows that $\sum a_n$ is finite. If $\int_1^{\infty} f(x) dx$ is infinite, the left-hand inequality shows that $\sum a_n$ is infinite. Hence the series and the integral are both finite or both infinite. ■

EXAMPLE 3 The p -Series

Show that the **p -series**

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} + \cdots$$

(p a real constant) converges if $p > 1$, and diverges if $p \leq 1$.

Solution If $p > 1$, then $f(x) = 1/x^p$ is a positive decreasing function of x . Since

$$\begin{aligned} \int_1^{\infty} \frac{1}{x^p} dx &= \int_1^{\infty} x^{-p} dx = \lim_{b \rightarrow \infty} \left[\frac{x^{-p+1}}{-p+1} \right]_1^b \\ &= \frac{1}{1-p} \lim_{b \rightarrow \infty} \left(\frac{1}{b^{p-1}} - 1 \right) \\ &= \frac{1}{1-p} (0 - 1) = \frac{1}{p-1}, \end{aligned}$$

$b^{p-1} \rightarrow \infty$ as $b \rightarrow \infty$
because $p - 1 > 0$.

the series converges by the Integral Test. We emphasize that the sum of the p -series is *not* $1/(p-1)$. The series converges, but we don't know the value it converges to.

If $p < 1$, then $1-p > 0$ and

$$\int_1^{\infty} \frac{1}{x^p} dx = \frac{1}{1-p} \lim_{b \rightarrow \infty} (b^{1-p} - 1) = \infty.$$

The series diverges by the Integral Test.

If $p = 1$, we have the (divergent) harmonic series

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \cdots.$$

We have convergence for $p > 1$ but divergence for every other value of p . ■

The p -series with $p = 1$ is the **harmonic series** (Example 1). The p -Series Test shows that the harmonic series is just *barely* divergent; if we increase p to 1.000000001, for instance, the series converges!

The slowness with which the partial sums of the harmonic series approaches infinity is impressive. For instance, it takes about 178,482,301 terms of the harmonic series to move the partial sums beyond 20. It would take your calculator several weeks to compute a sum with this many terms. (See also Exercise 33b.)

EXAMPLE 4 A Convergent Series

The series

$$\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$$

converges by the Integral Test. The function $f(x) = 1/(x^2 + 1)$ is positive, continuous, and decreasing for $x \geq 1$, and

$$\begin{aligned} \int_1^{\infty} \frac{1}{x^2 + 1} dx &= \lim_{b \rightarrow \infty} [\arctan x]_1^b \\ &= \lim_{b \rightarrow \infty} [\arctan b - \arctan 1] \\ &= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}. \end{aligned}$$

Again we emphasize that $\pi/4$ is *not* the sum of the series. The series converges, but we do not know the value of its sum. ■

Convergence of the series in Example 4 can also be verified by comparison with the series $\sum 1/n^2$. Comparison tests are studied in the next section.

EXERCISES 11.3

Determining Convergence or Divergence

Which of the series in Exercises 1–30 converge, and which diverge? Give reasons for your answers. (When you check an answer, remember that there may be more than one way to determine the series' convergence or divergence.)

1. $\sum_{n=1}^{\infty} \frac{1}{10^n}$

2. $\sum_{n=1}^{\infty} e^{-n}$

3. $\sum_{n=1}^{\infty} \frac{n}{n+1}$

4. $\sum_{n=1}^{\infty} \frac{5}{n+1}$

5. $\sum_{n=1}^{\infty} \frac{3}{\sqrt{n}}$

6. $\sum_{n=1}^{\infty} \frac{-2}{n\sqrt{n}}$

7. $\sum_{n=1}^{\infty} -\frac{1}{8^n}$

8. $\sum_{n=1}^{\infty} \frac{-8}{n}$

9. $\sum_{n=2}^{\infty} \frac{\ln n}{n}$

10. $\sum_{n=2}^{\infty} \frac{\ln n}{\sqrt{n}}$

11. $\sum_{n=1}^{\infty} \frac{2^n}{3^n}$

12. $\sum_{n=1}^{\infty} \frac{5^n}{4^n + 3}$

13. $\sum_{n=0}^{\infty} \frac{-2}{n+1}$

14. $\sum_{n=1}^{\infty} \frac{1}{2n-1}$

15. $\sum_{n=1}^{\infty} \frac{2^n}{n+1}$

16. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(\sqrt{n}+1)}$

17. $\sum_{n=2}^{\infty} \frac{\sqrt{n}}{\ln n}$

18. $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^n$

19. $\sum_{n=1}^{\infty} \frac{1}{(\ln 2)^n}$

20. $\sum_{n=1}^{\infty} \frac{1}{(\ln 3)^n}$

21. $\sum_{n=3}^{\infty} \frac{(1/n)}{(\ln n)\sqrt{\ln^2 n - 1}}$

22. $\sum_{n=1}^{\infty} \frac{1}{n(1 + \ln^2 n)}$

23. $\sum_{n=1}^{\infty} n \sin \frac{1}{n}$ 24. $\sum_{n=1}^{\infty} n \tan \frac{1}{n}$
25. $\sum_{n=1}^{\infty} \frac{e^n}{1 + e^{2n}}$ 26. $\sum_{n=1}^{\infty} \frac{2}{1 + e^n}$
27. $\sum_{n=1}^{\infty} \frac{8 \tan^{-1} n}{1 + n^2}$ 28. $\sum_{n=1}^{\infty} \frac{n}{n^2 + 1}$
29. $\sum_{n=1}^{\infty} \operatorname{sech} n$ 30. $\sum_{n=1}^{\infty} \operatorname{sech}^2 n$

Theory and Examples

For what values of a , if any, do the series in Exercises 31 and 32 converge?

31. $\sum_{n=1}^{\infty} \left(\frac{a}{n+2} - \frac{1}{n+4} \right)$ 32. $\sum_{n=3}^{\infty} \left(\frac{1}{n-1} - \frac{2a}{n+1} \right)$

33. a. Draw illustrations like those in Figures 11.7 and 11.8 to show that the partial sums of the harmonic series satisfy the inequalities

$$\begin{aligned} \ln(n+1) &= \int_1^{n+1} \frac{1}{x} dx \leq 1 + \frac{1}{2} + \cdots + \frac{1}{n} \\ &\leq 1 + \int_1^n \frac{1}{x} dx = 1 + \ln n. \end{aligned}$$

- T** b. There is absolutely no empirical evidence for the divergence of the harmonic series even though we know it diverges. The partial sums just grow too slowly. To see what we mean, suppose you had started with $s_1 = 1$ the day the universe was formed, 13 billion years ago, and added a new term every second. About how large would the partial sum s_n be today, assuming a 365-day year?

34. Are there any values of x for which $\sum_{n=1}^{\infty} (1/(nx))$ converges? Give reasons for your answer.
35. Is it true that if $\sum_{n=1}^{\infty} a_n$ is a divergent series of positive numbers then there is also a divergent series $\sum_{n=1}^{\infty} b_n$ of positive numbers with $b_n < a_n$ for every n ? Is there a “smallest” divergent series of positive numbers? Give reasons for your answers.
36. (Continuation of Exercise 35.) Is there a “largest” convergent series of positive numbers? Explain.
37. **The Cauchy condensation test** The Cauchy condensation test says: Let $\{a_n\}$ be a nonincreasing sequence ($a_n \geq a_{n+1}$ for all n) of positive terms that converges to 0. Then $\sum a_n$ converges if and only if $\sum 2^n a_{2^n}$ converges. For example, $\sum (1/n)$ diverges because $\sum 2^n \cdot (1/2^n) = \sum 1$ diverges. Show why the test works.
38. Use the Cauchy condensation test from Exercise 37 to show that

- a. $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ diverges;
- b. $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

39. Logarithmic p -series

- a. Show that

$$\int_2^{\infty} \frac{dx}{x(\ln x)^p} \quad (p \text{ a positive constant})$$

converges if and only if $p > 1$.

- b. What implications does the fact in part (a) have for the convergence of the series

$$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p}?$$

Give reasons for your answer.

40. (Continuation of Exercise 39.) Use the result in Exercise 39 to determine which of the following series converge and which diverge. Support your answer in each case.

a. $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)}$ b. $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^{1.01}}$

c. $\sum_{n=2}^{\infty} \frac{1}{n \ln(n^3)}$ d. $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^3}$

41. **Euler's constant** Graphs like those in Figure 11.8 suggest that as n increases there is little change in the difference between the sum

$$1 + \frac{1}{2} + \cdots + \frac{1}{n}$$

and the integral

$$\ln n = \int_1^n \frac{1}{x} dx.$$

To explore this idea, carry out the following steps.

- a. By taking $f(x) = 1/x$ in the proof of Theorem 9, show that

$$\ln(n+1) \leq 1 + \frac{1}{2} + \cdots + \frac{1}{n} \leq 1 + \ln n$$

or

$$0 < \ln(n+1) - \ln n \leq 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \ln n \leq 1.$$

Thus, the sequence

$$a_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \ln n$$

is bounded from below and from above.

- b. Show that

$$\frac{1}{n+1} < \int_n^{n+1} \frac{1}{x} dx = \ln(n+1) - \ln n,$$

and use this result to show that the sequence $\{a_n\}$ in part (a) is decreasing.

Since a decreasing sequence that is bounded from below converges (Exercise 107 in Section 11.1), the numbers a_n defined in part (a) converge:

$$1 + \frac{1}{2} + \cdots + \frac{1}{n} - \ln n \rightarrow \gamma.$$

The number γ , whose value is $0.5772\dots$, is called *Euler's constant*. In contrast to other special numbers like π and e , no other

expression with a simple law of formulation has ever been found for γ .

- 42.** Use the integral test to show that

$$\sum_{n=0}^{\infty} e^{-n^2}$$

converges.

11.4

Comparison Tests

We have seen how to determine the convergence of geometric series, p -series, and a few others. We can test the convergence of many more series by comparing their terms to those of a series whose convergence is known.

THEOREM 10 The Comparison Test

Let $\sum a_n$ be a series with no negative terms.

- (a) $\sum a_n$ converges if there is a convergent series $\sum c_n$ with $a_n \leq c_n$ for all $n > N$, for some integer N .
- (b) $\sum a_n$ diverges if there is a divergent series of nonnegative terms $\sum d_n$ with $a_n \geq d_n$ for all $n > N$, for some integer N .

Proof In Part (a), the partial sums of $\sum a_n$ are bounded above by

$$M = a_1 + a_2 + \cdots + a_N + \sum_{n=N+1}^{\infty} c_n.$$

They therefore form a nondecreasing sequence with a limit $L \leq M$.

In Part (b), the partial sums of $\sum a_n$ are not bounded from above. If they were, the partial sums for $\sum d_n$ would be bounded by

$$M^* = d_1 + d_2 + \cdots + d_N + \sum_{n=N+1}^{\infty} a_n$$

and $\sum d_n$ would have to converge instead of diverge. ■

EXAMPLE 1 Applying the Comparison Test

(a) The series

$$\sum_{n=1}^{\infty} \frac{5}{5n-1}$$

diverges because its n th term

$$\frac{5}{5n-1} = \frac{1}{n - \frac{1}{5}} > \frac{1}{n}$$

is greater than the n th term of the divergent harmonic series.

HISTORICAL BIOGRAPHY

Albert of Saxony
(ca. 1316–1390)

(b) The series

$$\sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots$$

converges because its terms are all positive and less than or equal to the corresponding terms of

$$1 + \sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots.$$

The geometric series on the left converges and we have

$$1 + \sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + \frac{1}{1 - (1/2)} = 3.$$

The fact that 3 is an upper bound for the partial sums of $\sum_{n=0}^{\infty} (1/n!)$ does not mean that the series converges to 3. As we will see in Section 11.9, the series converges to e .

(c) The series

$$5 + \frac{2}{3} + \frac{1}{7} + 1 + \frac{1}{2 + \sqrt{1}} + \frac{1}{4 + \sqrt{2}} + \frac{1}{8 + \sqrt{3}} + \cdots + \frac{1}{2^n + \sqrt{n}} + \cdots$$

converges. To see this, we ignore the first three terms and compare the remaining terms with those of the convergent geometric series $\sum_{n=0}^{\infty} (1/2^n)$. The term $1/(2^n + \sqrt{n})$ of the truncated sequence is less than the corresponding term $1/2^n$ of the geometric series. We see that term by term we have the comparison,

$$1 + \frac{1}{2 + \sqrt{1}} + \frac{1}{4 + \sqrt{2}} + \frac{1}{8 + \sqrt{3}} + \cdots \leq 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$$

So the truncated series and the original series converge by an application of the Comparison Test. ■

The Limit Comparison Test

We now introduce a comparison test that is particularly useful for series in which a_n is a rational function of n .

THEOREM 11 Limit Comparison Test

Suppose that $a_n > 0$ and $b_n > 0$ for all $n \geq N$ (N an integer).

1. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.
2. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.
3. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Proof We will prove Part 1. Parts 2 and 3 are left as Exercises 37(a) and (b).

Since $c/2 > 0$, there exists an integer N such that for all n

$$n > N \Rightarrow \left| \frac{a_n}{b_n} - c \right| < \frac{c}{2}. \quad \begin{array}{l} \text{Limit definition with} \\ \epsilon = c/2, L = c, \text{ and} \\ a_n \text{ replaced by } a_n/b_n \end{array}$$

Thus, for $n > N$,

$$-\frac{c}{2} < \frac{a_n}{b_n} - c < \frac{c}{2},$$

$$\frac{c}{2} < \frac{a_n}{b_n} < \frac{3c}{2},$$

$$\left(\frac{c}{2}\right)b_n < a_n < \left(\frac{3c}{2}\right)b_n.$$

If $\sum b_n$ converges, then $\sum (3c/2)b_n$ converges and $\sum a_n$ converges by the Direct Comparison Test. If $\sum b_n$ diverges, then $\sum (c/2)b_n$ diverges and $\sum a_n$ diverges by the Direct Comparison Test. ■

EXAMPLE 2 Using the Limit Comparison Test

Which of the following series converge, and which diverge?

$$(a) \quad \frac{3}{4} + \frac{5}{9} + \frac{7}{16} + \frac{9}{25} + \cdots = \sum_{n=1}^{\infty} \frac{2n+1}{(n+1)^2} = \sum_{n=1}^{\infty} \frac{2n+1}{n^2+2n+1}$$

$$(b) \quad \frac{1}{1} + \frac{1}{3} + \frac{1}{7} + \frac{1}{15} + \cdots = \sum_{n=1}^{\infty} \frac{1}{2^n - 1}$$

$$(c) \quad \frac{1+2\ln 2}{9} + \frac{1+3\ln 3}{14} + \frac{1+4\ln 4}{21} + \cdots = \sum_{n=2}^{\infty} \frac{1+n\ln n}{n^2+5}$$

Solution

(a) Let $a_n = (2n+1)/(n^2+2n+1)$. For large n , we expect a_n to behave like $2n/n^2 = 2/n$ since the leading terms dominate for large n , so we let $b_n = 1/n$. Since

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges}$$

and

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{2n^2 + n}{n^2 + 2n + 1} = 2,$$

$\sum a_n$ diverges by Part 1 of the Limit Comparison Test. We could just as well have taken $b_n = 2/n$, but $1/n$ is simpler.

- (b) Let $a_n = 1/(2^n - 1)$. For large n , we expect a_n to behave like $1/2^n$, so we let $b_n = 1/2^n$. Since

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{2^n} \text{ converges}$$

and

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \frac{2^n}{2^n - 1} \\ &= \lim_{n \rightarrow \infty} \frac{1}{1 - (1/2^n)} \\ &= 1, \end{aligned}$$

$\sum a_n$ converges by Part 1 of the Limit Comparison Test.

- (c) Let $a_n = (1 + n \ln n)/(n^2 + 5)$. For large n , we expect a_n to behave like $(n \ln n)/n^2 = (\ln n)/n$, which is greater than $1/n$ for $n \geq 3$, so we take $b_n = 1/n$. Since

$$\sum_{n=2}^{\infty} b_n = \sum_{n=2}^{\infty} \frac{1}{n} \text{ diverges}$$

and

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \frac{n + n^2 \ln n}{n^2 + 5} \\ &= \infty, \end{aligned}$$

$\sum a_n$ diverges by Part 3 of the Limit Comparison Test. ■

EXAMPLE 3 Does $\sum_{n=1}^{\infty} \frac{\ln n}{n^{3/2}}$ converge?

Solution Because $\ln n$ grows more slowly than n^c for any positive constant c (Section 11.1, Exercise 91), we would expect to have

$$\frac{\ln n}{n^{3/2}} < \frac{n^{1/4}}{n^{3/2}} = \frac{1}{n^{5/4}}$$

for n sufficiently large. Indeed, taking $a_n = (\ln n)/n^{3/2}$ and $b_n = 1/n^{5/4}$, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \frac{\ln n}{n^{1/4}} \\ &= \lim_{n \rightarrow \infty} \frac{1/n}{(1/4)n^{-3/4}} && \text{L'Hôpital's Rule} \\ &= \lim_{n \rightarrow \infty} \frac{4}{n^{1/4}} = 0. \end{aligned}$$

Since $\sum b_n = \sum (1/n^{5/4})$ (a p -series with $p > 1$) converges, $\sum a_n$ converges by Part 2 of the Limit Comparison Test. ■

EXERCISES 11.4

Determining Convergence or Divergence

Which of the series in Exercises 1–36 converge, and which diverge?

Give reasons for your answers.

1. $\sum_{n=1}^{\infty} \frac{1}{2\sqrt{n} + \sqrt[3]{n}}$
2. $\sum_{n=1}^{\infty} \frac{3}{n + \sqrt{n}}$
3. $\sum_{n=1}^{\infty} \frac{\sin^2 n}{2^n}$
4. $\sum_{n=1}^{\infty} \frac{1 + \cos n}{n^2}$
5. $\sum_{n=1}^{\infty} \frac{2n}{3n - 1}$
6. $\sum_{n=1}^{\infty} \frac{n + 1}{n^2 \sqrt{n}}$
7. $\sum_{n=1}^{\infty} \left(\frac{n}{3n + 1} \right)^n$
8. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3 + 2}}$
9. $\sum_{n=3}^{\infty} \frac{1}{\ln(\ln n)}$
10. $\sum_{n=2}^{\infty} \frac{1}{(\ln n)^2}$
11. $\sum_{n=1}^{\infty} \frac{(\ln n)^2}{n^3}$
12. $\sum_{n=1}^{\infty} \frac{(\ln n)^3}{n^3}$
13. $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n} \ln n}$
14. $\sum_{n=1}^{\infty} \frac{(\ln n)^2}{n^{3/2}}$
15. $\sum_{n=1}^{\infty} \frac{1}{1 + \ln n}$
16. $\sum_{n=1}^{\infty} \frac{1}{(1 + \ln n)^2}$
17. $\sum_{n=2}^{\infty} \frac{\ln(n + 1)}{n + 1}$
18. $\sum_{n=1}^{\infty} \frac{1}{(1 + \ln^2 n)}$
19. $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{n^2 - 1}}$
20. $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2 + 1}$
21. $\sum_{n=1}^{\infty} \frac{1 - n}{n2^n}$
22. $\sum_{n=1}^{\infty} \frac{n + 2^n}{n^2 2^n}$
23. $\sum_{n=1}^{\infty} \frac{1}{3^{n-1} + 1}$
24. $\sum_{n=1}^{\infty} \frac{3^{n-1} + 1}{3^n}$
25. $\sum_{n=1}^{\infty} \sin \frac{1}{n}$
26. $\sum_{n=1}^{\infty} \tan \frac{1}{n}$
27. $\sum_{n=1}^{\infty} \frac{10n + 1}{n(n + 1)(n + 2)}$
28. $\sum_{n=3}^{\infty} \frac{5n^3 - 3n}{n^2(n - 2)(n^2 + 5)}$
29. $\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n^{1.1}}$
30. $\sum_{n=1}^{\infty} \frac{\sec^{-1} n}{n^{1.3}}$
31. $\sum_{n=1}^{\infty} \frac{\coth n}{n^2}$
32. $\sum_{n=1}^{\infty} \frac{\tanh n}{n^2}$
33. $\sum_{n=1}^{\infty} \frac{1}{n\sqrt[n]{n}}$
34. $\sum_{n=1}^{\infty} \frac{\sqrt[n]{n}}{n^2}$
35. $\sum_{n=1}^{\infty} \frac{1}{1 + 2 + 3 + \cdots + n}$
36. $\sum_{n=1}^{\infty} \frac{1}{1 + 2^2 + 3^2 + \cdots + n^2}$

Theory and Examples

37. Prove (a) Part 2 and (b) Part 3 of the Limit Comparison Test.

38. If $\sum_{n=1}^{\infty} a_n$ is a convergent series of nonnegative numbers, can anything be said about $\sum_{n=1}^{\infty} (a_n/n)$? Explain.

39. Suppose that $a_n > 0$ and $b_n > 0$ for $n \geq N$ (N an integer). If $\lim_{n \rightarrow \infty} (a_n/b_n) = \infty$ and $\sum a_n$ converges, can anything be said about $\sum b_n$? Give reasons for your answer.

40. Prove that if $\sum a_n$ is a convergent series of nonnegative terms, then $\sum a_n^2$ converges.

COMPUTER EXPLORATION

41. It is not yet known whether the series

$$\sum_{n=1}^{\infty} \frac{1}{n^3 \sin^2 n}$$

converges or diverges. Use a CAS to explore the behavior of the series by performing the following steps.

a. Define the sequence of partial sums

$$s_k = \sum_{n=1}^k \frac{1}{n^3 \sin^2 n}.$$

What happens when you try to find the limit of s_k as $k \rightarrow \infty$? Does your CAS find a closed form answer for this limit?

b. Plot the first 100 points (k, s_k) for the sequence of partial sums. Do they appear to converge? What would you estimate the limit to be?

c. Next plot the first 200 points (k, s_k) . Discuss the behavior in your own words.

d. Plot the first 400 points (k, s_k) . What happens when $k = 355$? Calculate the number $355/113$. Explain from your calculation what happened at $k = 355$. For what values of k would you guess this behavior might occur again?

You will find an interesting discussion of this series in Chapter 72 of *Mazes for the Mind* by Clifford A. Pickover, St. Martin's Press, Inc., New York, 1992.

11.5

The Ratio and Root Tests

The Ratio Test measures the rate of growth (or decline) of a series by examining the ratio a_{n+1}/a_n . For a geometric series $\sum ar^n$, this rate is a constant ($(ar^{n+1})/(ar^n) = r$), and the series converges if and only if its ratio is less than 1 in absolute value. The Ratio Test is a powerful rule extending that result. We prove it on the next page using the Comparison Test.

THEOREM 12 The Ratio Test

Let $\sum a_n$ be a series with positive terms and suppose that

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \rho.$$

Then

- (a) the series *converges* if $\rho < 1$,
- (b) the series *diverges* if $\rho > 1$ or ρ is infinite,
- (c) the test is *inconclusive* if $\rho = 1$.

Proof

- (a) $\rho < 1$. Let r be a number between ρ and 1. Then the number $\epsilon = r - \rho$ is positive. Since

$$\frac{a_{n+1}}{a_n} \rightarrow \rho,$$

a_{n+1}/a_n must lie within ϵ of ρ when n is large enough, say for all $n \geq N$. In particular

$$\frac{a_{n+1}}{a_n} < \rho + \epsilon = r, \quad \text{when } n \geq N.$$

That is,

$$\begin{aligned} a_{N+1} &< ra_N, \\ a_{N+2} &< ra_{N+1} < r^2a_N, \\ a_{N+3} &< ra_{N+2} < r^3a_N, \\ &\vdots \\ a_{N+m} &< ra_{N+m-1} < r^ma_N. \end{aligned}$$

These inequalities show that the terms of our series, after the N th term, approach zero more rapidly than the terms in a geometric series with ratio $r < 1$. More precisely, consider the series $\sum c_n$, where $c_n = a_n$ for $n = 1, 2, \dots, N$ and $c_{N+1} = ra_N$, $c_{N+2} = r^2a_N$, $\dots$, $c_{N+m} = r^ma_N$, $\dots$. Now $a_n \leq c_n$ for all n , and

$$\begin{aligned} \sum_{n=1}^{\infty} c_n &= a_1 + a_2 + \dots + a_{N-1} + a_N + ra_N + r^2a_N + \dots \\ &= a_1 + a_2 + \dots + a_{N-1} + a_N(1 + r + r^2 + \dots). \end{aligned}$$

The geometric series $1 + r + r^2 + \dots$ converges because $|r| < 1$, so $\sum c_n$ converges. Since $a_n \leq c_n$, $\sum a_n$ also converges.

- (b) $1 < \rho \leq \infty$. From some index M on,

$$\frac{a_{n+1}}{a_n} > 1 \quad \text{and} \quad a_M < a_{M+1} < a_{M+2} < \dots.$$

The terms of the series do not approach zero as n becomes infinite, and the series diverges by the n th-Term Test.

(c) $\rho = 1$. The two series

$$\sum_{n=1}^{\infty} \frac{1}{n} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^2}$$

show that some other test for convergence must be used when $\rho = 1$.

$$\text{For } \sum_{n=1}^{\infty} \frac{1}{n}: \quad \frac{a_{n+1}}{a_n} = \frac{1/(n+1)}{1/n} = \frac{n}{n+1} \rightarrow 1.$$

$$\text{For } \sum_{n=1}^{\infty} \frac{1}{n^2}: \quad \frac{a_{n+1}}{a_n} = \frac{1/(n+1)^2}{1/n^2} = \left(\frac{n}{n+1}\right)^2 \rightarrow 1^2 = 1.$$

In both cases, $\rho = 1$, yet the first series diverges, whereas the second converges. ■

The Ratio Test is often effective when the terms of a series contain factorials of expressions involving n or expressions raised to a power involving n .

EXAMPLE 1 Applying the Ratio Test

Investigate the convergence of the following series.

$$(a) \sum_{n=0}^{\infty} \frac{2^n + 5}{3^n} \quad (b) \sum_{n=1}^{\infty} \frac{(2n)!}{n!n!} \quad (c) \sum_{n=1}^{\infty} \frac{4^n n!n!}{(2n)!}$$

Solution

(a) For the series $\sum_{n=0}^{\infty} (2^n + 5)/3^n$,

$$\frac{a_{n+1}}{a_n} = \frac{(2^{n+1} + 5)/3^{n+1}}{(2^n + 5)/3^n} = \frac{1}{3} \cdot \frac{2^{n+1} + 5}{2^n + 5} = \frac{1}{3} \cdot \left(\frac{2 + 5 \cdot 2^{-n}}{1 + 5 \cdot 2^{-n}}\right) \rightarrow \frac{1}{3} \cdot \frac{2}{1} = \frac{2}{3}.$$

The series converges because $\rho = 2/3$ is less than 1. This does *not* mean that $2/3$ is the sum of the series. In fact,

$$\sum_{n=0}^{\infty} \frac{2^n + 5}{3^n} = \sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n + \sum_{n=0}^{\infty} \frac{5}{3^n} = \frac{1}{1 - (2/3)} + \frac{5}{1 - (1/3)} = \frac{21}{2}.$$

(b) If $a_n = \frac{(2n)!}{n!n!}$, then $a_{n+1} = \frac{(2n+2)!}{(n+1)!(n+1)!}$ and

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{n!n!(2n+2)(2n+1)(2n)!}{(n+1)!(n+1)!(2n)!} \\ &= \frac{(2n+2)(2n+1)}{(n+1)(n+1)} = \frac{4n+2}{n+1} \rightarrow 4. \end{aligned}$$

The series diverges because $\rho = 4$ is greater than 1.

(c) If $a_n = 4^n n!n!/(2n)!$, then

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{4^{n+1}(n+1)!(n+1)!}{(2n+2)(2n+1)(2n)!} \cdot \frac{(2n)!}{4^n n!n!} \\ &= \frac{4(n+1)(n+1)}{(2n+2)(2n+1)} = \frac{2(n+1)}{2n+1} \rightarrow 1. \end{aligned}$$

Because the limit is $\rho = 1$, we cannot decide from the Ratio Test whether the series converges. When we notice that $a_{n+1}/a_n = (2n+2)/(2n+1)$, we conclude that a_{n+1} is always greater than a_n because $(2n+2)/(2n+1)$ is always greater than 1. Therefore, all terms are greater than or equal to $a_1 = 2$, and the n th term does not approach zero as $n \rightarrow \infty$. The series diverges. ■

The Root Test

The convergence tests we have so far for $\sum a_n$ work best when the formula for a_n is relatively simple. But consider the following.

EXAMPLE 2 Let $a_n = \begin{cases} n/2^n, & n \text{ odd} \\ 1/2^n, & n \text{ even.} \end{cases}$ Does $\sum a_n$ converge?

Solution We write out several terms of the series:

$$\begin{aligned} \sum_{n=1}^{\infty} a_n &= \frac{1}{2^1} + \frac{1}{2^2} + \frac{3}{2^3} + \frac{1}{2^4} + \frac{5}{2^5} + \frac{1}{2^6} + \frac{7}{2^7} + \cdots \\ &= \frac{1}{2} + \frac{1}{4} + \frac{3}{8} + \frac{1}{16} + \frac{5}{32} + \frac{1}{64} + \frac{7}{128} + \cdots \end{aligned}$$

Clearly, this is not a geometric series. The n th term approaches zero as $n \rightarrow \infty$, so we do not know if the series diverges. The Integral Test does not look promising. The Ratio Test produces

$$\frac{a_{n+1}}{a_n} = \begin{cases} \frac{1}{2n}, & n \text{ odd} \\ \frac{n+1}{2}, & n \text{ even.} \end{cases}$$

As $n \rightarrow \infty$, the ratio is alternately small and large and has no limit.

A test that will answer the question (the series converges) is the Root Test. ■

THEOREM 13 The Root Test

Let $\sum a_n$ be a series with $a_n \geq 0$ for $n \geq N$, and suppose that

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \rho.$$

Then

- (a) the series *converges* if $\rho < 1$,
- (b) the series *diverges* if $\rho > 1$ or ρ is infinite,
- (c) the test is *inconclusive* if $\rho = 1$.

Proof

- (a) $\rho < 1$. Choose an $\epsilon > 0$ so small that $\rho + \epsilon < 1$. Since $\sqrt[n]{a_n} \rightarrow \rho$, the terms $\sqrt[n]{a_n}$ eventually get closer than ϵ to ρ . In other words, there exists an index $M \geq N$ such that

$$\sqrt[n]{a_n} < \rho + \epsilon \quad \text{when } n \geq M.$$

Then it is also true that

$$a_n < (\rho + \epsilon)^n \quad \text{for } n \geq M.$$

Now, $\sum_{n=M}^{\infty} (\rho + \epsilon)^n$, a geometric series with ratio $(\rho + \epsilon) < 1$, converges. By comparison, $\sum_{n=M}^{\infty} a_n$ converges, from which it follows that

$$\sum_{n=1}^{\infty} a_n = a_1 + \cdots + a_{M-1} + \sum_{n=M}^{\infty} a_n$$

converges.

- (b) $1 < \rho \leq \infty$. For all indices beyond some integer M , we have $\sqrt[n]{a_n} > 1$, so that $a_n > 1$ for $n > M$. The terms of the series do not converge to zero. The series diverges by the n th-Term Test.
- (c) $\rho = 1$. The series $\sum_{n=1}^{\infty} (1/n)$ and $\sum_{n=1}^{\infty} (1/n^2)$ show that the test is not conclusive when $\rho = 1$. The first series diverges and the second converges, but in both cases $\sqrt[n]{a_n} \rightarrow 1$. ■

EXAMPLE 3 Applying the Root Test

Which of the following series converges, and which diverges?

(a) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ (b) $\sum_{n=1}^{\infty} \frac{2^n}{n^2}$ (c) $\sum_{n=1}^{\infty} \left(\frac{1}{1+n} \right)^n$

Solution

(a) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ converges because $\sqrt[n]{\frac{n^2}{2^n}} = \frac{\sqrt[n]{n^2}}{\sqrt[n]{2^n}} = \frac{(\sqrt[n]{n})^2}{2} \rightarrow \frac{1}{2} < 1$.

(b) $\sum_{n=1}^{\infty} \frac{2^n}{n^2}$ diverges because $\sqrt[n]{\frac{2^n}{n^2}} = \frac{2}{(\sqrt[n]{n})^2} \rightarrow \frac{2}{1} > 1$.

(c) $\sum_{n=1}^{\infty} \left(\frac{1}{1+n} \right)^n$ converges because $\sqrt[n]{\left(\frac{1}{1+n} \right)^n} = \frac{1}{1+n} \rightarrow 0 < 1$. ■

EXAMPLE 2 Revisited

Let $a_n = \begin{cases} n/2^n, & n \text{ odd} \\ 1/2^n, & n \text{ even.} \end{cases}$ Does $\sum a_n$ converge?

Solution We apply the Root Test, finding that

$$\sqrt[n]{a_n} = \begin{cases} \sqrt[n]{n/2}, & n \text{ odd} \\ 1/2, & n \text{ even.} \end{cases}$$

Therefore,

$$\frac{1}{2} \leq \sqrt[n]{a_n} \leq \frac{\sqrt[n]{n}}{2}.$$

Since $\sqrt[n]{n} \rightarrow 1$ (Section 11.1, Theorem 5), we have $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = 1/2$ by the Sandwich Theorem. The limit is less than 1, so the series converges by the Root Test. ■

EXERCISES 11.5

Determining Convergence or Divergence

Which of the series in Exercises 1–26 converge, and which diverge? Give reasons for your answers. (When checking your answers, remember there may be more than one way to determine a series' convergence or divergence.)

1. $\sum_{n=1}^{\infty} \frac{n^{\sqrt{2}}}{2^n}$
2. $\sum_{n=1}^{\infty} n^2 e^{-n}$
3. $\sum_{n=1}^{\infty} n! e^{-n}$
4. $\sum_{n=1}^{\infty} \frac{n!}{10^n}$
5. $\sum_{n=1}^{\infty} \frac{n^{10}}{10^n}$
6. $\sum_{n=1}^{\infty} \left(\frac{n-2}{n} \right)^n$
7. $\sum_{n=1}^{\infty} \frac{2 + (-1)^n}{1.25^n}$
8. $\sum_{n=1}^{\infty} \frac{(-2)^n}{3^n}$
9. $\sum_{n=1}^{\infty} \left(1 - \frac{3}{n} \right)^n$
10. $\sum_{n=1}^{\infty} \left(1 - \frac{1}{3n} \right)^n$
11. $\sum_{n=1}^{\infty} \frac{\ln n}{n^3}$
12. $\sum_{n=1}^{\infty} \frac{(\ln n)^n}{n^n}$
13. $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n^2} \right)$
14. $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n^2} \right)^n$
15. $\sum_{n=1}^{\infty} \frac{\ln n}{n}$
16. $\sum_{n=1}^{\infty} \frac{n \ln n}{2^n}$
17. $\sum_{n=1}^{\infty} \frac{(n+1)(n+2)}{n!}$
18. $\sum_{n=1}^{\infty} e^{-n}(n^3)$
19. $\sum_{n=1}^{\infty} \frac{(n+3)!}{3!n!3^n}$
20. $\sum_{n=1}^{\infty} \frac{n2^n(n+1)!}{3^n n!}$
21. $\sum_{n=1}^{\infty} \frac{n!}{(2n+1)!}$
22. $\sum_{n=1}^{\infty} \frac{n!}{n^n}$
23. $\sum_{n=2}^{\infty} \frac{n}{(\ln n)^n}$
24. $\sum_{n=2}^{\infty} \frac{n}{(\ln n)^{(n/2)}}$
25. $\sum_{n=1}^{\infty} \frac{n! \ln n}{n(n+2)!}$
26. $\sum_{n=1}^{\infty} \frac{3^n}{n^3 2^n}$

Which of the series $\sum_{n=1}^{\infty} a_n$ defined by the formulas in Exercises 27–38 converge, and which diverge? Give reasons for your answers.

27. $a_1 = 2, \quad a_{n+1} = \frac{1 + \sin n}{n} a_n$
28. $a_1 = 1, \quad a_{n+1} = \frac{1 + \tan^{-1} n}{n} a_n$
29. $a_1 = \frac{1}{3}, \quad a_{n+1} = \frac{3n-1}{2n+5} a_n$
30. $a_1 = 3, \quad a_{n+1} = \frac{n}{n+1} a_n$
31. $a_1 = 2, \quad a_{n+1} = \frac{2}{n} a_n$

32. $a_1 = 5, \quad a_{n+1} = \frac{\sqrt[n]{n}}{2} a_n$
33. $a_1 = 1, \quad a_{n+1} = \frac{1 + \ln n}{n} a_n$
34. $a_1 = \frac{1}{2}, \quad a_{n+1} = \frac{n + \ln n}{n + 10} a_n$
35. $a_1 = \frac{1}{3}, \quad a_{n+1} = \sqrt[n]{a_n}$
36. $a_1 = \frac{1}{2}, \quad a_{n+1} = (a_n)^{n+1}$
37. $a_n = \frac{2^n n! n!}{(2n)!}$
38. $a_n = \frac{(3n)!}{n!(n+1)!(n+2)!}$

Which of the series in Exercises 39–44 converge, and which diverge? Give reasons for your answers.

39. $\sum_{n=1}^{\infty} \frac{(n!)^n}{(n^n)^2}$
40. $\sum_{n=1}^{\infty} \frac{(n!)^n}{n^{(n^2)}}$
41. $\sum_{n=1}^{\infty} \frac{n^n}{2^{(n^2)}}$
42. $\sum_{n=1}^{\infty} \frac{n^n}{(2^n)^2}$
43. $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot \cdots \cdot (2n-1)}{4^n 2^n n!}$
44. $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot \cdots \cdot (2n-1)}{[2 \cdot 4 \cdot \cdots \cdot (2n)](3^n + 1)}$

Theory and Examples

45. Neither the Ratio nor the Root Test helps with p -series. Try them on

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

and show that both tests fail to provide information about convergence.

46. Show that neither the Ratio Test nor the Root Test provides information about the convergence of

$$\sum_{n=2}^{\infty} \frac{1}{(\ln n)^p} \quad (p \text{ constant}).$$

47. Let $a_n = \begin{cases} n/2^n, & \text{if } n \text{ is a prime number} \\ 1/2^n, & \text{otherwise.} \end{cases}$

Does $\sum a_n$ converge? Give reasons for your answer.

11.6

Alternating Series, Absolute and Conditional Convergence

A series in which the terms are alternately positive and negative is an **alternating series**.

Here are three examples:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots + \frac{(-1)^{n+1}}{n} + \cdots \quad (1)$$

$$-2 + 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots + \frac{(-1)^{n+1}4}{2^n} + \cdots \quad (2)$$

$$1 - 2 + 3 - 4 + 5 - 6 + \cdots + (-1)^{n+1}n + \cdots \quad (3)$$

Series (1), called the **alternating harmonic series**, converges, as we will see in a moment. Series (2) a geometric series with ratio $r = -1/2$, converges to $-2/[1 + (1/2)] = -4/3$. Series (3) diverges because the n th term does not approach zero.

We prove the convergence of the alternating harmonic series by applying the Alternating Series Test.

THEOREM 14 **The Alternating Series Test (Leibniz's Theorem)**

The series

$$\sum_{n=1}^{\infty} (-1)^{n+1} u_n = u_1 - u_2 + u_3 - u_4 + \cdots$$

converges if all three of the following conditions are satisfied:

1. The u_n 's are all positive.
2. $u_n \geq u_{n+1}$ for all $n \geq N$, for some integer N .
3. $u_n \rightarrow 0$.

Proof If n is an even integer, say $n = 2m$, then the sum of the first n terms is

$$\begin{aligned} s_{2m} &= (u_1 - u_2) + (u_3 - u_4) + \cdots + (u_{2m-1} - u_{2m}) \\ &= u_1 - (u_2 - u_3) - (u_4 - u_5) - \cdots - (u_{2m-2} - u_{2m-1}) - u_{2m}. \end{aligned}$$

The first equality shows that s_{2m} is the sum of m nonnegative terms, since each term in parentheses is positive or zero. Hence $s_{2m+2} \geq s_{2m}$, and the sequence $\{s_{2m}\}$ is nondecreasing. The second equality shows that $s_{2m} \leq u_1$. Since $\{s_{2m}\}$ is nondecreasing and bounded from above, it has a limit, say

$$\lim_{m \rightarrow \infty} s_{2m} = L. \quad (4)$$

If n is an odd integer, say $n = 2m + 1$, then the sum of the first n terms is $s_{2m+1} = s_{2m} + u_{2m+1}$. Since $u_n \rightarrow 0$,

$$\lim_{m \rightarrow \infty} u_{2m+1} = 0$$

and, as $m \rightarrow \infty$,

$$s_{2m+1} = s_{2m} + u_{2m+1} \rightarrow L + 0 = L. \quad (5)$$

Combining the results of Equations (4) and (5) gives $\lim_{n \rightarrow \infty} s_n = L$ (Section 11.1, Exercise 119). ■

EXAMPLE 1 The alternating harmonic series

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

satisfies the three requirements of Theorem 14 with $N = 1$; it therefore converges. ■

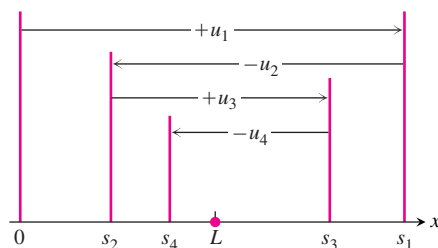


FIGURE 11.9 The partial sums of an alternating series that satisfies the hypotheses of Theorem 14 for $N = 1$ straddle the limit from the beginning.

A graphical interpretation of the partial sums (Figure 11.9) shows how an alternating series converges to its limit L when the three conditions of Theorem 14 are satisfied with $N = 1$. (Exercise 63 asks you to picture the case $N > 1$.) Starting from the origin of the x -axis, we lay off the positive distance $s_1 = u_1$. To find the point corresponding to $s_2 = u_1 - u_2$, we back up a distance equal to u_2 . Since $u_2 \leq u_1$, we do not back up any farther than the origin. We continue in this seesaw fashion, backing up or going forward as the signs in the series demand. But for $n \geq N$, each forward or backward step is shorter than (or at most the same size as) the preceding step, because $u_{n+1} \leq u_n$. And since the n th term approaches zero as n increases, the size of step we take forward or backward gets smaller and smaller. We oscillate across the limit L , and the amplitude of oscillation approaches zero. The limit L lies between any two successive sums s_n and s_{n+1} and hence differs from s_n by an amount less than u_{n+1} .

Because

$$|L - s_n| < u_{n+1} \quad \text{for } n \geq N,$$

we can make useful estimates of the sums of convergent alternating series.

THEOREM 15 The Alternating Series Estimation Theorem

If the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} u_n$ satisfies the three conditions of Theorem 14, then for $n \geq N$,

$$s_n = u_1 - u_2 + \cdots + (-1)^{n+1} u_n$$

approximates the sum L of the series with an error whose absolute value is less than u_{n+1} , the numerical value of the first unused term. Furthermore, the remainder, $L - s_n$, has the same sign as the first unused term.

We leave the verification of the sign of the remainder for Exercise 53.

EXAMPLE 2 We try Theorem 15 on a series whose sum we know:

$$\sum_{n=0}^{\infty} (-1)^n \frac{1}{2^n} = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \frac{1}{32} + \frac{1}{64} - \frac{1}{128} + \frac{1}{256} - \cdots$$

The theorem says that if we truncate the series after the eighth term, we throw away a total that is positive and less than $1/256$. The sum of the first eight terms is 0.6640625. The sum of the series is

$$\frac{1}{1 - (-1/2)} = \frac{1}{3/2} = \frac{2}{3}.$$

The difference, $(2/3) - 0.6640625 = 0.002604166\ldots$, is positive and less than $(1/256) = 0.00390625$. ■

Absolute and Conditional Convergence

DEFINITION Absolutely Convergent

A series $\sum a_n$ **converges absolutely** (is **absolutely convergent**) if the corresponding series of absolute values, $\sum |a_n|$, converges.

The geometric series

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots$$

converges absolutely because the corresponding series of absolute values

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$$

converges. The alternating harmonic series does not converge absolutely. The corresponding series of absolute values is the (divergent) harmonic series.

DEFINITION Conditionally Convergent

A series that converges but does not converge absolutely **converges conditionally**.

The alternating harmonic series converges conditionally.

Absolute convergence is important for two reasons. First, we have good tests for convergence of series of positive terms. Second, if a series converges absolutely, then it converges. That is the thrust of the next theorem.

THEOREM 16 The Absolute Convergence Test

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

Proof For each n ,

$$-|a_n| \leq a_n \leq |a_n|, \quad \text{so} \quad 0 \leq a_n + |a_n| \leq 2|a_n|.$$

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} 2|a_n|$ converges and, by the Direct Comparison Test, the nonnegative series $\sum_{n=1}^{\infty} (a_n + |a_n|)$ converges. The equality $a_n = (a_n + |a_n|) - |a_n|$ now lets us express $\sum_{n=1}^{\infty} a_n$ as the difference of two convergent series:

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (a_n + |a_n| - |a_n|) = \sum_{n=1}^{\infty} (a_n + |a_n|) - \sum_{n=1}^{\infty} |a_n|.$$

Therefore, $\sum_{n=1}^{\infty} a_n$ converges. ■

CAUTION We can rephrase Theorem 16 to say that every absolutely convergent series converges. However, the converse statement is false: Many convergent series do not converge absolutely (such as the alternating harmonic series in Example 1).

EXAMPLE 3 Applying the Absolute Convergence Test

- (a) For $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^2} = 1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \cdots$, the corresponding series of absolute values is the convergent series

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots.$$

The original series converges because it converges absolutely.

- (b) For $\sum_{n=1}^{\infty} \frac{\sin n}{n^2} = \frac{\sin 1}{1} + \frac{\sin 2}{4} + \frac{\sin 3}{9} + \cdots$, the corresponding series of absolute values is

$$\sum_{n=1}^{\infty} \left| \frac{\sin n}{n^2} \right| = \frac{|\sin 1|}{1} + \frac{|\sin 2|}{4} + \cdots,$$

which converges by comparison with $\sum_{n=1}^{\infty} (1/n^2)$ because $|\sin n| \leq 1$ for every n . The original series converges absolutely; therefore it converges. ■

EXAMPLE 4 Alternating p -Series

If p is a positive constant, the sequence $\{1/n^p\}$ is a decreasing sequence with limit zero. Therefore the alternating p -series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p} = 1 - \frac{1}{2^p} + \frac{1}{3^p} - \frac{1}{4^p} + \cdots, \quad p > 0$$

converges.

If $p > 1$, the series converges absolutely. If $0 < p \leq 1$, the series converges conditionally.

Conditional convergence: $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \cdots$

Absolute convergence: $1 - \frac{1}{2^{3/2}} + \frac{1}{3^{3/2}} - \frac{1}{4^{3/2}} + \cdots$ ■

Rearranging Series

THEOREM 17 The Rearrangement Theorem for Absolutely Convergent Series

If $\sum_{n=1}^{\infty} a_n$ converges absolutely, and $b_1, b_2, \dots, b_n, \dots$ is any arrangement of the sequence $\{a_n\}$, then $\sum b_n$ converges absolutely and

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n.$$

(For an outline of the proof, see Exercise 60.)

EXAMPLE 5 Applying the Rearrangement Theorem

As we saw in Example 3, the series

$$1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \cdots + (-1)^{n-1} \frac{1}{n^2} + \cdots$$

converges absolutely. A possible rearrangement of the terms of the series might start with a positive term, then two negative terms, then three positive terms, then four negative terms, and so on: After k terms of one sign, take $k + 1$ terms of the opposite sign. The first ten terms of such a series look like this:

$$1 - \frac{1}{4} - \frac{1}{16} + \frac{1}{9} + \frac{1}{25} + \frac{1}{49} - \frac{1}{36} - \frac{1}{64} - \frac{1}{100} - \frac{1}{144} + \cdots.$$

The Rearrangement Theorem says that both series converge to the same value. In this example, if we had the second series to begin with, we would probably be glad to exchange it for the first, if we knew that we could. We can do even better: The sum of either series is also equal to

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} - \sum_{n=1}^{\infty} \frac{1}{(2n)^2}.$$

(See Exercise 61.)

If we rearrange infinitely many terms of a conditionally convergent series, we can get results that are far different from the sum of the original series. Here is an example.

EXAMPLE 6 Rearranging the Alternating Harmonic Series

The alternating harmonic series

$$\frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \frac{1}{9} - \frac{1}{10} + \frac{1}{11} - \cdots$$

can be rearranged to diverge or to reach any preassigned sum.

- (a) *Rearranging $\sum_{n=1}^{\infty} (-1)^{n+1}/n$ to diverge.* The series of terms $\sum [1/(2n-1)]$ diverges to $+\infty$ and the series of terms $\sum (-1/2n)$ diverges to $-\infty$. No matter how far out in the sequence of odd-numbered terms we begin, we can always add enough positive terms to get an arbitrarily large sum. Similarly, with the negative terms, no matter how far out we start, we can add enough consecutive even-numbered terms to get a negative sum of arbitrarily large absolute value. If we wished to do so, we could start adding odd-numbered terms until we had a sum greater than $+3$, say, and then follow that with enough consecutive negative terms to make the new total less than -4 . We could then add enough positive terms to make the total greater than $+5$ and follow with consecutive unused negative terms to make a new total less than -6 , and so on. In this way, we could make the swings arbitrarily large in either direction.
- (b) *Rearranging $\sum_{n=1}^{\infty} (-1)^{n+1}/n$ to converge to 1.* Another possibility is to focus on a particular limit. Suppose we try to get sums that converge to 1. We start with the first term, $1/1$, and then subtract $1/2$. Next we add $1/3$ and $1/5$, which brings the total back to 1 or above. Then we add consecutive negative terms until the total is less than 1. We continue in this manner: When the sum is less than 1, add positive terms until the total is 1 or more; then subtract (add negative) terms until the total is again less than 1. This process can be continued indefinitely. Because both the odd-numbered

terms and the even-numbered terms of the original series approach zero as $n \rightarrow \infty$, the amount by which our partial sums exceed 1 or fall below it approaches zero. So the new series converges to 1. The rearranged series starts like this:

$$\begin{aligned} \frac{1}{1} - \frac{1}{2} + \frac{1}{3} + \frac{1}{5} - \frac{1}{4} + \frac{1}{7} + \frac{1}{9} - \frac{1}{6} + \frac{1}{11} + \frac{1}{13} - \frac{1}{8} + \frac{1}{15} + \frac{1}{17} - \frac{1}{10} \\ + \frac{1}{19} + \frac{1}{21} - \frac{1}{12} + \frac{1}{23} + \frac{1}{25} - \frac{1}{14} + \frac{1}{27} - \frac{1}{16} + \cdots \end{aligned}$$

The kind of behavior illustrated by the series in Example 6 is typical of what can happen with any conditionally convergent series. Therefore we must always add the terms of a conditionally convergent series in the order given.

We have now developed several tests for convergence and divergence of series. In summary:

1. **The n th-Term Test:** Unless $a_n \rightarrow 0$, the series diverges.
2. **Geometric series:** $\sum ar^n$ converges if $|r| < 1$; otherwise it diverges.
3. **p -series:** $\sum 1/n^p$ converges if $p > 1$; otherwise it diverges.
4. **Series with nonnegative terms:** Try the Integral Test, Ratio Test, or Root Test. Try comparing to a known series with the Comparison Test.
5. **Series with some negative terms:** Does $\sum |a_n|$ converge? If yes, so does $\sum a_n$, since absolute convergence implies convergence.
6. **Alternating series:** $\sum a_n$ converges if the series satisfies the conditions of the Alternating Series Test.

EXERCISES 11.6

Determining Convergence or Divergence

Which of the alternating series in Exercises 1–10 converge, and which diverge? Give reasons for your answers.

$$1. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^2}$$

$$2. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^{3/2}}$$

$$3. \sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{n}{10}\right)^n$$

$$4. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{10^n}{n^{10}}$$

$$5. \sum_{n=2}^{\infty} (-1)^{n+1} \frac{1}{\ln n}$$

$$6. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\ln n}{n}$$

$$7. \sum_{n=2}^{\infty} (-1)^{n+1} \frac{\ln n}{\ln n^2}$$

$$8. \sum_{n=1}^{\infty} (-1)^n \ln\left(1 + \frac{1}{n}\right)$$

$$9. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sqrt{n} + 1}{n + 1}$$

$$10. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{3\sqrt{n} + 1}{\sqrt{n} + 1}$$

Absolute Convergence

Which of the series in Exercises 11–44 converge absolutely, which converge, and which diverge? Give reasons for your answers.

$$11. \sum_{n=1}^{\infty} (-1)^{n+1} (0.1)^n$$

$$13. \sum_{n=1}^{\infty} (-1)^n \frac{1}{\sqrt{n}}$$

$$15. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{n^3 + 1}$$

$$17. \sum_{n=1}^{\infty} (-1)^n \frac{1}{n + 3}$$

$$19. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{3 + n}{5 + n}$$

$$21. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1 + n}{n^2}$$

$$23. \sum_{n=1}^{\infty} (-1)^n n^2 (2/3)^n$$

$$25. \sum_{n=1}^{\infty} (-1)^n \frac{\tan^{-1} n}{n^2 + 1}$$

$$12. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(0.1)^n}{n}$$

$$14. \sum_{n=1}^{\infty} \frac{(-1)^n}{1 + \sqrt{n}}$$

$$16. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n!}{2^n}$$

$$18. \sum_{n=1}^{\infty} (-1)^n \frac{\sin n}{n^2}$$

$$20. \sum_{n=2}^{\infty} (-1)^n \frac{1}{\ln(n^3)}$$

$$22. \sum_{n=1}^{\infty} \frac{(-2)^{n+1}}{n + 5^n}$$

$$24. \sum_{n=1}^{\infty} (-1)^{n+1} (\sqrt[n]{10})$$

$$26. \sum_{n=2}^{\infty} (-1)^{n+1} \frac{1}{n \ln n}$$

27. $\sum_{n=1}^{\infty} (-1)^n \frac{n}{n+1}$
28. $\sum_{n=1}^{\infty} (-1)^n \frac{\ln n}{n - \ln n}$
29. $\sum_{n=1}^{\infty} \frac{(-100)^n}{n!}$
30. $\sum_{n=1}^{\infty} (-5)^{-n}$
31. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2 + 2n + 1}$
32. $\sum_{n=2}^{\infty} (-1)^n \left(\frac{\ln n}{\ln n^2} \right)^n$
33. $\sum_{n=1}^{\infty} \frac{\cos n\pi}{n\sqrt{n}}$
34. $\sum_{n=1}^{\infty} \frac{\cos n\pi}{n}$
35. $\sum_{n=1}^{\infty} \frac{(-1)^n (n+1)^n}{(2n)^n}$
36. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} (n!)^2}{(2n)!}$
37. $\sum_{n=1}^{\infty} (-1)^n \frac{(2n)!}{2^n n! n}$
38. $\sum_{n=1}^{\infty} (-1)^n \frac{(n!)^2 3^n}{(2n+1)!}$
39. $\sum_{n=1}^{\infty} (-1)^n (\sqrt{n+1} - \sqrt{n})$
40. $\sum_{n=1}^{\infty} (-1)^n (\sqrt{n^2 + n} - n)$
41. $\sum_{n=1}^{\infty} (-1)^n (\sqrt{n+1} - \sqrt{n})$
42. $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n} + \sqrt{n+1}}$
43. $\sum_{n=1}^{\infty} (-1)^n \operatorname{sech} n$
44. $\sum_{n=1}^{\infty} (-1)^n \operatorname{csch} n$

Error Estimation

In Exercises 45–48, estimate the magnitude of the error involved in using the sum of the first four terms to approximate the sum of the entire series.

45. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$ It can be shown that the sum is $\ln 2$.
46. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{10^n}$
47. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{(0.01)^n}{n}$ As you will see in Section 11.7, the sum is $\ln(1.01)$.
48. $\frac{1}{1+t} = \sum_{n=0}^{\infty} (-1)^n t^n, \quad 0 < t < 1$

T Approximate the sums in Exercises 49 and 50 with an error of magnitude less than 5×10^{-6} .

49. $\sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n)!}$ As you will see in Section 11.9, the sum is $\cos 1$, the cosine of 1 radian.
50. $\sum_{n=0}^{\infty} (-1)^n \frac{1}{n!}$ As you will see in Section 11.9, the sum is e^{-1} .

Theory and Examples

51. a. The series

$$\frac{1}{3} - \frac{1}{2} + \frac{1}{9} - \frac{1}{4} + \frac{1}{27} - \frac{1}{8} + \cdots + \frac{1}{3^n} - \frac{1}{2^n} + \cdots$$

does not meet one of the conditions of Theorem 14. Which one?

b. Find the sum of the series in part (a).

T 52. The limit L of an alternating series that satisfies the conditions of Theorem 14 lies between the values of any two consecutive partial sums. This suggests using the average

$$\frac{s_n + s_{n+1}}{2} = s_n + \frac{1}{2} (-1)^{n+2} a_{n+1}$$

to estimate L . Compute

$$s_{20} + \frac{1}{2} \cdot \frac{1}{21}$$

as an approximation to the sum of the alternating harmonic series. The exact sum is $\ln 2 = 0.6931 \dots$

53. **The sign of the remainder of an alternating series that satisfies the conditions of Theorem 14** Prove the assertion in Theorem 15 that whenever an alternating series satisfying the conditions of Theorem 14 is approximated with one of its partial sums, then the remainder (sum of the unused terms) has the same sign as the first unused term. (*Hint:* Group the remainder's terms in consecutive pairs.)

54. Show that the sum of the first $2n$ terms of the series

$$1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \frac{1}{4} - \frac{1}{5} + \frac{1}{5} - \frac{1}{6} + \cdots$$

is the same as the sum of the first n terms of the series

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \frac{1}{5 \cdot 6} + \cdots$$

Do these series converge? What is the sum of the first $2n + 1$ terms of the first series? If the series converge, what is their sum?

55. Show that if $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} |a_n|$ diverges.

56. Show that if $\sum_{n=1}^{\infty} a_n$ converges absolutely, then

$$\left| \sum_{n=1}^{\infty} a_n \right| \leq \sum_{n=1}^{\infty} |a_n|.$$

57. Show that if $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both converge absolutely, then so does

- a. $\sum_{n=1}^{\infty} (a_n + b_n)$ b. $\sum_{n=1}^{\infty} (a_n - b_n)$
- c. $\sum_{n=1}^{\infty} k a_n$ (k any number)

58. Show by example that $\sum_{n=1}^{\infty} a_n b_n$ may diverge even if $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both converge.

T 59. In Example 6, suppose the goal is to arrange the terms to get a new series that converges to $-1/2$. Start the new arrangement with the first negative term, which is $-1/2$. Whenever you have a sum that is less than or equal to $-1/2$, start introducing positive terms, taken in order, until the new total is greater than $-1/2$. Then add negative terms until the total is less than or equal to $-1/2$ again. Continue this process until your partial sums have

been above the target at least three times and finish at or below it. If s_n is the sum of the first n terms of your new series, plot the points (n, s_n) to illustrate how the sums are behaving.

60. Outline of the proof of the Rearrangement Theorem (Theorem 17)

- a. Let ϵ be a positive real number, let $L = \sum_{n=1}^{\infty} a_n$, and let $s_k = \sum_{n=1}^k a_n$. Show that for some index N_1 and for some index $N_2 \geq N_1$,

$$\sum_{n=N_1}^{\infty} |a_n| < \frac{\epsilon}{2} \quad \text{and} \quad |s_{N_2} - L| < \frac{\epsilon}{2}.$$

Since all the terms $a_1, a_2, \dots, a_{N_2}$ appear somewhere in the sequence $\{b_n\}$, there is an index $N_3 \geq N_2$ such that if $n \geq N_3$, then $(\sum_{k=1}^n b_k) - s_{N_2}$ is at most a sum of terms a_m with $m \geq N_1$. Therefore, if $n \geq N_3$,

$$\begin{aligned} \left| \sum_{k=1}^n b_k - L \right| &\leq \left| \sum_{k=1}^n b_k - s_{N_2} \right| + |s_{N_2} - L| \\ &\leq \sum_{k=N_1}^{\infty} |a_k| + |s_{N_2} - L| < \epsilon. \end{aligned}$$

- b. The argument in part (a) shows that if $\sum_{n=1}^{\infty} a_n$ converges absolutely then $\sum_{n=1}^{\infty} b_n$ converges and $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n$. Now show that because $\sum_{n=1}^{\infty} |a_n|$ converges, $\sum_{n=1}^{\infty} |b_n|$ converges to $\sum_{n=1}^{\infty} |a_n|$.

61. Unzipping absolutely convergent series

- a. Show that if $\sum_{n=1}^{\infty} |a_n|$ converges and

$$b_n = \begin{cases} a_n, & \text{if } a_n \geq 0 \\ 0, & \text{if } a_n < 0, \end{cases}$$

then $\sum_{n=1}^{\infty} b_n$ converges.

- b. Use the results in part (a) to show likewise that if $\sum_{n=1}^{\infty} |a_n|$ converges and

$$c_n = \begin{cases} 0, & \text{if } a_n \geq 0 \\ a_n, & \text{if } a_n < 0, \end{cases}$$

then $\sum_{n=1}^{\infty} c_n$ converges.

In other words, if a series converges absolutely, its positive terms form a convergent series, and so do its negative terms. Furthermore,

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} b_n + \sum_{n=1}^{\infty} c_n$$

because $b_n = (a_n + |a_n|)/2$ and $c_n = (a_n - |a_n|)/2$.

- 62. What is wrong here?:**

Multiply both sides of the alternating harmonic series

$$S = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \frac{1}{9} - \frac{1}{10} + \frac{1}{11} - \frac{1}{12} + \dots$$

by 2 to get

$$2S = 2 - 1 + \frac{2}{3} - \frac{1}{2} + \frac{2}{5} - \frac{1}{3} + \frac{2}{7} - \frac{1}{4} + \frac{2}{9} - \frac{1}{5} + \frac{2}{11} - \frac{1}{6} + \dots$$

Collect terms with the same denominator, as the arrows indicate, to arrive at

$$2S = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$$

The series on the right-hand side of this equation is the series we started with. Therefore, $2S = S$, and dividing by S gives $2 = 1$. (Source: “Riemann’s Rearrangement Theorem” by Stewart Galanor, *Mathematics Teacher*, Vol. 80, No. 8, 1987, pp. 675–681.)

- 63. Draw a figure similar to Figure 11.9 to illustrate the convergence of the series in Theorem 14 when $N > 1$.**

11.7

Power Series

Now that we can test infinite series for convergence we can study the infinite polynomials mentioned at the beginning of this chapter. We call these polynomials power series because they are defined as infinite series of powers of some variable, in our case x . Like polynomials, power series can be added, subtracted, multiplied, differentiated, and integrated to give new power series.

Power Series and Convergence

We begin with the formal definition.

DEFINITIONS Power Series, Center, Coefficients

A **power series about $x = 0$** is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + \cdots \quad (1)$$

A **power series about $x = a$** is a series of the form

$$\sum_{n=0}^{\infty} c_n (x - a)^n = c_0 + c_1 (x - a) + c_2 (x - a)^2 + \cdots + c_n (x - a)^n + \cdots \quad (2)$$

in which the **center** a and the **coefficients** $c_0, c_1, c_2, \dots, c_n, \dots$ are constants.

Equation (1) is the special case obtained by taking $a = 0$ in Equation (2).

EXAMPLE 1 A Geometric Series

Taking all the coefficients to be 1 in Equation (1) gives the geometric power series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \cdots + x^n + \cdots$$

This is the geometric series with first term 1 and ratio x . It converges to $1/(1 - x)$ for $|x| < 1$. We express this fact by writing

$$\frac{1}{1 - x} = 1 + x + x^2 + \cdots + x^n + \cdots, \quad -1 < x < 1. \quad (3)$$

Up to now, we have used Equation (3) as a formula for the sum of the series on the right. We now change the focus: We think of the partial sums of the series on the right as polynomials $P_n(x)$ that approximate the function on the left. For values of x near zero, we need take only a few terms of the series to get a good approximation. As we move toward $x = 1$, or -1 , we must take more terms. Figure 11.10 shows the graphs of $f(x) = 1/(1 - x)$, and the approximating polynomials $y_n = P_n(x)$ for $n = 0, 1, 2$, and 8. The function $f(x) = 1/(1 - x)$ is not continuous on intervals containing $x = 1$, where it has a vertical asymptote. The approximations do not apply when $x \geq 1$.

EXAMPLE 2 A Geometric Series

The power series

$$1 - \frac{1}{2}(x - 2) + \frac{1}{4}(x - 2)^2 + \cdots + \left(-\frac{1}{2}\right)^n (x - 2)^n + \cdots \quad (4)$$

matches Equation (2) with $a = 2$, $c_0 = 1$, $c_1 = -1/2$, $c_2 = 1/4$, $\dots$, $c_n = (-1/2)^n$. This is a geometric series with first term 1 and ratio $r = -\frac{x - 2}{2}$. The series converges for

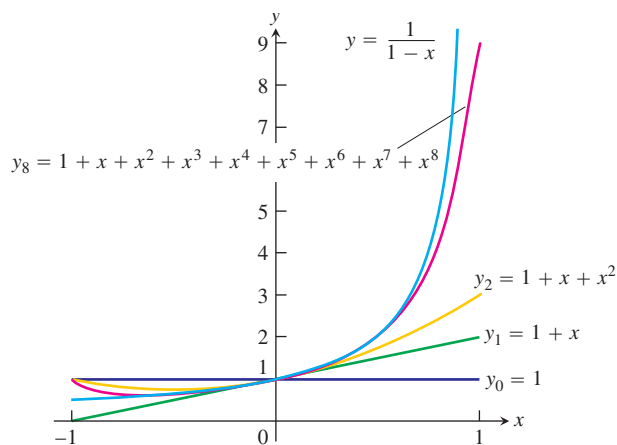


FIGURE 11.10 The graphs of $f(x) = 1/(1 - x)$ and four of its polynomial approximations (Example 1).

$\left| \frac{x-2}{2} \right| < 1$ or $0 < x < 4$. The sum is

$$\frac{1}{1-r} = \frac{1}{1 + \frac{x-2}{2}} = \frac{2}{x},$$

so

$$\frac{2}{x} = 1 - \frac{(x-2)}{2} + \frac{(x-2)^2}{4} - \cdots + \left(-\frac{1}{2}\right)^n (x-2)^n + \cdots, \quad 0 < x < 4.$$

Series (4) generates useful polynomial approximations of $f(x) = 2/x$ for values of x near 2:

$$P_0(x) = 1$$

$$P_1(x) = 1 - \frac{1}{2}(x-2) = 2 - \frac{x}{2}$$

$$P_2(x) = 1 - \frac{1}{2}(x-2) + \frac{1}{4}(x-2)^2 = 3 - \frac{3x}{2} + \frac{x^2}{4},$$

and so on (Figure 11.11).

EXAMPLE 3 Testing for Convergence Using the Ratio Test

For what values of x do the following power series converge?

(a) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$

(b) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{2n-1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$

(c) $\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

(d) $\sum_{n=0}^{\infty} n!x^n = 1 + x + 2!x^2 + 3!x^3 + \cdots$

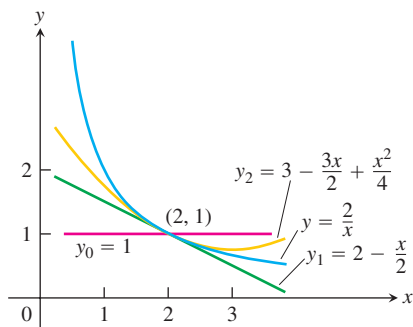
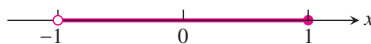


FIGURE 11.11 The graphs of $f(x) = 2/x$ and its first three polynomial approximations (Example 2).

Solution Apply the Ratio Test to the series $\sum |u_n|$, where u_n is the n th term of the series in question.

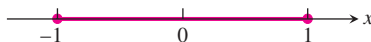
$$(a) \quad \left| \frac{u_{n+1}}{u_n} \right| = \frac{n}{n+1} |x| \rightarrow |x|.$$

The series converges absolutely for $|x| < 1$. It diverges if $|x| > 1$ because the n th term does not converge to zero. At $x = 1$, we get the alternating harmonic series $1 - 1/2 + 1/3 - 1/4 + \cdots$, which converges. At $x = -1$ we get $-1 - 1/2 - 1/3 - 1/4 - \cdots$, the negative of the harmonic series; it diverges. Series (a) converges for $-1 < x \leq 1$ and diverges elsewhere.



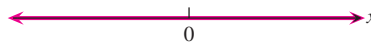
$$(b) \quad \left| \frac{u_{n+1}}{u_n} \right| = \frac{2n-1}{2n+1} x^2 \rightarrow x^2.$$

The series converges absolutely for $x^2 < 1$. It diverges for $x^2 > 1$ because the n th term does not converge to zero. At $x = 1$ the series becomes $1 - 1/3 + 1/5 - 1/7 + \cdots$, which converges by the Alternating Series Theorem. It also converges at $x = -1$ because it is again an alternating series that satisfies the conditions for convergence. The value at $x = -1$ is the negative of the value at $x = 1$. Series (b) converges for $-1 \leq x \leq 1$ and diverges elsewhere.



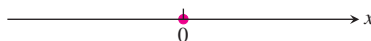
$$(c) \quad \left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| = \frac{|x|}{n+1} \rightarrow 0 \text{ for every } x.$$

The series converges absolutely for all x .



$$(d) \quad \left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{(n+1)!x^{n+1}}{n!x^n} \right| = (n+1)|x| \rightarrow \infty \text{ unless } x = 0.$$

The series diverges for all values of x except $x = 0$.



Example 3 illustrates how we usually test a power series for convergence, and the possible results.

THEOREM 18 The Convergence Theorem for Power Series

If the power series $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots$ converges for $x = c \neq 0$, then it converges absolutely for all x with $|x| < |c|$. If the series diverges for $x = d$, then it diverges for all x with $|x| > |d|$.

Proof Suppose the series $\sum_{n=0}^{\infty} a_n c^n$ converges. Then $\lim_{n \rightarrow \infty} a_n c^n = 0$. Hence, there is an integer N such that $|a_n c^n| < 1$ for all $n \geq N$. That is,

$$|a_n| < \frac{1}{|c|^n} \quad \text{for } n \geq N. \quad (5)$$

Now take any x such that $|x| < |c|$ and consider

$$|a_0| + |a_1 x| + \cdots + |a_{N-1} x^{N-1}| + |a_N x^N| + |a_{N+1} x^{N+1}| + \cdots.$$

There are only a finite number of terms prior to $|a_N x^N|$, and their sum is finite. Starting with $|a_N x^N|$ and beyond, the terms are less than

$$\left| \frac{x}{c} \right|^N + \left| \frac{x}{c} \right|^{N+1} + \left| \frac{x}{c} \right|^{N+2} + \cdots \quad (6)$$

because of Inequality (5). But Series (6) is a geometric series with ratio $r = |x/c|$, which is less than 1, since $|x| < |c|$. Hence Series (6) converges, so the original series converges absolutely. This proves the first half of the theorem.

The second half of the theorem follows from the first. If the series diverges at $x = d$ and converges at a value x_0 with $|x_0| > |d|$, we may take $c = x_0$ in the first half of the theorem and conclude that the series converges absolutely at d . But the series cannot converge absolutely and diverge at one and the same time. Hence, if it diverges at d , it diverges for all x with $|x| > |d|$. ■

To simplify the notation, Theorem 18 deals with the convergence of series of the form $\sum a_n x^n$. For series of the form $\sum a_n (x - a)^n$ we can replace $x - a$ by x' and apply the results to the series $\sum a_n (x')^n$.

The Radius of Convergence of a Power Series

The theorem we have just proved and the examples we have studied lead to the conclusion that a power series $\sum c_n (x - a)^n$ behaves in one of three possible ways. It might converge only at $x = a$, or converge everywhere, or converge on some interval of radius R centered at $x = a$. We prove this as a Corollary to Theorem 18.

COROLLARY TO THEOREM 18

The convergence of the series $\sum c_n (x - a)^n$ is described by one of the following three possibilities:

1. There is a positive number R such that the series diverges for x with $|x - a| > R$ but converges absolutely for x with $|x - a| < R$. The series may or may not converge at either of the endpoints $x = a - R$ and $x = a + R$.
2. The series converges absolutely for every x ($R = \infty$).
3. The series converges at $x = a$ and diverges elsewhere ($R = 0$).

Proof We assume first that $a = 0$, so that the power series is centered at 0. If the series converges everywhere we are in Case 2. If it converges only at $x = 0$ we are in Case 3. Otherwise there is a nonzero number d such that $\sum c_n d^n$ diverges. The set S of values of x for which the series $\sum c_n x^n$ converges is nonempty because it contains 0 and a positive number p as well. By Theorem 18, the series diverges for all x with $|x| > |d|$, so $|x| \leq |d|$ for all $x \in S$, and S is a bounded set. By the Completeness Property of the real numbers (see Appendix 4) a nonempty, bounded set has a least upper bound R . (The least upper bound is the smallest number with the property that the elements $x \in S$ satisfy $x \leq R$.) If $|x| > R \geq p$, then $x \notin S$ so the series $\sum c_n x^n$ diverges. If $|x| < R$, then $|x|$ is not an upper bound for S (because it's smaller than the least upper bound) so there is a number $b \in S$ such that $b > |x|$. Since $b \in S$, the series $\sum c_n b^n$ converges and therefore the series $\sum c_n |x|^n$ converges by Theorem 18. This proves the Corollary for power series centered at $a = 0$.

For a power series centered at $a \neq 0$, we set $x' = (x - a)$ and repeat the argument with x' . Since $x' = 0$ when $x = a$, a radius R interval of convergence for $\sum c_n (x')^n$ centered at $x' = 0$ is the same as a radius R interval of convergence for $\sum c_n (x - a)^n$ centered at $x = a$. This establishes the Corollary for the general case. ■

R is called the **radius of convergence** of the power series and the interval of radius R centered at $x = a$ is called the **interval of convergence**. The interval of convergence may be open, closed, or half-open, depending on the particular series. At points x with $|x - a| < R$, the series converges absolutely. If the series converges for all values of x , we say its radius of convergence is infinite. If it converges only at $x = a$, we say its radius of convergence is zero.

How to Test a Power Series for Convergence

1. Use the Ratio Test (or n th-Root Test) to find the interval where the series converges absolutely. Ordinarily, this is an open interval

$$|x - a| < R \quad \text{or} \quad a - R < x < a + R.$$

2. If the interval of absolute convergence is finite, test for convergence or divergence at each endpoint, as in Examples 3a and b. Use a Comparison Test, the Integral Test, or the Alternating Series Test.
3. If the interval of absolute convergence is $a - R < x < a + R$, the series diverges for $|x - a| > R$ (it does not even converge conditionally), because the n th term does not approach zero for those values of x .

Term-by-Term Differentiation

A theorem from advanced calculus says that a power series can be differentiated term by term at each interior point of its interval of convergence.

THEOREM 19 **The Term-by-Term Differentiation Theorem**

If $\sum c_n(x - a)^n$ converges for $a - R < x < a + R$ for some $R > 0$, it defines a function f :

$$f(x) = \sum_{n=0}^{\infty} c_n(x - a)^n, \quad a - R < x < a + R.$$

Such a function f has derivatives of all orders inside the interval of convergence. We can obtain the derivatives by differentiating the original series term by term:

$$f'(x) = \sum_{n=1}^{\infty} n c_n(x - a)^{n-1}$$

$$f''(x) = \sum_{n=2}^{\infty} n(n-1) c_n(x - a)^{n-2},$$

and so on. Each of these derived series converges at every interior point of the interval of convergence of the original series.

EXAMPLE 4 **Applying Term-by-Term Differentiation**

Find series for $f'(x)$ and $f''(x)$ if

$$\begin{aligned} f(x) &= \frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \cdots + x^n + \cdots \\ &= \sum_{n=0}^{\infty} x^n, \quad -1 < x < 1 \end{aligned}$$

Solution

$$\begin{aligned} f'(x) &= \frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + 4x^3 + \cdots + nx^{n-1} + \cdots \\ &= \sum_{n=1}^{\infty} nx^{n-1}, \quad -1 < x < 1 \\ f''(x) &= \frac{2}{(1-x)^3} = 2 + 6x + 12x^2 + \cdots + n(n-1)x^{n-2} + \cdots \\ &= \sum_{n=2}^{\infty} n(n-1)x^{n-2}, \quad -1 < x < 1 \end{aligned}$$

CAUTION Term-by-term differentiation might not work for other kinds of series. For example, the trigonometric series

$$\sum_{n=1}^{\infty} \frac{\sin(n!x)}{n^2}$$

converges for all x . But if we differentiate term by term we get the series

$$\sum_{n=1}^{\infty} \frac{n! \cos(n!x)}{n^2},$$

which diverges for all x . This is not a power series, since it is not a sum of positive integer powers of x .

Term-by-Term Integration

Another advanced calculus theorem states that a power series can be integrated term by term throughout its interval of convergence.

THEOREM 20 The Term-by-Term Integration Theorem

Suppose that

$$f(x) = \sum_{n=0}^{\infty} c_n (x - a)^n$$

converges for $a - R < x < a + R$ ($R > 0$). Then

$$\sum_{n=0}^{\infty} c_n \frac{(x - a)^{n+1}}{n + 1}$$

converges for $a - R < x < a + R$ and

$$\int f(x) dx = \sum_{n=0}^{\infty} c_n \frac{(x - a)^{n+1}}{n + 1} + C$$

for $a - R < x < a + R$.

EXAMPLE 5 A Series for $\tan^{-1} x$, $-1 \leq x \leq 1$

Identify the function

$$f(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots, \quad -1 \leq x \leq 1.$$

Solution We differentiate the original series term by term and get

$$f'(x) = 1 - x^2 + x^4 - x^6 + \cdots, \quad -1 < x < 1.$$

This is a geometric series with first term 1 and ratio $-x^2$, so

$$f'(x) = \frac{1}{1 - (-x^2)} = \frac{1}{1 + x^2}.$$

We can now integrate $f'(x) = 1/(1 + x^2)$ to get

$$\int f'(x) dx = \int \frac{dx}{1 + x^2} = \tan^{-1} x + C.$$

The series for $f(x)$ is zero when $x = 0$, so $C = 0$. Hence

$$f(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots = \tan^{-1} x, \quad -1 < x < 1. \quad (7)$$

In Section 11.10, we will see that the series also converges to $\tan^{-1} x$ at $x = \pm 1$. ■

Notice that the original series in Example 5 converges at both endpoints of the original interval of convergence, but Theorem 20 can guarantee the convergence of the differentiated series only inside the interval.

EXAMPLE 6 A Series for $\ln(1 + x)$, $-1 < x \leq 1$

The series

$$\frac{1}{1+t} = 1 - t + t^2 - t^3 + \dots$$

converges on the open interval $-1 < t < 1$. Therefore,

$$\begin{aligned} \ln(1+x) &= \int_0^x \frac{1}{1+t} dt = t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \dots \Bigg|_0^x && \text{Theorem 20} \\ &= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, && -1 < x < 1. \end{aligned}$$

It can also be shown that the series converges at $x = 1$ to the number $\ln 2$, but that was not guaranteed by the theorem. ■

USING TECHNOLOGY Study of Series

Series are in many ways analogous to integrals. Just as the number of functions with explicit antiderivatives in terms of elementary functions is small compared to the number of integrable functions, the number of power series in x that agree with explicit elementary functions on x -intervals is small compared to the number of power series that converge on some x -interval. Graphing utilities can aid in the study of such series in much the same way that numerical integration aids in the study of definite integrals. The ability to study power series at particular values of x is built into most Computer Algebra Systems.

If a series converges rapidly enough, CAS exploration might give us an idea of the sum. For instance, in calculating the early partial sums of the series $\sum_{k=1}^{\infty} [1/(2^{k-1})]$ (Section 11.4, Example 2b), Maple returns $S_n = 1.6066\ 95152$ for $31 \leq n \leq 200$. This suggests that the sum of the series is 1.6066 95152 to 10 digits. Indeed,

$$\sum_{k=201}^{\infty} \frac{1}{2^k - 1} = \sum_{k=201}^{\infty} \frac{1}{2^{k-1}(2 - (1/2^{k-1}))} < \sum_{k=201}^{\infty} \frac{1}{2^{k-1}} = \frac{1}{2^{199}} < 1.25 \times 10^{-60}.$$

The remainder after 200 terms is negligible.

However, CAS and calculator exploration cannot do much for us if the series converges or diverges very slowly, and indeed can be downright misleading. For example, try calculating the partial sums of the series $\sum_{k=1}^{\infty} [1/(10^{10}k)]$. The terms are tiny in comparison to the numbers we normally work with and the partial sums, even for hundreds of terms, are miniscule. We might well be fooled into thinking that the series converges. In fact, it diverges, as we can see by writing it as $(1/10^{10})\sum_{k=1}^{\infty} (1/k)$, a constant times the harmonic series.

We will know better how to interpret numerical results after studying error estimates in Section 11.9.

Multiplication of Power Series

Another theorem from advanced calculus states that absolutely converging power series can be multiplied the way we multiply polynomials. We omit the proof.

THEOREM 21 The Series Multiplication Theorem for Power Series

If $A(x) = \sum_{n=0}^{\infty} a_n x^n$ and $B(x) = \sum_{n=0}^{\infty} b_n x^n$ converge absolutely for $|x| < R$, and

$$c_n = a_0 b_n + a_1 b_{n-1} + a_2 b_{n-2} + \cdots + a_{n-1} b_1 + a_n b_0 = \sum_{k=0}^n a_k b_{n-k},$$

then $\sum_{n=0}^{\infty} c_n x^n$ converges absolutely to $A(x)B(x)$ for $|x| < R$:

$$\left(\sum_{n=0}^{\infty} a_n x^n \right) \cdot \left(\sum_{n=0}^{\infty} b_n x^n \right) = \sum_{n=0}^{\infty} c_n x^n.$$

EXAMPLE 7 Multiply the geometric series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \cdots + x^n + \cdots = \frac{1}{1-x}, \quad \text{for } |x| < 1,$$

by itself to get a power series for $1/(1-x)^2$, for $|x| < 1$.

Solution Let

$$A(x) = \sum_{n=0}^{\infty} a_n x^n = 1 + x + x^2 + \cdots + x^n + \cdots = 1/(1-x)$$

$$B(x) = \sum_{n=0}^{\infty} b_n x^n = 1 + x + x^2 + \cdots + x^n + \cdots = 1/(1-x)$$

and

$$\begin{aligned} c_n &= \underbrace{a_0 b_n + a_1 b_{n-1} + \cdots + a_k b_{n-k} + \cdots + a_n b_0}_{n+1 \text{ terms}} \\ &= \underbrace{1 + 1 + \cdots + 1}_{n+1 \text{ ones}} = n + 1. \end{aligned}$$

Then, by the Series Multiplication Theorem,

$$\begin{aligned} A(x) \cdot B(x) &= \sum_{n=0}^{\infty} c_n x^n = \sum_{n=0}^{\infty} (n+1)x^n \\ &= 1 + 2x + 3x^2 + 4x^3 + \cdots + (n+1)x^n + \cdots \end{aligned}$$

is the series for $1/(1-x)^2$. The series all converge absolutely for $|x| < 1$.

Notice that Example 4 gives the same answer because

$$\frac{d}{dx} \left(\frac{1}{1-x} \right) = \frac{1}{(1-x)^2}.$$

■

EXERCISES 11.7

Intervals of Convergence

In Exercises 1–32, **(a)** find the series' radius and interval of convergence. For what values of x does the series converge **(b)** absolutely, **(c)** conditionally?

1. $\sum_{n=0}^{\infty} x^n$
2. $\sum_{n=0}^{\infty} (x + 5)^n$
3. $\sum_{n=0}^{\infty} (-1)^n (4x + 1)^n$
4. $\sum_{n=1}^{\infty} \frac{(3x - 2)^n}{n}$
5. $\sum_{n=0}^{\infty} \frac{(x - 2)^n}{10^n}$
6. $\sum_{n=0}^{\infty} (2x)^n$
7. $\sum_{n=0}^{\infty} \frac{nx^n}{n + 2}$
8. $\sum_{n=1}^{\infty} \frac{(-1)^n (x + 2)^n}{n}$
9. $\sum_{n=1}^{\infty} \frac{x^n}{n\sqrt{n} 3^n}$
10. $\sum_{n=1}^{\infty} \frac{(x - 1)^n}{\sqrt{n}}$
11. $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$
12. $\sum_{n=0}^{\infty} \frac{3^n x^n}{n!}$
13. $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{n!}$
14. $\sum_{n=0}^{\infty} \frac{(2x + 3)^{2n+1}}{n!}$
15. $\sum_{n=0}^{\infty} \frac{x^n}{\sqrt{n^2 + 3}}$
16. $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{\sqrt{n^2 + 3}}$
17. $\sum_{n=0}^{\infty} \frac{n(x + 3)^n}{5^n}$
18. $\sum_{n=0}^{\infty} \frac{nx^n}{4^n(n^2 + 1)}$
19. $\sum_{n=0}^{\infty} \frac{\sqrt{nx}^n}{3^n}$
20. $\sum_{n=1}^{\infty} \sqrt[n]{n} (2x + 5)^n$
21. $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^n x^n$
22. $\sum_{n=1}^{\infty} (\ln n) x^n$
23. $\sum_{n=1}^{\infty} n^n x^n$
24. $\sum_{n=0}^{\infty} n! (x - 4)^n$
25. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} (x + 2)^n}{n2^n}$
26. $\sum_{n=0}^{\infty} (-2)^n (n + 1) (x - 1)^n$

27. $\sum_{n=2}^{\infty} \frac{x^n}{n(\ln n)^2}$ Get the information you need about $\sum 1/(n(\ln n)^2)$ from Section 11.3, Exercise 39.

28. $\sum_{n=2}^{\infty} \frac{x^n}{n \ln n}$ Get the information you need about $\sum 1/(n \ln n)$ from Section 11.3, Exercise 38.

29. $\sum_{n=1}^{\infty} \frac{(4x - 5)^{2n+1}}{n^{3/2}}$

30. $\sum_{n=1}^{\infty} \frac{(3x + 1)^{n+1}}{2n + 2}$

31. $\sum_{n=1}^{\infty} \frac{(x + \pi)^n}{\sqrt{n}}$

32. $\sum_{n=0}^{\infty} \frac{(x - \sqrt{2})^{2n+1}}{2^n}$

In Exercises 33–38, find the series' interval of convergence and, within this interval, the sum of the series as a function of x .

33. $\sum_{n=0}^{\infty} \frac{(x - 1)^{2n}}{4n}$
34. $\sum_{n=0}^{\infty} \frac{(x + 1)^{2n}}{9^n}$
35. $\sum_{n=0}^{\infty} \left(\frac{\sqrt{x}}{2} - 1\right)^n$
36. $\sum_{n=0}^{\infty} (\ln x)^n$
37. $\sum_{n=0}^{\infty} \left(\frac{x^2 + 1}{3}\right)^n$
38. $\sum_{n=0}^{\infty} \left(\frac{x^2 - 1}{2}\right)^n$

Theory and Examples

39. For what values of x does the series

$$1 - \frac{1}{2}(x - 3) + \frac{1}{4}(x - 3)^2 + \cdots + \left(-\frac{1}{2}\right)^n (x - 3)^n + \cdots$$

converge? What is its sum? What series do you get if you differentiate the given series term by term? For what values of x does the new series converge? What is its sum?

40. If you integrate the series in Exercise 39 term by term, what new series do you get? For what values of x does the new series converge, and what is another name for its sum?

41. The series

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!} + \cdots$$

converges to $\sin x$ for all x .

- a. Find the first six terms of a series for $\cos x$. For what values of x should the series converge?
- b. By replacing x by $2x$ in the series for $\sin x$, find a series that converges to $\sin 2x$ for all x .
- c. Using the result in part (a) and series multiplication, calculate the first six terms of a series for $2 \sin x \cos x$. Compare your answer with the answer in part (b).

42. The series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots$$

converges to e^x for all x .

- a. Find a series for $(d/dx)e^x$. Do you get the series for e^x ? Explain your answer.
- b. Find a series for $\int e^x dx$. Do you get the series for e^x ? Explain your answer.
- c. Replace x by $-x$ in the series for e^x to find a series that converges to e^{-x} for all x . Then multiply the series for e^x and e^{-x} to find the first six terms of a series for $e^{-x} \cdot e^x$.

43. The series

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \frac{62x^9}{2835} + \cdots$$

converges to $\tan x$ for $-\pi/2 < x < \pi/2$.

- Find the first five terms of the series for $\ln|\sec x|$. For what values of x should the series converge?
- Find the first five terms of the series for $\sec^2 x$. For what values of x should this series converge?
- Check your result in part (b) by squaring the series given for $\sec x$ in Exercise 44.

44. The series

$$\sec x = 1 + \frac{x^2}{2} + \frac{5}{24}x^4 + \frac{61}{720}x^6 + \frac{277}{8064}x^8 + \cdots$$

converges to $\sec x$ for $-\pi/2 < x < \pi/2$.

- Find the first five terms of a power series for the function $\ln|\sec x + \tan x|$. For what values of x should the series converge?
- Find the first four terms of a series for $\sec x \tan x$. For what values of x should the series converge?

- Check your result in part (b) by multiplying the series for $\sec x$ by the series given for $\tan x$ in Exercise 43.

45. Uniqueness of convergent power series

- Show that if two power series $\sum_{n=0}^{\infty} a_n x^n$ and $\sum_{n=0}^{\infty} b_n x^n$ are convergent and equal for all values of x in an open interval $(-c, c)$, then $a_n = b_n$ for every n . (Hint: Let $f(x) = \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} b_n x^n$. Differentiate term by term to show that a_n and b_n both equal $f^{(n)}(0)/(n!)$.)
- Show that if $\sum_{n=0}^{\infty} a_n x^n = 0$ for all x in an open interval $(-c, c)$, then $a_n = 0$ for every n .

46. **The sum of the series $\sum_{n=0}^{\infty} (n^2/2^n)$** To find the sum of this series, express $1/(1-x)$ as a geometric series, differentiate both sides of the resulting equation with respect to x , multiply both sides of the result by x , differentiate again, multiply by x again, and set x equal to $1/2$. What do you get? (Source: David E. Dobbs' letter to the editor, *Illinois Mathematics Teacher*, Vol. 33, Issue 4, 1982, p. 27.)

47. **Convergence at endpoints** Show by examples that the convergence of a power series at an endpoint of its interval of convergence may be either conditional or absolute.

48. Make up a power series whose interval of convergence is

- $(-3, 3)$
- $(-2, 0)$
- $(1, 5)$.

11.8

Taylor and Maclaurin Series

This section shows how functions that are infinitely differentiable generate power series called Taylor series. In many cases, these series can provide useful polynomial approximations of the generating functions.

Series Representations

We know from Theorem 19 that within its interval of convergence the sum of a power series is a continuous function with derivatives of all orders. But what about the other way around? If a function $f(x)$ has derivatives of all orders on an interval I , can it be expressed as a power series on I ? And if it can, what will its coefficients be?

We can answer the last question readily if we assume that $f(x)$ is the sum of a power series

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} a_n(x-a)^n \\ &= a_0 + a_1(x-a) + a_2(x-a)^2 + \cdots + a_n(x-a)^n + \cdots \end{aligned}$$

with a positive radius of convergence. By repeated term-by-term differentiation within the interval of convergence I we obtain

$$\begin{aligned} f'(x) &= a_1 + 2a_2(x-a) + 3a_3(x-a)^2 + \cdots + na_n(x-a)^{n-1} + \cdots \\ f''(x) &= 1 \cdot 2a_2 + 2 \cdot 3a_3(x-a) + 3 \cdot 4a_4(x-a)^2 + \cdots \\ f'''(x) &= 1 \cdot 2 \cdot 3a_3 + 2 \cdot 3 \cdot 4a_4(x-a) + 3 \cdot 4 \cdot 5a_5(x-a)^2 + \cdots, \end{aligned}$$